Polynomial mixing for infinite-energy solutions of the white-forced Navier–Stokes equation in an infinite pipe

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Abstract

We study the long-time behavior of the infinite-energy solutions to the white-forced Navier–Stokes equation in an infinite pipe $\mathbb{R} \times (-1, 1)$ with no-slip boundary conditions. The equation is globally well-posed in uniformly local spaces satisfying the divergence-free condition. Under suitable degeneracy and spatial decay assumptions on the noise, we prove the uniqueness of the invariant measure and establish polynomial mixing in the dual-Lipschitz metric. This provides an affirmative answer to Open Problem 1 posed by Kuksin and Shirikyan [KS17] in the case of zero cross-sectional flux.

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Contents

1 Introduction **2**
1.1 The main results . . . 3
1.1.1 Global well-posedness in the uniformly local spaces . . . 3
1.1.2 Mixing for the finite-energy solutions . . . 4
1.1.3 Mixing for the infinite-energy solutions . . . 6
1.2 Literature review and the strategy of the proof . . . 6
1.2.1 Literature review . . . 6
1.2.2 The strategy of the proof . . . 7
1.3 Extension to the damped NS equation in the whole space . . . 9

2 Finite-dimensional stabilization **12**
2.1 Asymptotic compactness and finite-dimensional stabilization . . . 12
2.1.1 The Foiaş–Prodi estimate . . . 13
2.1.2 Asymptotic compactness . . . 14
2.1.3 Growth of distance functionals . . . 19
2.2 Uniform finite-dimensional stabilization via compactness of noise coefficients 24

3 Mixing for finite-energy solutions **28**
3.1 Construction of the coupling . . . 28
3.2 Girsanov estimate . . . 29
3.3 Recurrence . . . 31
3.4 Polynomial squeezing and proof of Theorem 1.4 . . . 32

4 Decay estimates for infinite-energy solutions of the homogeneous NS equation **35**
4.1 Estimate on unit time interval . . . 36
4.2 Long-time estimate . . . 38

5 Mixing for infinite-energy solutions **41**
5.1 Probabilistic estimates for the remainder 41
5.2 The asymptotic behavior of the remainder 43
5.3 Proof of Theorem 1.5 44

A A priori energy estimates **46**

B Proof of Proposition 4.2 **48**

C Estimate for the Stokes semigroup in the uniformly local spaces **49**

D Global well-posedness in the uniformly local spaces **54**

1 Introduction

Since the 1990s, substantial progress has been made in the ergodic theory of stochastic dissipative PDEs, particularly concerning the existence and uniqueness of invariant measures and convergence to equilibrium. The 2D Navier–Stokes (NS) equation has played a central role in this development because of its importance in fluid dynamics and mathematical turbulence. Foundational contributions include [FM95, EMS01, KS02, HM06, HM08]; comprehensive accounts and further references can be found in the monograph [KS12] and the surveys [KS17, Deb13]. Much of the classical theory concerns bounded domains or spatially periodic settings, where compact Sobolev embeddings and the discrete spectrum of the Stokes operator provide effective tools for establishing the existence of invariant measures and mixing.

The study of ergodic properties on unbounded domains presents additional difficulties. On domains such as an infinite pipe, Sobolev embeddings are no longer compact and the Stokes operator has continuous spectrum, so the classical compactness arguments and stabilization methods cannot be applied directly. Even the existence of an invariant measure in the finite-energy phase space requires substantial additional analysis; see [BL06, BMO17]. Moreover, unbounded domains naturally accommodate infinite-energy solutions, which need not decay at spatial infinity. These solutions are not only mathematically more challenging, as the standard global energy estimates are no longer directly available, but also necessary for describing physically relevant flows. Indeed, the finite-energy assumption excludes every nontrivial shear flow of the form $(U(t,y),0)$. The importance of understanding the long-time behavior of infinite-energy solutions was highlighted by Kuksin and Shirikyan in [KS17], who posed the following open problem.

Open problem 1. Consider the 2D NS equation in the infinite pipe $\mathbb{R}\times(-1,1)$ with no-slip boundary conditions. The problem is well posed in spaces of bounded velocity fields without decay conditions at infinity; see [AM05, AZ14]. Moreover, it is dissipative and possesses a global attractor in the deterministic setting. It follows that, at least in the case of a bounded stochastic forcing, the random dynamics has an invariant measure. A challenging problem is to prove its uniqueness and mixing.

In this paper, we give an affirmative answer in the case of zero cross-sectional flux by establishing uniqueness of the invariant measure and polynomial mixing in a uniformly local phase space that includes infinite-energy velocity fields, under suitable degeneracy and spatial decay assumptions on the additive white-in-time noise. To be precise, we study the infinite-energy solutions to the NS equation perturbed by an additive white-in-time noise in an infinite

pipe $\mathbb{R}\times(-1,1)$ with the no-slip boundary conditions

$$\begin{cases}\partial_t u+(u\cdot\nabla)u+\nabla\mathfrak{p}-\nu\Delta u=h+\eta, & (x,y)\in D:=\mathbb{R}\times(-1,1),\\ \operatorname{div}u=0,\\ u|_{t=0}=u_0, \qquad u|_{\partial D}=0,\end{cases}\tag{1.1}$$

where $u=(u_1,u_2)$ is the velocity field, $\mathfrak{p}$ is the pressure, and ν is the kinematic viscosity. Define the space

$$H:=\{u\in L^2(D)|\operatorname{div}u=0,\ (u\cdot n)|_{\partial D}=0\}\tag{1.2}$$

endowed with the usual L^2-norm $\|\cdot\|$. The driving process is composed of two parts: $h=h(x)\in H$ is a deterministic forcing term and η is a white-in-time noise in the form

$$\eta(t,x):=\partial_t W(t,x),\qquad W(t,x):=\textstyle\sum_{j\geq1}b_j e_j(x)\beta_j(t),\tag{1.3}$$

where $\{e_j\}_{j\geq1}$ is an orthonormal basis in H, $\{b_j\}_{j\geq1}$ is a sequence of real numbers satisfying $\mathfrak{B}_0:=\sum_{j\geq1}b_j^2<\infty$, and $\{\beta_j\}_{j\geq1}$ is a sequence of mutually independent standard Brownian motions defined on a filtered probability space $(\Omega,\mathcal{F},\mathcal{F}_t,\mathbb{P})$ satisfying the usual conditions; see Definition 2.25 in [KS14].

1.1 The main results

1.1.1 Global well-posedness in the uniformly local spaces

For $1\leq p<\infty$, we define the uniformly local Lebesgue space

$$L^p_b:=\left\{u\in L^p_{\mathrm{loc}}\middle|\|u\|_{L^p_b}:=\sup_{\mathfrak{z}\in\mathbb{R}}\|u\|_{L^p(D_{\mathfrak{z}})}<\infty\right\},\qquad D_{\mathfrak{z}}:=(\mathfrak{z},\mathfrak{z}+1)\times(-1,1),\tag{1.4}$$

and the uniformly local velocity class

$$L^p_{b,\mathrm{div}}:=\left\{u\in L^p_b\middle|\operatorname{div}u=0,\ (u\cdot n)|_{\partial D}=0,\ \int_{-1}^{1}u_1\mathrm{d}x_2=0\text{ for a.e. }x_1\right\}\tag{1.5}$$

with the norm inherited from L^p_b. For $p=2$, we write

$$H_b:=L^2_{b,\mathrm{div}}.\tag{1.6}$$

The condition on the cross-sectional flux $\int_{-1}^{1}u_1\mathrm{d}x_2$ is prescribed, in order to ensure the uniqueness of the infinite-energy solutions; see Remark 5.3 in [Zel07]. In this paper, we focus on the case of zero cross-sectional flux (zero-flux, for short). Introduce the phase space

$$X:=(H_b,d_{\mathrm{loc}}),\tag{1.7}$$

where

$$d_{\mathrm{loc}}(u,v):=\textstyle\sum_{k\geq1}\dfrac{1\wedge\|u-v\|_{L^2(I_k)}}{2^k},\qquad I_k:=(-k,k)\times(-1,1).\tag{1.8}$$

Throughout, X is equipped with its Borel σ-algebra $\mathcal{B}(X)$. It should be noted that although X is not a Polish space, it is a standard measurable space; see Definition A.2.1 in [KS12] and Appendix D. As a result, regular conditional probabilities are well-defined, which provides the measure-theoretic framework for studying X-valued Markov families.[1]

The notion of infinite-energy solutions is stated as follows.

Definition 1.1. A progressively measurable process $u:\Omega\times[0,\infty)\times D\to\mathbb{R}^2$ is said to be an infinite-energy solution of (1.1), if the following conditions are satisfied.

[1] See Section 1.3 in [KS12] for the definition of a Markov family.

1. $u(\cdot) \in C([0,\infty); X)$ holds $\mathbb{P}$-almost surely.
2. For any $T > 0$, it follows that

$$\sup_{t\in[0,T]} \|u(t)\|_{L^2_b}^2 + \sup_{\mathfrak{z}\in\mathbb{R}} \int_0^T \|\nabla u(t)\|_{L^2(D_\mathfrak{z})}^2 \mathrm{d}t < \infty, \qquad \mathbb{P}\text{-a.s.}$$

3. $u|_{\partial D} = 0$ holds for almost every (ω, t).
4. For any test function $\varphi \in C_0^\infty \cap H$, it follows that

$$\langle u(t), \varphi\rangle - \langle u_0, \varphi\rangle - \int_0^t (\langle u\otimes u, \nabla\varphi\rangle + \nu\langle u, \Delta\varphi\rangle + \langle h, \varphi\rangle)\,\mathrm{d}s = \sum_{j\geq 1} \int_0^t b_j \langle e_j, \varphi\rangle \mathrm{d}\beta_j$$

for any $t \geq 0$ $\mathbb{P}$-almost surely.

For the deterministic NS equation, infinite-energy solutions are defined analogously.

Now we present the well-posedness result for equation (1.1) in the infinite-energy space X; see Appendix D for its proof.

Theorem 1.2. *Suppose that $h \in H$,*

$$\mathfrak{B}_1 := \sum_{j\geq 1} b_j^2 \|e_j\|_{H^1}^2 < \infty, \tag{1.9}$$

and the orthonormal basis $\{e_j\}_{j\geq 1}$ satisfies (1.13). Then, for any $u_0 \in X$, there is a unique global infinite-energy solution of (1.1). Moreover, the following statements hold.

1. *For any $\delta, T > 0$, $\sup_{t\in[\delta,T]} \|u(t)\|_\infty < \infty$ holds $\mathbb{P}$-almost surely.*
2. *If $u_0 \in X \cap L^\infty$, then the following decomposition holds*

$$u(t) = w(t) + u_{\mathrm{rem}}(t),$$

where $w(t)$ is an infinite-energy solution of the homogeneous NS equation

$$\begin{cases} \partial_t w + (w\cdot\nabla)w + \nabla \mathfrak{q} - \nu\Delta w = 0, \\ \operatorname{div} w = 0, \\ w|_{t=0} = u_0, \qquad w|_{\partial D} = 0, \end{cases} \tag{1.10}$$

and $u_{\mathrm{rem}}(t)$ satisfies the path regularity

$$u_{\mathrm{rem}}(\cdot) \in C([0,\infty); H) \cap L^2_{\mathrm{loc}}(\mathbb{R}_+; H_0^1 \cap H) \tag{1.11}$$

$\mathbb{P}$-almost surely.

3. *The infinite-energy solutions define a Markov family $(u_t, \mathbb{P}_u^X)$.*

1.1.2 Mixing for the finite-energy solutions

We next establish polynomial mixing for finite-energy solutions. This result has independent interest and provides the basis for our extension to the infinite-energy case. To formulate the result, we introduce the admissible weight functions.

Definition 1.3. A smooth function $\mathfrak{w}(x) : \mathbb{R} \to [1, +\infty)$ is called an admissible weight, if

$$\lim_{x\to\infty} \mathfrak{w}(x) = \lim_{x\to-\infty} \mathfrak{w}(x) = +\infty. \tag{1.12}$$

Throughout the paper, we shall fix an admissible weight $\mathfrak{w}$ and assume that the orthonormal basis $\{e_j\}_{j\geq 1}$ in (1.3) satisfies

$$e_j \in H^2 \cap H_0^1 \cap H, \qquad \mathfrak{w} e_j \in L^2. \tag{1.13}$$

Such an orthonormal basis can be constructed through standard density and Gram–Schmidt arguments; see Section 1.1 in [NZ24b]. The mixing result of the finite-energy solutions is presented below.

Theorem 1.4. *For any $\mathcal{R} > 0$, there exists a positive integer $N = N(\mathcal{R}, \nu, h)$ such that if*

$$\mathfrak{B}_{\mathfrak{w}} := \sum_{j\geq 1} b_j^2(\|\mathfrak{w}e_j\|^2 + \|e_j\|_{H^1}^2) \leq \mathcal{R}^2, \tag{1.14}$$

and

$$b_j \neq 0, \qquad \forall j = 1, \dots, N, \tag{1.15}$$

then the stochastic NS equation (1.1) has a unique invariant measure $\mu \in \mathcal{P}(H)$, where $\mathcal{P}(H)$ denotes the space of all probability measures on H endowed with the weak topology. Moreover, the following statements hold.

1. *The invariant measure μ satisfies*

$$\int_H \|v\|^2 \mu(\mathrm{d}v) < \infty, \qquad \mu(H \cap L^\infty) = 1. \tag{1.16}$$

2. *For any $q \geq 1$, it follows that*

$$\left|\mathbb{E}(f(u(t))) - \int_H f(u)\mu(\mathrm{d}u)\right| \leq C_q \|f\|_{\mathrm{Lip}} (1+t)^{-q} \left(1 + \|u_0\|^2\right) \tag{1.17}$$

   for any $t \geq 0$, $u_0 \in H$, and bounded Lipschitz-continuous function $f : H \to \mathbb{R}$, where C_q depends on q, ν, h and $\{b_j\}_{j\geq 1}$.

3. *In particular, for any probability measure $\lambda \in \mathcal{P}(H)$,*

$$\lim_{t\to\infty} \|\mathcal{D}(u(t;\lambda)) - \mu\|^*_{\mathrm{Lip}(H)} = 0, \tag{1.18}$$

   where $\|\cdot\|^*_{\mathrm{Lip}(H)}$ is the dual-Lipschitz metric defined in (1.28) and $\mathcal{D}(u(t;\lambda))$ denotes the distribution of the solution $u(t)$ issued from the initial data u_0 distributed as λ.

Several remarks are in order. First, an important feature of Theorem 1.4 is that it deals with the no-slip boundary conditions without relying on probabilistic growth of vorticity. Indeed, in [NZ24b], Nersesyan and the author proved polynomial mixing for the damped white-forced NS equation, through a weighted version of the Foiaş–Prodi estimate combined with a probabilistic growth estimate for the vorticity; see Propositions 3.5 and 4.3 therein. However, due to the presence of the no-slip boundary conditions, the resulting boundary terms obstruct such estimates; see [KV14] for similar difficulties in obtaining strong-norm moment bounds. Consequently, Theorem 1.4 is not only an intermediate step towards the infinite-energy case, but also constitutes a substantial improvement of [NZ24b].

As we shall see in Sections 2 and 3, the proof of Theorem 1.4 does not rely essentially on the particular shape of the underlying domain D. In fact, for any unbounded Poincaré-type domain, one can define an admissible weight function $\mathfrak{w} : D \to [1, +\infty)$ by replacing the condition (1.12) with

$$\mathfrak{w}(x, y) \to \infty, \qquad \text{for any } (x, y) \in D \text{ satisfying } (x, y) \to \infty.$$

This, together with the proof presented in Sections 2 and 3, yields polynomial mixing for finite-energy solutions of (1.1) in arbitrary Poincaré-type domain.

It is noteworthy that no size restriction is imposed on the viscosity ν. Moreover, conditions (1.14) and (1.15) are compatible. Indeed, fix $\mathcal{R} > 0$ and let $N_{\mathcal{R}}$ be the integer given by Theorem 1.4. For any given sequence $\{b_j\}_{j\geq 1}$ satisfying (1.14), replace each vanishing coefficient among $b_1, \dots, b_N$ by a nonzero number. Since only finitely many coefficients are modified, the weighted sum in (1.14) remains finite. Multiplying the resulting sequence by a sufficiently small positive constant then ensures that (1.14) holds, while preserving the degeneracy condition (1.15).

The fact that the number N in (1.15) depends on the parameter $\mathcal{R}$, instead of each noise coefficient b_j, plays an essential role in the above adding and rescaling argument. Such parameter dependence is ensured by the admissible weight function $\mathfrak{w}$. Indeed, it is shown in Lemma 2.8 that the set of coefficients $\{b_j\}_{j\geq 1}$ satisfying (1.14) is a compact subset of the sequence space ℓ^2. Consequently, we are able to choose a uniform integer N on this set; see Section 2.2.

For the non-degenerate case, i.e., $b_j \neq 0$ for all $j \geq 1$, the above parameter dependence is not required, and therefore the assumption can be simplified: Theorem 1.4 holds for any orthonormal basis $\{e_j\}_{j\geq 1}$ in H satisfying $e_j \in H_0^1 \cap H$ for all $j \geq 1$ and any noise coefficients $\{b_j\}_{j\geq 1}$ satisfying $\mathfrak{B}_1 < \infty$.

1.1.3 Mixing for the infinite-energy solutions

Finally, we present the mixing result for the infinite-energy solutions.

Theorem 1.5. *For any $\mathcal{R} > 0$, there exists a positive integer $N = N(\mathcal{R}, \nu, h)$ such that if (1.14) and (1.15) are satisfied, then the following statements hold for the invariant measure μ constructed in Theorem 1.4.*

1. *For any $q \geq 1$ and $R > 0$, it follows that*

$$\sup_{\|u_0\|_{L_b^2}\leq R} \left| \mathbb{E}(f(u(t))) - \int_X f(u)\mu(\mathrm{d}u) \right| \leq C_{q,R}\|f\|_{\mathrm{Lip}}\,(1+t)^{-q} \tag{1.19}$$

   for any $t \geq 0$ and bounded Lipschitz-continuous function $f : X \to \mathbb{R}$, where $C_{q,R}$ depends on $q, R, \nu, h, \{b_j\}_{j\geq 1}$.
2. *The probability measure μ is the unique invariant measure of the stochastic NS equation (1.1) in $\mathcal{P}(X)$, where $\mathcal{P}(X)$ denotes the space of all probability measures on X endowed with the weak topology.*
3. *In particular, for any probability measure $\lambda \in \mathcal{P}(X)$,*

$$\lim_{t\to\infty} \|\mathcal{D}(u(t;\lambda)) - \mu\|^*_{\mathrm{Lip}(X)} = 0, \tag{1.20}$$

   where $\mathcal{D}(u(t;\lambda))$ denotes the distribution of the solution $u(t)$ issued from the initial data u_0 distributed as λ.

We make several comments below. Notice that

$$L^\infty_{\mathrm{div}} := \left\{ u \in L^\infty \middle| \operatorname{div} u = 0,\ (u\cdot n)|_{\partial D} = 0,\ \int_{-1}^{1} u_1 \mathrm{d}x_2 = 0 \text{ for a.e. } x_1 \right\} \subset X,$$

so each probability measure on L^∞_{div} can be extended to $\mathcal{P}(X)$. Consequently, Theorem 1.5 provides an affirmative answer to Open problem 1 in the zero-flux case. The case of non-zero flux requires additional treatments, which shall be addressed in a forthcoming paper.

The conditions (1.14) and (1.15) for the degeneracy of the noise are posed for the same reason as discussed in the finite-energy case. Similarly, for the non-degenerate noise, the assumption can be simplified: Theorem 1.5 holds for any orthonormal basis $\{e_j\}_{j\geq 1}$ in H such that $e_j \in H_0^1 \cap H$ for all $j \geq 1$ and any non-degenerate noise coefficients $\{b_j\}_{j\geq 1}$ satisfying $\mathfrak{B}_1 < \infty$.

1.2 Literature review and the strategy of the proof

1.2.1 Literature review

In the finite-energy setting, Brzeźniak and his collaborators [BL06, BMO17] established the existence of invariant measures for the stochastic NS equation in unbounded domains under suitable assumptions. Concerning uniqueness and mixing, Nersesyan [Ner22] proved

uniqueness of the invariant measure and exponential mixing for the NS equation in unbounded Poincaré domains driven by bounded random forcing. More recently, Nersesyan and the author [NZ24b] established uniqueness and polynomial mixing for the damped NS equation on $\mathbb{R}^2$ driven by additive white-in-time noise. However, as discussed previously, their argument does not directly apply to the pipe-flow setting considered in the present paper, due to the presence of the no-slip boundary conditions.

In the infinite-energy setting, the existence of invariant measures was established by Eckmann–Hairer [EH01] and Rougemont [Rou02] for the Ginzburg–Landau equations driven by localized noise and spatially homogeneous noise, respectively. Results on uniqueness and convergence to the invariant measure have also been obtained for Burgers equations and scalar conservation laws on $\mathbb{R}$. For the inviscid Burgers equation, existence of invariant measures and convergence to equilibrium were established under spatially homogeneous Poisson noise in [BCK14]. Analogous results for the viscous Burgers equation were obtained in [BL19, DGR21] under spatially homogeneous random kick forcing and white-in-time Gaussian potential forcing, respectively. For more general viscous scalar conservation laws, [DDG+23] established existence and uniqueness of ergodic, spatially homogeneous invariant measures for each mean in a certain unbounded set. The scalar structure plays an essential role in these arguments, and extending the analysis of uniqueness and mixing to infinite-energy NS flows presents essential difficulties.

1.2.2 The strategy of the proof

Mixing for the finite-energy solutions

The key analytical ingredient in the proof of Theorem 1.4 is a finite-dimensional stabilization estimate. We obtain this estimate by combining local compactness with a probabilistic form of asymptotic compactness. The argument relies only on the estimates for the velocity field and avoids using bounds for the vorticity growth required by the approach of [NZ24b].

Let $u(t)$ be a finite-energy solution, and let $v(t)$ solve the same equation with an additional feedback forcing $aP_N(u-v)$, where P_N denotes the projection operator onto the span of $\{e_1,\dots,e_N\}$. For any finite set $K \subset C_0^\infty \cap H$, we show that there are parameters N, a such that

$$\|u(t)-v(t)\|^2 \le \|u(0)-v(0)\|^2 \exp\left(-ct + C\int_0^t \operatorname{dist}_V(u(s),K)^2 \mathrm{d}s\right), \tag{1.21}$$

where

$$V := H_0^1 \cap H, \qquad \|u\|_V := \|\nabla u\|, \qquad \operatorname{dist}_V(u,K) := \inf_{u'\in K}\|u-u'\|_V;$$

see Proposition 2.1. To obtain this estimate, we bound the convection terms in the equation of $u(t)-v(t)$ via approximating the reference solution $u(t)$ by elements in K. The approximation error is measured by $\operatorname{dist}_V(u,K)$, while the convection terms involving members in K are controlled by a localized compactness estimate and the finite-dimensional feedback; see Lemma 2.3. This provides a substitute for the global high-mode estimate used in the classical Foiaş–Prodi estimate.

The remaining task is to choose K so that the time-integrated distance in (1.21) grows sufficiently slowly. More precisely, for any sufficiently small $\epsilon > 0$, we construct a finite set K such that

$$\sup_{t\ge 0}\left(\int_0^t \operatorname{dist}_V(u(s),K)^2\, ds - \epsilon t\right) \le \rho \tag{1.22}$$

with a failure probability that decays at any polynomial speed, as $\rho \to \infty$; see Proposition 2.2.

The construction of K is based on the asymptotic compactness property established in Proposition 2.4. This property indicates that as $t \to \infty$, the finite-energy solution $u(t)$ approaches a compact subset of H with a high probability and the distance measured by the H-norm $\|\cdot\|$. To connect this H-based distance to the V-based distance in (1.22), we choose a dense subset $\{\mathfrak{u}_j\}_{j\geq 1} \subset C_0^\infty \cap H$ in V, set $K_m = \{0, \mathfrak{u}_1, \dots, \mathfrak{u}_m\}$, and consider

$$\Phi_m(u) = \mathbb{E}_u \int_0^1 \operatorname{dist}_V(u(s), K_m)^2 \mathrm{d}s,$$

where $\mathbb{E}_u$ denotes the expectation with respect to the distribution of the solution $u(t)$ issued from u. These functionals converge to 0 for any $u \in H$ and are uniformly bounded and equicontinuous on bounded subsets of H. Consequently, they become uniformly small near the compact sets indicated in the asymptotic compactness property, as $m \to \infty$. For any $R > 0$, by choosing m sufficiently large, this yields the existence of a finite set K and an integer $\mathfrak{n}_0 \geq 1$ satisfying

$$\sup_{\|u\|\leq R} \mathbb{E}_u \int_{\mathfrak{n}_0}^{\mathfrak{n}_0+1} \operatorname{dist}_V(u(t), K)^2 \mathrm{d}t \leq \epsilon.$$

The Markov property and martingale moment estimates then turn the above estimate into the high-probability growth bound required in (1.22).

Choosing ϵ sufficiently small leads to exponential stabilization on the corresponding high-probability events. Together with the polynomial bounds on the tail probability, this verifies the stabilization component of the coupling argument and leads to polynomial mixing; see Section 3 and [KS12].

Finally, we make some remarks on the asymptotic compactness property. This property was established by Rosa [Ros98] for the deterministic 2D NS equation on unbounded Poincaré domains, in order to prove the existence of a global attractor. This approach was subsequently extended to the stochastic setting by Brzeźniak and his collaborators [BL06, BCL$^+$13], who employed asymptotic compactness to establish the existence of invariant measures and random attractors in unbounded Poincaré domains. More recently, this property was introduced by Nersesyan [Ner22] to the study of mixing for the NS equation with bounded noise in unbounded Poincaré domains. Beyond the NS setting, Xiang and his collaborators [LWX$^+$24, CXZ26, CXZZ26] used control-theoretic methods combined with the asymptotic compactness property to establish exponential mixing for randomly forced dispersive equations in bounded domains, including the nonlinear wave and Schrödinger equations.

Mixing for the infinite-energy solutions

For the infinite-energy case, we first consider $u_0 \in H_b \cap L^\infty$. The key observation is the following decomposition

$$u(t) = w(t) + u_{\mathrm{rem}}(t),$$

where $w(t)$ is an infinite-energy solution of the homogeneous NS equation (1.10) issued from u_0, and the remainder u_{rem} satisfies the stochastic perturbed NS equation around the background flow w:

$$\begin{cases} \partial_t u_{\mathrm{rem}} + (u_{\mathrm{rem}} \cdot \nabla) u_{\mathrm{rem}} + (w \cdot \nabla) u_{\mathrm{rem}} + (u_{\mathrm{rem}} \cdot \nabla) w + \nabla \mathfrak{p}_{\mathrm{rem}} - \nu \Delta u_{\mathrm{rem}} = h + \eta, \\ \operatorname{div} u_{\mathrm{rem}} = 0, \\ u_{\mathrm{rem}}|_{t=0} = 0, \qquad u_{\mathrm{rem}}|_{\partial D} = 0. \end{cases} \tag{1.23}$$

The main analytical task is to control the homogeneous component strongly enough to obtain uniform moment bounds for u_{rem} and a quantitative comparison with finite-energy

dynamics. For the homogeneous component, we establish the following decay estimates: for any $R > 0$ and $\|u_0\|_{L^2_b} \le R$, it follows that

$$\int_0^\infty \|w(s)\|_\infty^2 \mathrm{d}s \lesssim_R \big(1 + \log(1 + \|u_0\|_\infty)\big), \tag{1.24}$$

and

$$\|w(t)\|_\infty^2 \lesssim_R e^{-\alpha_1 t}, \qquad \forall t \ge 1; \tag{1.25}$$

see Proposition 4.1. The first estimate is used to control the cumulative effect of the background flow w on u_{rem}, while the second makes the perturbation quantitatively small at large times.

These two bounds are obtained by using stronger control than the dissipative and smoothing estimates in [AZ14]. Around $t = 0$, we establish analyticity for the Stokes semigroup in suitable weighted spaces and derive estimates in uniformly local spaces. Combined with a fixed-point argument and interpolation, these estimates yield the required square-integrability in time of $\|w\|_\infty$. For large times, a refined weighted energy estimate gives exponential decay to 0 for the background flow w.

The logarithmic dependence on $\|w_0\|_\infty$ in (1.24) is particularly useful. The moment bounds for u_{rem} contain exponential factors involving $\int_0^\infty \|w(s)\|_\infty^2 \mathrm{d}s$. The logarithmic bound therefore yields polynomial dependence on $1 + \|u_0\|_\infty$, together with uniform-in-time moment estimates for u_{rem}; see Proposition 5.1.

To transfer finite-energy mixing to u_{rem}, we introduce a family of finite-energy solutions $u_{\mathrm{app}}(t; T)$ of the stochastic NS equation (1.1) satisfying

$$u_{\mathrm{app}}(T; T) = u_{\mathrm{rem}}(T).$$

We compare the two processes on the interval $[T, T + \sqrt{T}]$. By combining the moment estimates for u_{rem} with the exponential decay (1.25) of the background flow w, we obtain a quantitative approximation for u_{rem} at time $T + \sqrt{T}$, with bounds of any polynomial order; see Proposition 5.2. The length of the comparison interval tends to infinity, allowing the finite-energy dynamics to mix, while the background flow w is already exponentially small at the beginning of the interval. An application of Theorem 1.4 to u_{app} combined with the approximation bounds yields polynomial mixing for $u_0 \in H_b \cap L^\infty$.

Finally, quantitative positive-time smoothing and the Markov property extend this argument from $u_0 \in H_b \cap L^\infty$ to all $u_0 \in H_b$; see Lemma 5.3. This yields polynomial mixing and uniqueness of the invariant measure in $\mathcal{P}(X)$.

1.3 Extension to the damped NS equation in the whole space

The decomposition and comparison argument also extends the finite-energy mixing result in [NZ24b] to infinite-energy solutions of the damped NS equation on $\mathbb{R}^2$. Under the assumptions stated below, this yields uniqueness of the invariant measure and polynomial mixing in a uniformly local phase space, providing an affirmative answer to Open Problem 2 in [KS17]. To be precise, we consider the damped NS equation in the whole space

$$\begin{cases} \partial_t u + (u \cdot \nabla)u - \nu \Delta u + au + \nabla p = h + \eta, & x = (x_1, x_2) \in \mathbb{R}^2, \\ \operatorname{div} u = 0, \\ u|_{t=0} = u_0, \end{cases} \tag{1.26}$$

where $a, \nu > 0$. In this subsection, we take

$$H := \{u \in L^2(\mathbb{R}^2; \mathbb{R}^2) | \operatorname{div} u = 0\},$$

and

$$H_b := \{u \in L^2_{\mathrm{loc}}(\mathbb{R}^2; \mathbb{R}^2) | \operatorname{div} u = 0,\ \|u\|_{L^2_b} < \infty\}, \qquad \|u\|_{L^2_b} := \sup_{x \in \mathbb{R}^2} \|u\|_{L^2(B(x,1))},$$

where $B(x,1) := \{x' \in \mathbb{R}^2 | |x'-x| < 1\}$. The phase space is $X := (H_b, d_{\mathrm{loc}})$, where d_{loc} is defined in (1.8) with I_k replaced by the rectangle $(-k,k)^2$. No spatial decay or zero-mean condition is imposed on the initial velocity and the phase space. The pressure gradient is fixed by the canonical uniformly local extension of the usual pressure-gradient operator; see Definition 3.2 and Remark 3.3 in [Zel13]. In particular, no additional spatially constant and time-dependent pressure gradient is allowed.

We adopt the assumptions on the forcing term introduced in [NZ24b]. More precisely, $\eta := \partial_t \sum_{j\geq 1} b_j e_j \beta_j$, where $\{e_j\}_{j\geq 1}$ is a fixed orthonormal basis in H satisfying

$$e_j \in H^2, \qquad \varphi e_j \in H^1, \qquad \varphi(x) := (1+|x|^2)^{1/4}$$

for all $j \geq 1$. The deterministic force and the noise coefficients satisfy

$$h \in H, \qquad \varphi h \in L^2, \qquad \mathfrak{B}_\varphi := \sum_{j\geq 1} b_j^2 \big(\|\varphi e_j\|^2 + \|\varphi \operatorname{curl} e_j\|^2\big) < \infty,$$

and $b_j \neq 0$ for $j = 1, \dots, N_0$ with N_0 given in Main Theorem in [NZ24b]. The global well-posedness in X and smoothing effect can be established by applying the decomposition strategy in Appendix D combined with the whole-space theory in [Zel13]. For $u_0 \in H_b \cap L^\infty$, let $u = w + u_{\mathrm{rem}}$, where w solves the homogeneous damped NS equation with $w(0) = u_0$, and u_{rem} is the finite-energy remainder. The homogeneous estimates in [Zel13], combined with positive-time smoothing, give

$$\|w(t)\|_\infty \leq C(u_0) e^{-ct}, \qquad \forall t \geq 1.$$

Consequently, by applying the comparison argument in Section 5 combined with the mixing result in [NZ24b] and the smoothing property of (1.26), one gets polynomial mixing in the dual-Lipschitz metric associated with d_{loc}, which gives uniqueness of the invariant measure in $\mathcal{P}(X)$.

Organization of the paper

The paper is organized as follows. Section 2 establishes finite-dimensional stabilization estimates through the asymptotic compactness property of the NS dynamics. This yields polynomial mixing for finite-energy solutions in Section 3. Section 4 derives decay estimates for infinite-energy solutions of the homogeneous NS equation, which are used in Section 5 to establish mixing for infinite-energy solutions. The Appendix collects a priori energy estimates, global well-posedness in uniformly local spaces, estimates for the Stokes semigroup in uniformly local spaces, and proofs of auxiliary results.

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Notations

We shall use the following standard notations.

- For $1 \leq p \leq \infty$, $L^p := L^p(D)$ is the classical Lebesgue space with the norm denoted by $\|\cdot\|_p$, L^p_b is the uniformly local Lebesgue space defined in (1.4), and $L^p_{b,\mathrm{div}}$ is the uniformly local velocity class introduced in (1.5). For $p = 2$, we write $H_b := L^2_{b,\mathrm{div}}$.
- Let H denote the L^2-based velocity class introduced in (1.2). For $p \in (1,\infty)$, we define

$$L^p_{\mathrm{div}} := \{u \in L^p(D) | \operatorname{div} u = 0,\ (u\cdot n)|_{\partial D} = 0\}.$$

- For $k \geq 1$ and $p \in [1, \infty]$, $W^{k,p}$ is the classical Sobolev space with the norm denoted by $\|\cdot\|_{k,p}$, $W_0^{1,p}$ is the space of all functions in $W^{1,p}$ with zero trace, and $W^{-1,p} := (W_0^{1,p'})^*$ is the dual of $W_0^{1,p'}$ with its norm denoted by $\|\cdot\|_{-1,p}$. Here, p' is the Hölder conjugate exponent of p.
- For $k \geq 1$ and $p \in [1, \infty)$, we define the uniformly local Sobolev space

$$W_b^{k,p} := \left\{ u \in W_{\mathrm{loc}}^{k,p} \middle| \|u\|_{W_b^{k,p}} := \sup_{\mathfrak{z} \in \mathbb{R}} \|u\|_{W^{k,p}(D_{\mathfrak{z}})} < \infty \right\}, \quad D_{\mathfrak{z}} := (\mathfrak{z}, \mathfrak{z}+1) \times (-1, 1).$$

  The fractional uniformly local Sobolev space $W_b^{s,p}$ is defined in the same way.
- For $p = 2$, we write $H^s := W^{s,2}$, $H_b^s := W_b^{s,2}$, and $H_0^1 := W_0^{1,2}$. Define $V := H_0^1 \cap H$ endowed with the norm $\|u\|_V := \|\nabla u\|$. Let V^* denote the dual space of V.
- Let $X := (H_b, d_{\mathrm{loc}})$, where the metric d_{loc} is defined in (1.8). Let

$$H_{\mathrm{loc}} = \left\{ u \in L_{\mathrm{loc}}^2 \middle| \operatorname{div} u = 0,\ (u \cdot n)|_{\partial D} = 0,\ \int_{-1}^{1} u_1 \mathrm{d}x_2 = 0 \text{ for a.e. } x_1 \right\}. \tag{1.27}$$

  The space $(H_{\mathrm{loc}}, d_{\mathrm{loc}})$ is Polish.
- For a metric space $(\mathfrak{X}, d_{\mathfrak{X}})$, $\mathcal{A} \subset \mathfrak{X}$, and $x \in \mathfrak{X}$, we define

$$\operatorname{dist}_{\mathfrak{X}}(x, \mathcal{A}) := \inf_{y \in \mathcal{A}} d_{\mathfrak{X}}(x, y).$$

- For a separable metric space $(\mathfrak{X}, d_{\mathfrak{X}})$, $\mathcal{P}(\mathfrak{X})$ denotes the space of all probability measures on $\mathfrak{X}$ endowed with the weak topology. For $f \in C_b(\mathfrak{X})$ and $\mu \in \mathcal{P}(\mathfrak{X})$, we write

$$\langle f, \mu \rangle = \int_{\mathfrak{X}} f(y) \mu(\mathrm{d}y),$$

  where $C_b(\mathfrak{X})$ is the space of all bounded continuous functions. The weak topology in $\mathcal{P}(\mathfrak{X})$ is metrized by the dual-Lipschitz metric

$$\|\mu_1 - \mu_2\|_{\mathrm{Lip}(\mathfrak{X})}^* := \sup_{\substack{f \in \mathrm{Lip}(\mathfrak{X}) \\ \|f\|_{\mathrm{Lip}(\mathfrak{X})} \leq 1}} |\langle f, \mu_1 \rangle - \langle f, \mu_2 \rangle|. \tag{1.28}$$

  Here, $\mathrm{Lip}(\mathfrak{X})$ is the space of all bounded Lipschitz functions endowed with the norm

$$\|f\|_{\mathrm{Lip}(\mathfrak{X})} := \|f\|_\infty + \sup_{x, y \in \mathfrak{X},\ x \neq y} \frac{|f(x) - f(y)|}{d_{\mathfrak{X}}(x, y)}.$$

- For a metric space $(\mathfrak{X}, d_{\mathfrak{X}})$,

$$B_{\mathfrak{X}}(x, R) := \{ y \in \mathfrak{X} | d_{\mathfrak{X}}(x, y) < R \}, \qquad B_{\mathfrak{X}}(R) := B_{\mathfrak{X}}(0, R).$$

- In the current domain $D := \mathbb{R} \times (-1, 1)$, one has the following Poincaré inequality

$$\|\nabla u\|^2 \geq \frac{\pi^2}{4} \|u\|^2, \qquad \forall u \in H_0^1.$$

  To track the dependence of our estimates on the Poincaré inequality, we write $\alpha_0 := \frac{\pi^2}{4}$.
- We write

$$\mathfrak{B}_0 := \textstyle\sum_{j \geq 1} b_j^2, \qquad \mathfrak{B}_{\mathfrak{w}} := \textstyle\sum_{j \geq 1} b_j^2 (\|\mathfrak{w} e_j\|^2 + \|e_j\|_{H^1}^2).$$

2 Finite-dimensional stabilization

In this section, we study the finite-dimensional stabilization problem for the finite-energy solutions of (1.1). By applying the Leray projection Π and writing

$$L := \Pi\Delta, \qquad B(u) := B(u,u), \qquad B(u,v) = \Pi(u\cdot\nabla)v, \tag{2.1}$$

we eliminate the pressure and therefore consider (1.1) as an abstract evolutionary equation

$$\partial_t u - \nu L u + B(u) = h + \eta \tag{2.2}$$

in the energy space H. It is known that (2.2) defines a Markov family $(u_t^H, \mathbb{P}_u^H)$ parameterized by the initial data $u \in H$. In particular, under $\mathbb{P}_u^H$, the process $\{u_t^H\}_{t\geq 0}$ is distributed as the solution of (2.2) issued from u. Throughout Sections 2 and 3, we shall drop the superscript and denote this Markov family by $(u_t, \mathbb{P}_u)$. This will not lead to any confusion. Define the transition operators associated with this Markov family by

$$\mathfrak{P}_t f(u) := \int_H f(v) P_t(u, \mathrm{d}v), \qquad \mathfrak{P}_t : C_b(H) \to C_b(H), \tag{2.3}$$

$$\mathfrak{P}_t^* \lambda(\Gamma) := \int_H P_t(u,\Gamma)\lambda(\mathrm{d}u), \qquad \mathfrak{P}_t^* : \mathcal{P}(H) \to \mathcal{P}(H), \tag{2.4}$$

where

$$P_t(u,\Gamma) := \mathbb{P}_u\{u_t \in \Gamma\} \tag{2.5}$$

is the transition function.

2.1 Asymptotic compactness and finite-dimensional stabilization

The goal of this subsection is to establish finite-dimensional stabilization for (2.2), where the dimension N depends on the noise coefficients $\{b_j\}_{j\geq 1}$. Such dependence shall be removed in the next subsection.

Let $\{v(t)\}_{t\geq 0}$ be the solution of the auxiliary equation

$$\partial_t v - \nu L v + B(v) + a\mathrm{P}_N(v-u) = h + \eta, \qquad v|_{t=0} = u' \in H, \tag{2.6}$$

where $a > 0$, P_N is the projection operator onto the space spanned by $\{e_1, \dots, e_N\}$, and $\{u(t)\}_{t\geq 0}$ is the solution of (2.2) issued from $u \in H$. The main results are presented below.

Proposition 2.1. *For any finite set*

$$K := \{\mathfrak{u}_1, \dots, \mathfrak{u}_l\} \subset C_0^\infty \cap H,$$

there is an integer $N \geq 1$ and a constant $a > 0$ such that for any $u, u' \in H$,

$$\|u(t) - v(t)\|^2 \leq \|u - u'\|^2 \exp\left(-\nu\alpha_0 t + \mathcal{C}_1 \int_0^t \mathrm{dist}_V(u(s), K)^2 \mathrm{d}s\right), \tag{2.7}$$

where $\mathcal{C}_1$ depends only on ν, α_0 is the constant in the Poincaré inequality, and

$$\mathrm{dist}_V(u, K) := \inf_{w\in K} \|u - w\|_V \tag{2.8}$$

with $V := H_0^1 \cap H$.

Proposition 2.2. *Suppose that $\mathfrak{B}_0 := \sum_{j\geq 1} b_j^2 < \infty$. Then, for any $\epsilon > 0$, there is a finite set $K \subset C_0^\infty \cap H$ depending on $\epsilon, \nu, h, \{b_j\}_{j\geq 1}$ such that for any $T, \rho \geq 0$, $q \geq 2$, and $u \in H$,*

$$\mathbb{P}\left\{\sup_{t\geq T}\left(\int_0^t \mathrm{dist}_V(u(s), K)^2 \mathrm{d}s - \epsilon t\right) \geq \rho\right\} \leq C_q \frac{1 + \|u\|^{8q}}{(1 + T + \rho)^q}, \tag{2.9}$$

where C_q depends on $q, \epsilon, \nu, h, \{b_j\}_{j\geq 1}$.

2.1.1 *The Foiaş–Prodi estimate*

This part is devoted to the proof of Proposition 2.1, which is based on the following lemma.

Lemma 2.3. *For any $\epsilon > 0$ and $\mathcal{O} \Subset D$, there is an integer $N \geq 1$ and a constant $C_\epsilon > 0$ such that*

$$\|g\|_{L^2(\mathcal{O})} \leq \epsilon \|\nabla g\| + C_\epsilon \|\mathrm{P}_N g\| \tag{2.10}$$

for any $g \in V$.

Proof. **Step I.** By contradiction, suppose that there is $\epsilon_0 > 0$ such that for any integer $N \geq 1$, one can find a function $g_N \in V$ satisfying

$$\|g_N\|_{L^2(\mathcal{O})} > \epsilon_0 \|\nabla g_N\| + N \|\mathrm{P}_N g_N\|.$$

Let $h_N := \frac{g_N}{\|g_N\|_{L^2(\mathcal{O})}}$. Then, one has

$$\epsilon_0 \|\nabla h_N\| + N \|\mathrm{P}_N h_N\| < 1, \tag{2.11}$$

which, together with the Poincaré inequality, implies that there is a function $h \in V$ such that, up to a subsequence, h_N converges weakly to h, as $N \to \infty$. In particular, for each $j \geq 1$, it follows that

$$\langle h, e_j \rangle = \lim_{N \to \infty} \langle h_N, e_j \rangle.$$

Combining this with (2.11), one gets

$$|\langle h, e_j \rangle| = \lim_{N \to \infty} |\langle h_N, e_j \rangle| \leq \lim_{N \to \infty} \|\mathrm{P}_N h_N\| = 0,$$

and thus $h \equiv 0$.

Step II. Since $\mathcal{O} \Subset D$, by compact Sobolev embedding, it follows

$$h_N \to h$$

strongly in $L^2(\mathcal{O})$, as $N \to \infty$. Combining this with the fact that $\|h_N\|_{L^2(\mathcal{O})} = 1$, one has

$$\|h\|_{L^2(\mathcal{O})} = 1,$$

which is a contradiction. Therefore, estimate (2.10) holds. □

Now we turn to the proof of Proposition 2.1.

Proof of Proposition 2.1. Let $g(t) := u(t) - v(t)$. The equation for g is given by

$$\partial_t g - \nu L g + B(g, u) + B(v, g) + a \mathrm{P}_N g = 0. \tag{2.12}$$

Therefore, by testing against g on both sides, one has

$$\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|g\|^2 + \nu \|\nabla g\|^2 + a \|\mathrm{P}_N g\|^2 = -\langle B(g, u), g \rangle. \tag{2.13}$$

Let $\mathcal{O} := \cup_{i=1}^{l} \operatorname{supp} \mathfrak{u}_i$. For any $i = 1, \ldots, l$, it follows that

$$\langle B(g, u), g \rangle = \langle B(g, u - \mathfrak{u}_i), g \rangle + \langle B(g, \mathfrak{u}_i), g \rangle, \tag{2.14}$$

where, by the Ladyzhenskaya inequality, the Poincaré inequality, and Lemma 2.3, one has

$$|\langle B(g, u - \mathfrak{u}_i), g \rangle| \leq C \|g\| \|\nabla g\| \|\nabla (u - \mathfrak{u}_i)\| \leq \frac{\nu}{4} \|\nabla g\|^2 + \mathcal{C}_1 \|g\|^2 \|u - \mathfrak{u}_i\|_V^2, \tag{2.15}$$

and

$$|\langle B(g,u_i),g\rangle| \le \left(\max_{i=1,\dots,l}\|\nabla \mathfrak{u}_i\|_\infty\right)\|g\|^2_{L^2(\mathcal{O})} \le \epsilon^2 C_K\|\nabla g\|^2 + C_{K,\epsilon}\|\mathrm{P}_N g\|^2. \tag{2.16}$$

By plugging (2.14)–(2.16) into (2.13) and choosing ϵ sufficiently small so that $\epsilon^2 C_K \le \frac{\nu}{8}$, it follows that

$$\frac{\mathrm{d}}{\mathrm{d}t}\|g\|^2 + \nu\|\nabla g\|^2 + 2a\|\mathrm{P}_N g\|^2 \le \mathcal{C}_1\|g\|^2\|u-\mathfrak{u}_i\|_V^2 + C_{\nu,K}\|\mathrm{P}_N g\|^2,$$

which, together with setting $2a = C_{\nu,K}$, taking minimum in i, and applying the Poincaré and Gronwall inequalities, yields (2.7). □

2.1.2 Asymptotic compactness

We establish the following asymptotic compactness property for equation (2.2), which plays an important role in the proof of Proposition 2.2.

Proposition 2.4. *For any $R,\epsilon>0$, there exists a compact subset $\mathcal{K}$ such that for any $\delta>0$, there is an integer $\mathfrak{n}_0 \ge 1$ satisfying*

$$\sup_{\|u\|\le R}\mathbb{P}\{\mathrm{dist}_H(u(t),\mathcal{K})>\delta\} \le \epsilon, \qquad \forall t \ge \mathfrak{n}_0, \tag{2.17}$$

where $\{u(t)\}_{t\ge0}$ is the solution of (2.2) issued from u. Moreover, the compact set $\mathcal{K}$ depends on $R,\epsilon,\nu,h,\{b_j\}_{j\ge1}$, and the integer $\mathfrak{n}_0$ additionally depends on δ and the choice of $\mathcal{K}$.

To prove the above proposition, we need the following lemma, which is a generalization of Proposition 2.4 in [Ner22] to the case of white-in-time noise.

Lemma 2.5. *For any $\{u_n\}_{n\ge1}$ and $\{t_n\}_{n\ge1}$ satisfying*

$$\sup_{n\ge1}\|u_n\|<\infty, \qquad \lim_{n\to\infty} t_n = \infty,$$

the sequence $\{P_{t_n}(u_n,\cdot)\}_{n\ge1}$ is tight, where $P_t(u,\cdot)$ is the transition function defined in (2.5).

Proof. **Step I.** In this step, we establish an auxiliary estimate. Let $\{v_n(t)\}_{t\ge0}$ be the solution of (2.2) issued from $v_n(0)$. Suppose that

$$\mathcal{M} := \sup_n \mathbb{E}\|v_n(0)\|^4 < \infty, \tag{2.18}$$

and

$$v_n(0) \rightharpoonup v(0) \qquad \text{in } H \tag{2.19}$$

almost surely, as $n\to\infty$. Let $\{v(t)\}_{t\ge0}$ be the solution of (2.2) issued from $v(0)$. We claim that

$$\limsup_{n\to\infty}\mathbb{E}\|v_n(t)-v(t)\|^2 \le e^{-\nu\alpha_0 t}\limsup_{n\to\infty}\mathbb{E}\|v_n(0)-v(0)\|^2, \tag{2.20}$$

where α_0 is the constant in the Poincaré inequality.

First, we identify the limit of $\{v_n(\cdot)\}_{n\ge1}$. Let $W(t) := \int_0^t \eta(s)\mathrm{d}s$ denote the driving Wiener process in (2.2). Fix arbitrary $T>0$. By applying (A.6) and (A.7) with $q=1$ and $\gamma\in(0,\frac12)$, one has

$$\sup_{n\ge1}\mathbb{E}\left(\|v_n(\cdot)\|_{L^2(0,T;V)} + \|v_n(\cdot)\|_{W^{\gamma,2}(0,T;V^*)}\right) \lesssim_{\gamma,T,\nu,h,\mathfrak{B}_0} 1+\mathcal{M}.$$

Together with (2.18), Theorem 2.1 in [FG95], and the Prokhorov theorem, this implies that the sequence $\mathcal{D}(v_n(\cdot), v(\cdot), v_n(0), v(0), W)$ is tight in

$$\mathcal{P}\left([L^2(0,T;H_{\rm loc})\cap L^2(0,T;V)_{\rm w}]^2\times H_{\rm w}\times H_{\rm w}\times C([0,T];H)\right),$$

where $H_{\rm w}$ stands for the space H endowed with the weak topology and $L^2(0,T;V)_{\rm w}$ is understood in the same way. Thus, by applying the Jakubowski–Skorohod theorem, see Theorem 1.1 in [Jak98], and changing the probability space if necessary, there exists a sequence $(\tilde v_{n_j}(\cdot), \tilde v'_{n_j}(\cdot), \tilde v_{n_j}(0), \tilde v'_{n_j}(0), \tilde W_{n_j})$, which has the same law as $(v_{n_j}(\cdot), v(\cdot), v_{n_j}(0), v(0), W)$ and converges $\mathbb{P}$-almost surely. Such convergence is sufficient to pass to the limit and therefore indicates that the limit of $\tilde v_{n_j}(\cdot)$ is a solution of (2.2) driven by $\tilde W$ with the initial data distributed as $\mathcal{D}(v(0))$. This, together with the moment estimate

$$\mathbb{E}\|v(0)\|^4\le \mathbb{E}\liminf_{n\to\infty}\|v_n(0)\|^4\le \liminf_{n\to\infty}\mathbb{E}\|v_n(0)\|^4\le \mathcal{M}, \tag{2.21}$$

implies that the limit is a finite-energy solution. On the other hand, by (2.19), the limits of $\tilde v_{n_j}(0)$ and $\tilde v'_{n_j}(0)$ coincide. Together with the pathwise uniqueness of (2.2), one has

$$(\tilde v_{n_j}(\cdot), \tilde v'_{n_j}(\cdot))\to(\tilde v(\cdot),\tilde v(\cdot))\qquad \text{in } [L^2(0,T;H_{\rm loc})]^2 \text{ a.s.}$$

for some process $\tilde v$, which implies

$$v_{n_j}(\cdot)\to v(\cdot)\qquad \text{in } L^2(0,T;H_{\rm loc}) \text{ and in probability,} \tag{2.22}$$

as $j\to\infty$. Consequently, any subsequence of $\{v_n(\cdot)\}_{n\ge1}$ has a subsequence converging to the same limit. Thus, (2.22) holds for the whole sequence $\{v_n(\cdot)\}_{n\ge1}$. Furthermore, by (A.4) with $q=2$, the sequence $\{\|v_n-v\|^2_{L^2((0,T);L^2(\mathcal{O}))}\}_{n\ge1}$ is uniformly integrable. An application of the Vitali convergence theorem, see Theorem 4.5.4 in [Bog07], then implies

$$v_n(\cdot)\to v(\cdot)\qquad \text{in } L^2(\Omega\times(0,T);L^2(\mathcal{O})), \tag{2.23}$$

as $n\to\infty$ for any $\mathcal{O}\Subset D$.

Next, by applying (A.6) with $q=1$, $\{v_n(\cdot)\}_{n\ge1}$ is uniformly bounded in $L^2(\Omega\times(0,T);V)$. Thus, there exists a subsequence n_k and a process $\bar v(\cdot)$ such that

$$v_{n_k}(\cdot)\rightharpoonup \bar v(\cdot)\qquad \text{in } L^2(\Omega\times(0,T);V),$$

as $k\to\infty$. Since $V\subset L^2(\mathcal{O})$ for any $\mathcal{O}\Subset D$, one has

$$v_{n_k}(\cdot)\rightharpoonup \bar v(\cdot)\qquad \text{in } L^2(\Omega\times(0,T);L^2(\mathcal{O})),$$

which, together with (2.23), implies that $v(\cdot)=\bar v(\cdot)$. As $v(\cdot)$ is the only accumulation point of $\{v_n(\cdot)\}_{n\ge1}$ in the weak topology of $L^2(\Omega\times(0,T);V)$, one gets

$$v_n(\cdot)\rightharpoonup v(\cdot)\qquad \text{in } L^2(\Omega\times(0,T);V), \tag{2.24}$$

as $n\to\infty$.

Finally, we claim that for any $t>0$,

$$v_n(t)\rightharpoonup v(t)\qquad \text{in } L^2(\Omega;H), \tag{2.25}$$

as $n\to\infty$. Indeed, by density argument, to establish (2.25), it suffices to show

$$\lim_{n\to\infty}\mathbb{E}|\langle v_n(t)-v(t),\varphi\rangle|=0 \tag{2.26}$$

for any test function $\varphi\in C_0^\infty\cap H$. For simplicity, we write $z_n(t):=v_n(t)-v(t)$ and set $\mathcal{O}:=\operatorname{supp}\varphi$. Then, for any $s\le t\le T$, it follows

$$|\langle z_n(t)-z_n(s),\varphi\rangle|\le\nu\int_s^t|\langle\nabla z_n(r),\nabla\varphi\rangle|\mathrm{d}r+\int_s^t|\langle v_n\otimes v_n(r)-v\otimes v(r),\nabla\varphi\rangle|\mathrm{d}r,$$

where, by applying (A.6) with $q = 1$, one has

$$\mathbb{E}\int_s^t |\langle \nabla z_n(r), \nabla\varphi\rangle| \mathrm{d}r \le \sqrt{t-s}\|\nabla\varphi\| \left(\mathbb{E}\int_0^T \|\nabla z_n(r)\|^2 \mathrm{d}r\right)^{\frac{1}{2}} \lesssim_{T,\nu,h,\mathfrak{B}_0} \sqrt{t-s}\|\nabla\varphi\|(1+\mathcal{M}),$$

and

$$\begin{aligned}
&\mathbb{E}\int_s^t |\langle v_n \otimes v_n(r) - v\otimes v(r), \nabla\varphi\rangle| \mathrm{d}r \\
&\quad \le C\sqrt{\alpha_0}\sqrt{t-s}\|\nabla\varphi\|_\infty \left(\mathbb{E}\int_0^T \|\nabla z_n(r)\|^2 \mathrm{d}r\right)^{\frac{1}{2}} \left(\mathbb{E}\sup_{r\le T}\left(\|v_n(r)\|^2 + \|v(r)\|^2\right)\right)^{\frac{1}{2}} \\
&\quad \lesssim_{T,\nu,h,\mathfrak{B}_0} \sqrt{t-s}\|\nabla\varphi\|_\infty(1+\mathcal{M}).
\end{aligned}$$

Consequently,

$$\mathbb{E}|\langle z_n(t) - z_n(s), \varphi\rangle| \lesssim_{\mathcal{M},T,\nu,h,\mathfrak{B}_0} \sqrt{t-s}\left(\|\nabla\varphi\| + \|\nabla\varphi\|_\infty\right).$$

Together with (2.23), this implies

$$\begin{aligned}
\mathbb{E}|\langle z_n(t),\varphi\rangle| &\le \frac{1}{\delta}\int_{t-\delta}^t \mathbb{E}|\langle z_n(t) - z_n(s),\varphi\rangle| \mathrm{d}s + \frac{1}{\delta}\mathbb{E}\int_{t-\delta}^t |\langle z_n(s),\varphi\rangle| \mathrm{d}s \\
&\lesssim_{\varphi,\mathcal{M},T,\nu,h,\mathfrak{B}_0} \frac{1}{\delta}\int_{t-\delta}^t \sqrt{t-s}\mathrm{d}s + \frac{1}{\sqrt{\delta}}\left(\mathbb{E}\int_{t-\delta}^t \|z_n(s)\|^2_{L^2(\mathcal{O})}\mathrm{d}s\right)^{\frac{1}{2}} \\
&\lesssim_{\varphi,\mathcal{M},T,\nu,h,\mathfrak{B}_0} \sqrt{\delta} + \frac{1}{\sqrt{\delta}}\left(\mathbb{E}\int_0^T \|z_n(s)\|^2_{L^2(\mathcal{O})}\mathrm{d}s\right)^{\frac{1}{2}},
\end{aligned}$$

which ensures (2.26) and thus (2.25).

Now we are in a position to prove (2.20). From (A.1), it follows

$$\begin{aligned}
\mathbb{E}\|v_n(t)\|^2 = e^{-\nu\alpha_0 t}\mathbb{E}\|v_n(0)\|^2 &+ \mathbb{E}\int_0^t e^{-\nu\alpha_0(t-s)}\left(2\langle v_n(s), h\rangle + \mathfrak{B}_0\right)\mathrm{d}s \\
&- \mathbb{E}\int_0^t e^{-\nu\alpha_0(t-s)}\left(2\nu\|\nabla v_n(s)\|^2 - \nu\alpha_0\|v_n(s)\|^2\right)\mathrm{d}s. \qquad (2.27)
\end{aligned}$$

By applying (2.24), one has

$$\lim_{n\to\infty}\mathbb{E}\int_0^t e^{-\nu\alpha_0(t-s)}\langle v_n(s), h\rangle \mathrm{d}s = \mathbb{E}\int_0^t e^{-\nu\alpha_0(t-s)}\langle v(s), h\rangle \mathrm{d}s. \qquad (2.28)$$

Furthermore, by the Poincaré inequality, it follows that

$$\nu\|\nabla v\|^2 \le 2\nu\|\nabla v\|^2 - \nu\alpha_0\|v\|^2 \le 2\nu\|\nabla v\|^2.$$

Consequently, the functional

$$u \mapsto \left(\mathbb{E}\int_0^t e^{-\nu\alpha_0(t-s)}\left(2\nu\|\nabla u(s)\|^2 - \nu\alpha_0\|u(s)\|^2\right)\mathrm{d}s\right)^{\frac{1}{2}}$$

defines an equivalent norm on $L^2((0,t)\times\Omega; V)$, which implies

$$\begin{aligned}
&\mathbb{E}\int_0^t e^{-\nu\alpha_0(t-s)}\left(2\nu\|\nabla v(s)\|^2 - \nu\alpha_0\|v(s)\|^2\right)\mathrm{d}s \\
&\qquad \le \liminf_{n\to\infty}\mathbb{E}\int_0^t e^{-\nu\alpha_0(t-s)}\left(2\nu\|\nabla v_n(s)\|^2 - \nu\alpha_0\|v_n(s)\|^2\right)\mathrm{d}s. \qquad (2.29)
\end{aligned}$$

Since the energy equality (2.27) also holds for $\{v(t)\}_{t\geq 0}$, by applying (2.27)–(2.29), one obtains

$$\limsup_{n\to\infty} \mathbb{E}\left(\|v_n(t)\|^2 - \|v(t)\|^2\right) \leq e^{-\nu\alpha_0 t} \limsup_{n\to\infty} \mathbb{E}\left(\|v_n(0)\|^2 - \|v(0)\|^2\right). \tag{2.30}$$

To conclude the proof of (2.20), notice that $v(t) \in L^2(\Omega; H)$. By combining this with (2.25), one gets

$$\begin{aligned}\limsup_{n\to\infty} \mathbb{E}\|v_n(t) - v(t)\|^2 &= \limsup_{n\to\infty} \mathbb{E}\left(\|v_n(t)\|^2 + \|v(t)\|^2 - 2\langle v_n(t), v(t)\rangle\right) \\ &= \limsup_{n\to\infty} \mathbb{E}\left(\|v_n(t)\|^2 - \|v(t)\|^2\right). \end{aligned}\tag{2.31}$$

On the other hand, by using (2.19) and (2.21), the sequence $\{\langle v_n(0), v(0)\rangle\}_{n\geq 1}$ is uniformly integrable. Since

$$\langle v_n(0), v(0)\rangle \to \|v(0)\|^2,$$

almost surely, as $n \to \infty$, an application of the Vitali convergence theorem then gives

$$\lim_{n\to\infty} \mathbb{E}\langle v_n(0), v(0)\rangle = \mathbb{E}\|v(0)\|^2.$$

As shown in (2.31), this yields

$$\limsup_{n\to\infty} \mathbb{E}\|v_n(0) - v(0)\|^2 = \limsup_{n\to\infty} \mathbb{E}\left(\|v_n(0)\|^2 - \|v(0)\|^2\right).$$

By plugging this and (2.31) into (2.30), one obtains (2.20).

Step II. Fix an arbitrary integer $m \geq 1$. For $t_n \geq m$, we write

$$\mu_n := P_{t_n}(u_n, \cdot), \qquad \lambda_{n,m} := P_{t_n - m}(u_n, \cdot).$$

By applying (A.4) with $q = 2$, it follows that

$$\int_H \|v\|^4 \lambda_{n,m}(\mathrm{d}v) \leq \|u_n\|^4 + C_{\nu,h,\mathfrak{B}_0} \leq \sup_{n\geq 1} \|u_n\|^4 + C_{\nu,h,\mathfrak{B}_0}. \tag{2.32}$$

Together with the Jakubowski–Skorohod representation theorem, this implies that there are H-valued random variables $\{\mathfrak{v}_{n_k,m}\}_{k\geq 1}$ and $\mathfrak{v}_m$ satisfying

$$\mathcal{D}(\mathfrak{v}_{n_k,m}) = \lambda_{n_k,m}, \qquad \forall k \geq 1, \tag{2.33}$$

and

$$\mathfrak{v}_{n_k,m} \rightharpoonup \mathfrak{v}_m \qquad \text{in } H \tag{2.34}$$

almost surely, as $k \to \infty$. After a diagonal argument, one may find a common subsequence (still denoted by) n_k such that (2.33) and (2.34) hold for all $m \geq 1$. Let $\{\mathfrak{v}_{n_k,m}(t)\}_{t\geq 0}$ and $\{\mathfrak{v}_m(t)\}_{t\geq 0}$ denote the solutions of (2.2) issued from $\mathfrak{v}_{n_k,m}$ and $\mathfrak{v}_m$, respectively. Since (2.18) and (2.19) hold for $\{\mathfrak{v}_{n_k,m}\}_{k\geq 1}$ and $\mathfrak{v}_m$, due to (2.32) and (2.34), by applying (2.20), it follows

$$\begin{aligned}\limsup_{k\to\infty} \mathbb{E}\|\mathfrak{v}_{n_k,m}(t) - \mathfrak{v}_m(t)\|^2 &\leq e^{-\nu\alpha_0 t} \limsup_{k\to\infty} \mathbb{E}\|\mathfrak{v}_{n_k,m}(0) - \mathfrak{v}_m(0)\|^2 \\ &\leq e^{-\nu\alpha_0 t} \left(\sup_{n\geq 1} \|u_n\|^4 + C_{\nu,h,\mathfrak{B}_0}\right)^{\frac{1}{2}}.\end{aligned}$$

Combining this with the fact

$$\mathcal{D}(\mathfrak{v}_{n_k,m}(m)) = \mu_{n_k}, \qquad \forall k \geq 1$$

and the definition (1.28) of the dual-Lipschitz metric, one gets

$$\limsup_{k\to\infty} \|\mu_{n_k} - \lambda_m\|^*_{\mathrm{Lip}(H)} \le \left(\sup_{n\ge 1} \|u_n\|^4 + C_{\nu,h,\mathfrak{B}_0}\right)^{\frac{1}{4}} \exp\left(-\frac{\nu\alpha_0}{2} m\right), \tag{2.35}$$

where $\lambda_m := \mathcal{D}(\mathfrak{v}_m(m))$.

Step III. Since m is arbitrary in (2.35), by the triangle inequality, one has

$$\limsup_{k,j\to\infty} \|\mu_{n_k} - \mu_{n_j}\|^*_{\mathrm{Lip}(H)} \le 2\left(\sup_{n\ge 1} \|u_n\|^4 + C_{\nu,h,\mathfrak{B}_0}\right)^{\frac{1}{4}} \exp\left(-\frac{\nu\alpha_0}{2} m\right).$$

Together with sending $m \to \infty$, this shows that $\{\mu_{n_k}\}_{k\ge 1}$ is a Cauchy sequence and thus that $\{\mu_n\}_{n\ge 1}$ is compact. An application of the Prokhorov theorem then completes the proof. □

Now we turn to the proof of Proposition 2.4.

Proof of Proposition 2.4. **Step I.** Fix $R, \epsilon > 0$. Define $\mathscr{K}_R$ as the set of probability measures λ such that there are $\{u_n\}_{n\ge 1}$ and $\{t_n\}_{n\ge 1}$ satisfying

$$\sup_{n\ge 1} \|u_n\| \le R, \qquad \lim_{n\to\infty} t_n = \infty,$$

and

$$\lambda = \lim_{n\to\infty} P_{t_n}(u_n, \cdot).$$

We claim that $\mathscr{K}_R$ is compact. To see this, for any sequence $\{\lambda_k\}_{k\ge 1} \subset \mathscr{K}_R$, choose

$$\tilde{\lambda}_k := P_{t_k}(u_k, \cdot)$$

such that $t_k \ge k$, $\|u_k\| \le R$, and

$$\|\lambda_k - \tilde{\lambda}_k\|^*_{\mathrm{Lip}(H)} \le \frac{1}{k}. \tag{2.36}$$

By Lemma 2.5, $\{\tilde{\lambda}_k\}_{k\ge 1}$ has a convergent subsequence $\{\tilde{\lambda}_{k_j}\}_{j\ge 1}$. This, together with (2.36), implies the convergence of $\{\lambda_{k_j}\}_{j\ge 1}$ and thus the compactness of $\mathscr{K}_R$.

Step II. By applying the Prokhorov theorem, there exists a compact subset $\mathcal{K} = \mathcal{K}_{R,\epsilon}$ of H such that

$$\sup_{\lambda\in\mathscr{K}_R} \lambda(\mathcal{K}^c) \le \frac{\epsilon}{2}.$$

We claim that (2.17) holds for the above set $\mathcal{K}$. Indeed, by contradiction, suppose that there is $\delta_0 > 0$, $\{u_n\}_{n\ge 1}$, and $\{t_n\}_{n\ge 1}$ satisfying

$$\sup_{n\ge 1} \|u_n\| \le R, \qquad \lim_{n\to\infty} t_n = \infty,$$

and

$$\mathbb{P}\{\mathrm{dist}_H(u(t_n; u_n), \mathcal{K}) > \delta_0\} > \epsilon,$$

where $\{u(t;u)\}_{t\ge 0}$ denotes the solution of (2.2) issued from u. Choose $\lambda \in \mathscr{K}_R$ such that, up to a subsequence,

$$\lambda = \lim_{n\to\infty} P_{t_n}(u_n, \cdot).$$

Since $F_\delta := \{v| \operatorname{dist}_H(v, \mathcal{K}) \geq \delta\}$ is closed and $F_\delta \subset \mathcal{K}^c$, by the Portmanteau theorem, one has

$$\begin{aligned}\epsilon &\leq \limsup_{n\to\infty} \mathbb{P}\{\operatorname{dist}_H(u(t_n; u_n), \mathcal{K}) \geq \delta_0\} = \limsup_{n\to\infty} P_{t_n}(u_n, F_\delta)\\ &\leq \lambda(F_\delta) \leq \lambda(\mathcal{K}^c) \leq \frac{\epsilon}{2},\end{aligned}$$

which is a contradiction. This gives the validity of (2.17). □

2.1.3 Growth of distance functionals

The goal of this part is to establish Proposition 2.2. To this end, we introduce the following lemma, which provides an estimate on the expected growth for the distance functional in (2.9) over a unit time interval.

Lemma 2.6. *For any $R, \epsilon > 0$, there is an integer $\mathfrak{n}_0 \geq 1$ and a finite set*

$$K := \{0, \mathfrak{u}_1, \ldots, \mathfrak{u}_l\} \subset C_0^\infty \cap H$$

such that

$$\sup_{\|u\|\leq R} \mathbb{E} \int_{\mathfrak{n}_0}^{\mathfrak{n}_0+1} \operatorname{dist}_V(u(t), K)^2 \mathrm{d}t \leq \epsilon, \tag{2.37}$$

where $\{u(t)\}_{t\geq 0}$ is the solution of (2.2) issued from $u \in H$. Moreover, the finite set K and the integer $\mathfrak{n}_0$ depend on $R, \epsilon, \nu, h, \{b_j\}_{j\geq 1}$.

Proof. **Step I.** First, we construct the finite set K in (2.37). Let $\{\mathfrak{u}_j\}_{j\geq 1} \subset C_0^\infty \cap H$ be a dense sequence in V. Define

$$K_m := \{0, \mathfrak{u}_1, \ldots, \mathfrak{u}_m\},$$

and

$$\Phi_m(u) := \mathbb{E}_u \int_0^1 \operatorname{dist}_V(u_t, K_m)^2 \mathrm{d}t,$$

where $(u_t, \mathbb{P}_u)$ is the Markov family associated with (2.2). We claim that for any $R > 0$, the sequence $\{\Phi_m\}_{m\geq 1}$ is uniformly bounded and equicontinuous on $B_H(R)$. Indeed, since $0 \in K_m$, by applying (A.6) with $q = 1$, one has

$$\Phi_m(u) \leq \mathbb{E}_u \int_0^1 \|u_t\|_V^2 \mathrm{d}t \lesssim_{\nu,h,\mathfrak{B}_0} 1 + \|u\|^2, \tag{2.38}$$

which ensures the uniform boundedness on $B_H(R)$. On the other hand, by (A.8), (2.38), and the Minkowski inequality, for any $u, u' \in B_H(R)$, it follows that

$$\begin{aligned}|\Phi_m(u) - \Phi_m(u')| &\lesssim_{\nu,h,\mathfrak{B}_0} (1 + \|u\| + \|u'\|) \left|\sqrt{\Phi_m(u)} - \sqrt{\Phi_m(u')}\right|\\ &\lesssim_{\nu,h,\mathfrak{B}_0,R} \left(\mathbb{E} \int_0^1 \|u(t) - u'(t)\|_V^2 \mathrm{d}t\right)^{\frac{1}{2}}\\ &\lesssim_{\nu,h,\mathfrak{B}_0,R} \|u - u'\|^{\frac{1}{2}} + \|u - u'\|^{\frac{c_0}{2}}.\end{aligned}$$

This gives the uniform equicontinuity of $\{\Phi_m\}_{m\geq 1}$ on $B_H(R)$.

Now fix $R, \epsilon > 0$. Let $\mathcal{K}$ be the compact subset in Proposition 2.4 corresponding to R, ϵ', where ϵ' is to be determined in the next step. Since $\{\mathfrak{u}_j\}_{j\geq 1}$ is dense in V, by the dominated convergence theorem combined with (2.38), it follows

$$\lim_{m\to\infty} \Phi_m(u) = 0$$

for any $u \in H$. This, together with the uniform equicontinuity of $\{\Phi_m\}_{m\geq 1}$, gives

$$\lim_{m\to\infty} \sup_{u\in\mathcal{K}} \Phi_m(u) = 0.$$

In particular, choose m sufficiently large so that

$$\sup_{u\in\mathcal{K}} \Phi_m(u) < \frac{\epsilon}{4}, \tag{2.39}$$

and set $K := K_m$.

Step II. Since Φ_m is continuous and $\mathcal{K}$ is compact, there is $\delta > 0$ such that if $\mathrm{dist}_H(v, \mathcal{K}) \leq \delta$, then $\Phi_m(v) < \frac{\epsilon}{2}$. Let $\mathfrak{n}_0 \geq 1$ be the integer in Proposition 2.4 corresponding to δ. Introduce

$$\mathcal{G}_{\mathfrak{n}_0} := \{\mathrm{dist}_H(u_{\mathfrak{n}_0}, \mathcal{K}) \leq \delta\}.$$

By the Markov property, Proposition 2.4, (2.38), and (A.4) with $q = 2$, for any $u \in B_H(R)$, it follows

$$\begin{aligned}\mathbb{E}_u \int_{\mathfrak{n}_0}^{\mathfrak{n}_0+1} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t = \mathbb{E}_u \Phi_m(u_{\mathfrak{n}_0}) &= \mathbb{E}_u \left(\mathbb{I}_{\mathcal{G}_{\mathfrak{n}_0}} \Phi_m(u_{\mathfrak{n}_0})\right) + \mathbb{E}_u \left(\mathbb{I}_{\mathcal{G}^c_{\mathfrak{n}_0}} \Phi_m(u_{\mathfrak{n}_0})\right) \\ &< \frac{\epsilon}{2} + C_{\nu,h,\mathfrak{B}_0} \sqrt{\epsilon'} \left(\mathbb{E}_u \left(1 + \|u_{\mathfrak{n}_0}\|^4\right)\right)^{\frac{1}{2}} < \frac{\epsilon}{2} + C_{\nu,h,\mathfrak{B}_0,R} \sqrt{\epsilon'}\end{aligned}$$

Together with choosing ϵ' sufficiently small, this implies (2.37). □

Now we turn to the proof of Proposition 2.2.

Proof. **Step I.** We derive some auxiliary estimates. Fix arbitrary $\epsilon > 0$. Applying Lemma 2.6 with ϵ replaced by $\frac{\epsilon}{16}$ and R to be chosen in the next step, there is an integer $\mathfrak{n}_0 \geq 1$ and a finite set $K := \{0, \mathfrak{u}_1, \ldots, \mathfrak{u}_l\} \subset C_0^\infty \cap H$ such that (2.37) holds. On the other hand, for $u \notin \overline{B_H(R)}$, by the Markov property, (A.4), and (A.6) with $q = 1$, it follows

$$\begin{aligned}\mathbb{E}_u \int_{\mathfrak{n}_0}^{\mathfrak{n}_0+1} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \leq \mathbb{E}_u \int_{\mathfrak{n}_0}^{\mathfrak{n}_0+1} \|u_t\|_V^2 \mathrm{d}t &= \mathbb{E}_u \left(\mathbb{E}_{u_{\mathfrak{n}_0}} \left(\int_0^1 \|u_t\|_V^2 \mathrm{d}t\right)\right) \\ &\lesssim_{\nu,h,\mathfrak{B}_0} \mathbb{E}_u (1 + \|u_{\mathfrak{n}_0}\|^2) \lesssim_{\nu,h,\mathfrak{B}_0} 1 + \|u\|^2 \lesssim_{\nu,h,\mathfrak{B}_0} \frac{(1 + \|u\|^2)^2}{R^2}.\end{aligned}$$

Consequently, by combining this with (2.37), for any $u \in H$, one has

$$\mathbb{E}_u \int_{\mathfrak{n}_0}^{\mathfrak{n}_0+1} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \leq \frac{\epsilon}{16} + \mathcal{C}_2 \frac{(1 + \|u\|^2)^2}{R^2}, \tag{2.40}$$

where the constant $\mathcal{C}_2$ depends on $\nu, h, \mathfrak{B}_0$.

Step II. In this step, we derive an estimate for the expected growth of the distance functional in (2.9) over larger time-blocks. Let $M \geq \mathfrak{n}_0$ be an integer. It follows that

$$\begin{aligned}\mathbb{E}_u \int_0^M \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t = \mathbb{E}_u \int_0^{\mathfrak{n}_0} &\mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \\ &+ \textstyle\sum_{j=0}^{M-1-\mathfrak{n}_0} \mathbb{E}_u \int_{\mathfrak{n}_0+j}^{\mathfrak{n}_0+1+j} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t,\end{aligned} \tag{2.41}$$

where, by the Markov property, (2.40), and (A.4) with $q = 2$, it follows

$$\begin{aligned}\mathbb{E}_u \int_{\mathfrak{n}_0+j}^{\mathfrak{n}_0+1+j} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t &= \mathbb{E}_u \left(\mathbb{E}_{u_j} \int_{\mathfrak{n}_0}^{\mathfrak{n}_0+1} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \right) \\ &\leq \frac{\epsilon}{16} + \frac{\mathcal{C}_2}{R^2} \mathbb{E}_u \left(1 + \|u_j\|^2\right)^2 \\ &\leq \frac{\epsilon}{16} + \frac{C_{\nu,h,\mathfrak{B}_0}}{R^2} + \frac{\mathcal{C}_2}{R^2} e^{-\nu\alpha_0 j} \left(1 + \|u\|^2\right)^2.\end{aligned}$$

On the other hand, by (A.6) with $q = 1$, one has

$$\mathbb{E}_u \int_0^{\mathfrak{n}_0} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \leq C_{\nu,h,\mathfrak{B}_0}(\mathfrak{n}_0 + \|u\|^2) \leq C_{\nu,h,\mathfrak{B}_0} \mathfrak{n}_0 \left(1 + \|u\|^2\right)^2.$$

Plugging the above estimates into (2.41), one gets

$$\mathbb{E}_u \int_0^M \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \leq \mathcal{C}_3 \left(\mathfrak{n}_0 + R^{-2}\right) \left(1 + \|u\|^2\right)^2 + \left(\frac{\epsilon}{16} + \mathcal{C}_3 R^{-2}\right) M \tag{2.42}$$

for some constant $\mathcal{C}_3$ depending on $\nu, h, \mathfrak{B}_0$. Choosing R sufficiently large so that

$$\frac{\epsilon}{16} + \mathcal{C}_3 R^{-2} \leq \frac{\epsilon}{8},$$

and writing

$$\mathcal{V}(u) := \left(1 + \|u\|^2\right)^2,$$

it follows

$$\mathbb{E}_u \int_0^M \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \leq \mathcal{C}_4 \mathcal{V}(u) + \frac{\epsilon}{8} M, \tag{2.43}$$

where $\mathcal{C}_4$ is a constant depending on $\epsilon, \nu, h, \{b_j\}_{j\geq 0}$.

Step Ⅲ. We shall divide the whole time interval into blocks of length M and apply growth estimate on $[0, M]$ combined with the Markov property. However, estimate (2.43) is not sufficient, since this would yield a sum of Lyapunov functionals $\sum_{j\geq 0} \mathcal{V}(u_{jM})$. To this end, we aim to improve (2.43). Let $\mathfrak{P}_t$ and $\mathfrak{P}_t^*$ be the Markov operators introduced in (2.3) and (2.4), respectively. Applying (A.4) combined with the Markov property, it follows that

$$\mathfrak{P}_M \mathcal{V}(u) \leq \mathfrak{c}^M \mathcal{V}(u) + \frac{1 - \mathfrak{c}^M}{1 - \mathfrak{c}} C_{\nu,h,\mathfrak{B}_0}, \qquad \mathfrak{c} := e^{-\nu\alpha_0}. \tag{2.44}$$

By multiplying both sides by $\mathfrak{a}_M := (1 - \mathfrak{c}^M)^{-1} \mathcal{C}_4$, this implies

$$\mathfrak{a}_M \left(\mathfrak{P}_M \mathcal{V}(u) - \mathcal{V}(u)\right) \leq -\mathcal{C}_4 \mathcal{V}(u) + \frac{\mathcal{C}_4 C_{\nu,h,\mathfrak{B}_0}}{1 - \mathfrak{c}}.$$

Adding the above estimate to (2.43), one eliminates the Lyapunov functional $\mathcal{C}_4 \mathcal{V}(u)$ on the right-hand side of (2.43) and thereby obtains

$$\mathbb{E}_u \int_0^M \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t + \mathfrak{a}_M \left(\mathfrak{P}_M \mathcal{V}(u) - \mathcal{V}(u)\right) \leq \frac{\epsilon}{8} M + \frac{\mathcal{C}_4 C_{\nu,h,\mathfrak{B}_0}}{1 - \mathfrak{c}}.$$

Consequently, by choosing $M \geq 2$ sufficiently large, it follows

$$\mathbb{E}_u \int_0^M \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t + \mathfrak{a}_M \left(\mathfrak{P}_M \mathcal{V}(u) - \mathcal{V}(u)\right) \leq \frac{\epsilon}{4} M. \tag{2.45}$$

Step IV. In this step, we derive a pathwise estimate for the distance functional

$$\mathfrak{I}(kM) := \int_0^{kM} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t.$$

Define

$$\Gamma_j := \mathfrak{I}((j+1)M) - \mathfrak{I}(jM) + \mathfrak{a}_M \left(\mathcal{V}(u_{(j+1)M}) - \mathcal{V}(u_{jM})\right). \tag{2.46}$$

Applying the Markov property combined with (2.45), it follows

$$\mathbb{E}_u(\Gamma_j|\mathcal{F}_{jM}) = \mathbb{E}_{u_{jM}} \int_0^M \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t + \mathfrak{a}_M(\mathfrak{P}_M \mathcal{V}(u_{jM}) - \mathcal{V}(u_{jM})) \le \frac{\epsilon}{4} M. \tag{2.47}$$

By summing over j from 0 to $k-1$ on both sides of (2.46) and using (2.47), one gets

$$\mathfrak{I}(kM) + \mathfrak{a}_M \left(\mathcal{V}(u_{kM}) - \mathcal{V}(u)\right) = \textstyle\sum_{j=0}^{k-1} \Gamma_j \le \frac{\epsilon k M}{4} + \sum_{j=0}^{k-1} d_j \tag{2.48}$$

where

$$d_j := \Gamma_j - \mathbb{E}_u(\Gamma_j|\mathcal{F}_{jM})$$

is a sequence of martingale differences.

Step V. We establish moment estimates for the martingale differences. Fix $q \ge 2$. By the conditional Jensen inequality, (A.4), (A.6), the Markov property, and the fact that $0 \in K$, one has

$$\begin{aligned}
\mathbb{E}_u(|d_j|^q|\mathcal{F}_{jM}) &\lesssim_q \mathbb{E}_u(|\Gamma_j|^q|\mathcal{F}_{jM}) \\
&\lesssim_q \mathbb{E}_u \left(\left(\int_{jM}^{(j+1)M} \|u_t\|_V^2 \mathrm{d}t \right)^q \middle| \mathcal{F}_{jM} \right) \\
&\quad + \mathfrak{a}_M^q \mathbb{E}_u \left(\mathcal{V}(u_{(j+1)M})^q \middle| \mathcal{F}_{jM} \right) + \mathfrak{a}_M^q \mathcal{V}(u_{jM})^q \\
&\lesssim_{q,\epsilon,\nu,h,\{b_j\}_{j\ge1}} \mathcal{V}(u_{jM})^q.
\end{aligned}$$

By taking expectation on both sides and using (A.4), it follows

$$\sup_{j\ge0} \mathbb{E}_u |d_j|^q \lesssim_{q,\epsilon,\nu,h,\{b_j\}_{j\ge1}} \mathcal{V}(u)^q. \tag{2.49}$$

As a result, by the discrete-time Burkholder–Gundy–Davis inequality (Theorem 6.3.6 in [Str10]) and the generalized Minkowski inequality, one obtains

$$\begin{aligned}
\mathbb{E}_u \left(\max_{l\le k} \left| \textstyle\sum_{j=0}^{l-1} d_j \right|^q \right) &\lesssim_q \mathbb{E}_u \left(\textstyle\sum_{j=0}^{k-1} |d_j|^2 \right)^{\frac{q}{2}} \lesssim_q \left(\textstyle\sum_{j=0}^{k-1} \left(\mathbb{E}_u |d_j|^q\right)^{\frac{2}{q}} \right)^{\frac{q}{2}} \\
&\lesssim_{q,\epsilon,\nu,h,\{b_j\}_{j\ge1}} k^{\frac{q}{2}} \mathcal{V}(u)^q.
\end{aligned} \tag{2.50}$$

Step VI. Finally, we are ready to prove (2.9). For $t \ge 0$, define $k(t) := \lceil \frac{t}{M} \rceil$. Then, from (2.48), it follows that

$$\begin{aligned}
\mathfrak{I}(t) &\le \mathfrak{I}(k(t)M) \le \mathfrak{a}_M \mathcal{V}(u) + \frac{\epsilon}{4} k(t) M + \textstyle\sum_{j=0}^{k(t)-1} d_j \\
&\le \mathcal{C}_5 \mathcal{V}(u) + \frac{\epsilon}{4} t + \textstyle\sum_{j=0}^{k(t)-1} d_j
\end{aligned}$$

for some constant $\mathcal{C}_5$ depending on $\epsilon, \nu, h, \{b_j\}_{j\ge1}$. This implies

$$\mathfrak{I}(t) - \epsilon t \le \mathcal{C}_5 \mathcal{V}(u) + \textstyle\sum_{j=0}^{k(t)-1} d_j - \frac{3\epsilon t}{4}. \tag{2.51}$$

Introduce a constant $\mathcal{C}_6$, which is determined in (2.53). We distinguish two different cases.
Case I. Suppose that $1+T+\rho \le \mathcal{C}_6\mathcal{V}(u)$. Then, since $\mathcal{V}(u)\ge 1$, one has

$$\frac{\mathcal{V}(u)^{2q}}{(1+T+\rho)^q} \ge \mathcal{C}_6^{-q}.$$

This implies

$$\mathbb{P}_u\left\{\sup_{t\ge T}\left(\int_0^t \mathrm{dist}_V(u(s),K)^2\mathrm{d}s-\epsilon t\right)\ge \rho\right\} \le 1 \lesssim_{q,\epsilon,\nu,h,\{b_j\}_{j\ge 1}} \frac{\mathcal{V}(u)^{2q}}{(1+T+\rho)^q}. \tag{2.52}$$

Case II. Now assume that $1+T+\rho > \mathcal{C}_6\mathcal{V}(u)$. Choose $\mathcal{C}_6$ sufficiently large so that

$$\mathcal{C}_6 \ge 2, \qquad \mathcal{C}_5\mathcal{C}_6^{-1} \le \frac{1}{8}\min\left\{1,\frac{3\epsilon}{4}\right\}. \tag{2.53}$$

This yields

$$\mathcal{C}_5\mathcal{V}(u) \le \frac{\mathfrak{c}_\epsilon}{8}\mathscr{L}, \qquad \mathfrak{c}_\epsilon := \min\left\{1,\frac{3\epsilon}{4}\right\}, \qquad \mathscr{L} := 1+T+\rho. \tag{2.54}$$

Notice that $\mathscr{L} > T$. We introduce the following dyadic decomposition

$$I_0 := [T, 2\mathscr{L}), \qquad I_j := [2^j\mathscr{L}, 2^{j+1}\mathscr{L}), \qquad j\ge 1.$$

Suppose that there is $t\in I_0$ satisfying

$$\mathfrak{I}(t)-\epsilon t \ge \rho. \tag{2.55}$$

Then, by applying (2.51) and (2.54), it follows

$$\textstyle\sum_{j=0}^{k(t)-1} d_j \ge \rho - \mathcal{C}_5\mathcal{V}(u) + \frac{3\epsilon t}{4} \ge \rho + \frac{3\epsilon T}{4} - \frac{\mathfrak{c}_\epsilon\mathscr{L}}{8}. \tag{2.56}$$

On the other hand, since $1+T+\rho > \mathcal{C}_6\mathcal{V}(u)\ge 2$, one has

$$T+\rho > \frac{\mathscr{L}}{2},$$

which implies

$$\rho + \frac{3\epsilon T}{4} \ge \mathfrak{c}_\epsilon(\rho+T) > \frac{\mathfrak{c}_\epsilon\mathscr{L}}{2}.$$

Plugging the above estimate into (2.56) gives

$$\textstyle\sum_{j=0}^{k(t)-1} d_j \ge \frac{3\mathfrak{c}_\epsilon\mathscr{L}}{8}.$$

Consequently, by using

$$k(t) \le \frac{t}{M}+1 < \frac{2\mathscr{L}}{M}+1 \le \mathcal{C}_6\mathscr{L},$$

one gets

$$\left\{\sup_{t\in I_0}(\mathfrak{I}(t)-\epsilon t)\ge\rho\right\} \subset \left\{\max_{l\le \mathcal{C}_6\mathscr{L}}\left|\textstyle\sum_{j=0}^{l-1} d_j\right| \ge \frac{3\mathfrak{c}_\epsilon\mathscr{L}}{8}\right\}.$$

An application of the Chebyshev inequality combined with (2.50) yields

$$\begin{aligned}\mathbb{P}_u\left\{\sup_{t\in I_0}(\mathfrak{I}(t)-\epsilon t)\ge\rho\right\} &\lesssim_{q,\epsilon} \mathscr{L}^{-2q}\mathbb{E}_u\max_{l\le\mathcal{C}_6\mathscr{L}}\left|\textstyle\sum_{j=0}^{l-1} d_j\right|^{2q}\\ &\lesssim_{q,\epsilon,\nu,h,\{b_j\}_{j\ge1}} \mathscr{L}^{-q}\mathcal{V}(u)^{2q}.\end{aligned} \tag{2.57}$$

Now assume that for $j \geq 1$, there is $t \in I_j$ satisfying (2.55). Then, one has

$$\sum_{j=0}^{k(t)-1} d_j \geq \mathfrak{c}_\epsilon 2^j \mathscr{L} - \frac{\mathfrak{c}_\epsilon \mathscr{L}}{8} \geq 2^{j-1} \mathscr{L} \mathfrak{c}_\epsilon,$$

which, together with the estimate

$$k(t) \leq \frac{t}{M} + 1 < \frac{2^{j+1}\mathscr{L}}{M} + 1 \leq 2^{j+1}\mathscr{L},$$

implies

$$\left\{ \sup_{t \in I_j} (\mathfrak{I}(t) - \epsilon t) \geq \rho \right\} \subset \left\{ \max_{l \leq 2^{j+1}\mathscr{L}} \left| \sum_{j=0}^{l-1} d_j \right| \geq 2^{j-1} \mathscr{L} \mathfrak{c}_\epsilon \right\}$$

and thus

$$\begin{aligned} \mathbb{P}_u \left\{ \sup_{t \in I_j} (\mathfrak{I}(t) - \epsilon t) \geq \rho \right\} &\lesssim_{q,\epsilon} \mathscr{L}^{-2q} 2^{-2q(j-1)} \mathbb{E}_u \max_{l \leq 2^{j+1}\mathscr{L}} \left| \sum_{j=0}^{l-1} d_j \right|^{2q} \\ &\lesssim_{q,\epsilon,\nu,h,\{b_j\}_{j\geq 1}} \mathscr{L}^{-q} 2^{-qj} \mathcal{V}(u)^{2q}. \end{aligned}$$

Combining this with (2.52) and (2.57), one obtains (2.9). □

2.2 *Uniform finite-dimensional stabilization via compactness of noise coefficients*

It is noteworthy that the number N of perturbed modes in (2.6) depends on the noise coefficients $\{b_j\}_{j\geq 1}$. Indeed, the finite set K in Proposition 2.2 relies on $\{b_j\}_{j\geq 1}$, since the proof is based on Lemma 2.5, namely the asymptotic compactness property for the transition kernels. Such dependence is not sufficient to establish mixing in the degenerate case through the coupling method.

In this subsection, we aim to remove this dependence. More precisely, we show that N can be chosen uniformly over each bounded subset of

$$\ell^2_{\mathfrak{w}} := \{b_\cdot \in \ell^2 | \|b_\cdot\|^2_{\ell^2_{\mathfrak{w}}} := \mathfrak{B}_{\mathfrak{w}} < \infty\},$$

where $\mathfrak{B}_{\mathfrak{w}}$ is introduced in (1.14). Together with a Girsanov-type estimate carried out later, this is sufficient to prove Theorem 1.4. The main result of this subsection is presented below.

Proposition 2.7. *For any $\epsilon, \mathcal{R} > 0$, there is a finite set $K \subset C_0^\infty \cap H$ depending only on $\epsilon, \mathcal{R}, \nu, h$ such that for any $T, \rho \geq 0$, $q \geq 2$, $u \in H$, and $\mathcal{B} := \{b_j\}_{j\geq 1} \in \overline{B}_{\ell^2_{\mathfrak{w}}}(\mathcal{R})$,*

$$\mathbb{P}_u^{\mathcal{B}} \left\{ \sup_{t \geq T} \left(\int_0^t \operatorname{dist}_V(u_s, K)^2 \mathrm{d}s - \epsilon t \right) \geq \rho \right\} \lesssim_{q,\mathcal{R},\epsilon,\nu,h} \frac{1 + \|u\|^{8q}}{(1 + T + \rho)^q}, \tag{2.58}$$

where $(u_t, \mathbb{P}_u^{\mathcal{B}})$ denotes the Markov family associated with (2.2) whose noise coefficients are given by $\mathcal{B}$.

The proof is based on the following lemmas.

Lemma 2.8. *$\ell^2_{\mathfrak{w}}$ is compactly embedded into ℓ^2.*

Proof. **Step I.** We claim that

$$\lim_{j \to \infty} \left(\|e_j\|^2_{H^1} + \|\mathfrak{w} e_j\|^2 \right) = \infty. \tag{2.59}$$

By contradiction, suppose that there is a subsequence j_k such that

$$\sup_{k \geq 1} \left(\|e_{j_k}\|^2_{H^1} + \|\mathfrak{w} e_{j_k}\|^2 \right) < \infty.$$

Since $\mathfrak{w}$ is an admissible weight function, for any $\epsilon > 0$, one may choose $R > 0$ sufficiently large so that

$$\int_{-1}^{1}\int_{|x|>R} e_{j_k}^2 \mathrm{d}x\mathrm{d}y \le \frac{1}{\inf_{|x|\ge R}\mathfrak{w}(x)^2}\|\mathfrak{w}e_{j_k}\|^2 < \frac{\epsilon^2}{16}, \qquad \forall k \ge 1. \tag{2.60}$$

On the other hand, by applying the compact Sobolev embedding theorem, $\{e_{j_k}\}_{k\ge1}$ is compact in $L^2(\mathcal{O}_R)$ with $\mathcal{O}_R := (-R, R)\times(-1, 1)$. Consequently, up to a subsequence, there is a function e satisfying

$$e_{j_k} \to e, \qquad \text{in } L^2(\mathcal{O}_R),$$

as $k \to \infty$. In particular, there exists $k_0 \ge 1$ such that

$$\|e_{j_k} - e_{j_l}\|_{L^2(\mathcal{O}_R)} < \frac{\epsilon}{2}, \qquad \forall k, l \ge k_0.$$

Together with (2.60), this implies that

$$\|e_{j_k} - e_{j_l}\| < \epsilon, \qquad \forall k, l \ge k_0.$$

However, as $\{e_j\}_{j\ge1}$ is an orthonormal basis in H, one has

$$\|e_j - e_k\|^2 = \|e_j\|^2 + \|e_k\|^2 = 2, \qquad \forall j \ne k.$$

This is a contradiction.

Step II. Now let $\{b_\cdot^{(n)}\}_{n\ge1}$ be a bounded sequence of elements in $\ell^2_{\mathfrak{w}}$. For any integer $J \ge 1$, one has

$$\begin{aligned}\textstyle\sum_{j\ge J}(b_j^{(n)})^2 &\le \frac{1}{\inf_{j\ge J}(\|\mathfrak{w}e_j\|^2 + \|e_j\|^2_{H^1})}\textstyle\sum_{j\ge J}(b_j^{(n)})^2(\|\mathfrak{w}e_j\|^2 + \|e_j\|^2_{H^1})\\ &\le \frac{\sup_{n\ge1}\|b_\cdot^{(n)}\|^2_{\ell^2_{\mathfrak{w}}}}{\inf_{j\ge J}(\|\mathfrak{w}e_j\|^2 + \|e_j\|^2_{H^1})},\end{aligned}$$

which, together with (2.59), implies that for any $\epsilon > 0$, there is $J \ge 1$ such that

$$\textstyle\sum_{j\ge J}(b_j^{(n)})^2 \le \frac{\epsilon^2}{16}, \qquad \forall n \ge 1. \tag{2.61}$$

For $j < J$, since

$$\sup_{n\ge1}(b_j^{(n)})^2 \le \sup_{n\ge1}\|b_\cdot^{(n)}\|^2_{\ell^2_{\mathfrak{w}}},$$

by compactness and a diagonal argument, up to a subsequence, $\{(b_1^{(n)}, \dots, b_{J-1}^{(n)})\}_{n\ge1}$ is convergent. In particular, there is $n_0 \ge 1$ such that

$$\textstyle\sum_{j<J}(b_j^{(n)} - b_j^{(m)})^2 \le \frac{\epsilon^2}{4}, \qquad \forall n, m \ge n_0.$$

Combining this with (2.61), $\{b_\cdot^{(n)}\}_{n\ge1}$ is Cauchy. This yields the compact embedding of $\ell^2_{\mathfrak{w}}$ into ℓ^2. □

Next, we present a uniform version of Lemma 2.6.

Lemma 2.9. *For any $\mathcal{R}, R, \epsilon > 0$, there exist integers $\mathfrak{n}_0, \dots, \mathfrak{n}_k \ge 1$ and a finite set*

$$K := \{0, \mathfrak{u}_1, \dots, \mathfrak{u}_l\} \subset C_0^\infty \cap H$$

such that for any $\mathcal{B} = \{b_j\}_{j\ge1} \in \overline{B}_{\ell^2_{\mathfrak{w}}}(\mathcal{R})$,

$$\min_{i=0,\dots,k}\sup_{\|u\|\le R}\mathbb{E}_u^{\mathcal{B}}\int_{\mathfrak{n}_i}^{\mathfrak{n}_i+1}\mathrm{dist}_V(u_t, K)^2\mathrm{d}t \le \epsilon. \tag{2.62}$$

Proof. **Step I.** Fix $\mathcal{R}, R, \epsilon > 0$. Let $u \in B_H(R)$ and

$$\mathcal{B} := \{b_j\}_{j\geq 1}, \qquad \mathcal{B}^{(0)} := \{b_j^{(0)}\}_{j\geq 1}$$

belong to $\overline{B}_{\ell^2_{\mathfrak{w}}}(\mathcal{R})$. Denote by $\{u^{(0)}(t)\}_{t\geq 0}$ and $\{u(t)\}_{t\geq 0}$ the solutions of (2.2) issued from the same initial data u with the noise coefficients given by $\mathcal{B}^{(0)}$ and $\mathcal{B}$, respectively. By applying Lemma 2.6, there is an integer $\mathfrak{n}_0 \geq 1$ and a finite set

$$K_0 := \{0, \mathfrak{u}_1^{(0)}, \dots, \mathfrak{u}_{l_0}^{(0)}\} \subset C_0^\infty \cap H$$

depending on $R, \epsilon, \mathcal{B}^{(0)}$ such that

$$\sup_{\|u\|\leq R} \mathbb{E} \int_{\mathfrak{n}_0}^{\mathfrak{n}_0+1} \mathrm{dist}_V(u^{(0)}(t), K_0)^2 \mathrm{d}t \leq \frac{\epsilon}{4}. \tag{2.63}$$

On the other hand, by using (A.8), there is $\delta_0 > 0$ depending on $R, \mathcal{R}, \epsilon, \nu, h, \mathcal{B}^{(0)}$ such that for any $\mathcal{B} \in \ell^2$ satisfying

$$\|\mathcal{B} - \mathcal{B}^{(0)}\|_{\ell^2} \leq \delta_0,$$

one has

$$\mathbb{E} \int_0^{\mathfrak{n}_0+1} \|u(t) - u^{(0)}(t)\|_V^2 \mathrm{d}t \leq \frac{\epsilon}{4}.$$

Together with (2.63), this gives

$$\sup_{\|u\|\leq R} \mathbb{E}_u^{\mathcal{B}} \int_{\mathfrak{n}_0}^{\mathfrak{n}_0+1} \mathrm{dist}_V(u_t, K_0)^2 \mathrm{d}t \leq \epsilon, \qquad \forall \mathcal{B} \in B_{\ell^2}(\mathcal{B}^{(0)}, \delta_0). \tag{2.64}$$

Step II. By the definition (1.14) of $\mathfrak{B}_{\mathfrak{w}}$, the set $\overline{B}_{\ell^2_{\mathfrak{w}}}(\mathcal{R})$ is closed in ℓ^2. This, together with Lemma 2.8, implies the existence of $\mathcal{B}^{(0)}, \dots, \mathcal{B}^{(k)} \in \overline{B}_{\ell^2_{\mathfrak{w}}}(\mathcal{R})$ and $\delta_0, \dots, \delta_k > 0$ such that

$$\overline{B}_{\ell^2_{\mathfrak{w}}}(\mathcal{R}) \subset \cup_{i=0}^k B_{\ell^2}(\mathcal{B}^{(i)}, \delta_i).$$

For $i = 0, \dots, k$, we denote by $\mathfrak{n}_i$ and K_i the integer and the finite set corresponding to $R, \epsilon, \mathcal{B}^{(i)}$, as indicated in Lemma 2.6. After relabeling, we write

$$K := \cup_{i=0}^k K_i = \{0, \mathfrak{u}_1, \dots, \mathfrak{u}_l\} \subset C_0^\infty \cap H.$$

Now for any $\mathcal{B} \in \overline{B}_{\ell^2_{\mathfrak{w}}}(\mathcal{R})$, there is $\mathfrak{n}_i$ such that

$$\sup_{\|u\|\leq R} \mathbb{E}_u^{\mathcal{B}} \int_{\mathfrak{n}_i}^{\mathfrak{n}_i+1} \mathrm{dist}_V(u_t, K_i)^2 \mathrm{d}t \leq \epsilon.$$

Since $\mathrm{dist}_V(u_t, K) \leq \mathrm{dist}_V(u_t, K_i)$, this ensures (2.62). □

Now we turn to the proof of Proposition 2.7.

Proof of Proposition 2.7. The main strategy is to proceed as in the proof of Proposition 2.2 and avoid the dependence of the finite set K in (2.58) on the specific noise coefficients, through an application of Lemma 2.9. Thus, we only indicate the necessary modifications.

Fix $\epsilon, \mathcal{R} > 0$ and $\mathcal{B} \in \overline{B}_{\ell^2_{\mathfrak{w}}}(\mathcal{R})$. Applying Lemma 2.9 with ϵ replaced by $\frac{\epsilon}{16}$ and R determined in (2.65), one may find integers $\mathfrak{n}_0, \dots, \mathfrak{n}_m \geq 1$ and a finite set

$$K := \{0, \mathfrak{u}_1, \dots, \mathfrak{u}_l\} \subset C_0^\infty \cap H$$

depending only on $\mathcal{R}, \epsilon, \nu, h$ such that (2.62) holds. In particular, there is $\mathfrak{n}_i$ such that

$$\sup_{\|u\|\le R} \mathbb{E}_u^{\mathcal{B}} \int_{\mathfrak{n}_i}^{\mathfrak{n}_i+1} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \le \frac{\epsilon}{16}.$$

Together with the Markov property, (A.4), and (A.6), this implies that for any $u \in H$,

$$\mathbb{E}_u^{\mathcal{B}} \int_{\mathfrak{n}_i}^{\mathfrak{n}_i+1} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \le \frac{\epsilon}{16} + \mathcal{C}_7 \frac{\mathcal{V}(u)}{R^2},$$

where $\mathcal{C}_7$ depends on $\mathcal{R}, \nu, h$ and $\mathcal{V}(u) := (1 + \|u\|^2)^2$. Let $M \ge \max_{i=0,\dots,m} \mathfrak{n}_i$. Proceeding as in the proof of (2.42), it follows

$$\mathbb{E}_u^{\mathcal{B}} \int_0^M \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \le \mathcal{C}_8(\mathfrak{n}_i + R^{-2})\mathcal{V}(u) + \left(\frac{\epsilon}{16} + \mathcal{C}_8 R^{-2}\right) M$$

for some constant $\mathcal{C}_8$ depending on $\mathcal{R}, \nu, h$. By taking R large enough so that

$$\frac{\epsilon}{16} + \mathcal{C}_8 R^{-2} \le \frac{\epsilon}{8}, \tag{2.65}$$

one has

$$\mathbb{E}_u^{\mathcal{B}} \int_0^M \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t \le \mathcal{C}_9 \mathcal{V}(u) + \frac{\epsilon}{8} M, \tag{2.66}$$

where $\mathcal{C}_9$ depends on $\mathcal{R}, \epsilon, \nu, h$.

Let $\mathfrak{P}_t^{\mathcal{B}}$ and $\mathfrak{P}_t^{\mathcal{B}*}$ denote the Markov operators associated with the Markov family $(u_t, \mathbb{P}_u^{\mathcal{B}})$. Applying (A.4) combined with the Markov property, one has

$$\mathfrak{P}_M^{\mathcal{B}} \mathcal{V}(u) \le \mathfrak{c}^M \mathcal{V}(u) + \frac{1 - \mathfrak{c}^M}{1 - \mathfrak{c}} C_{\mathcal{R},\nu,h}, \qquad \mathfrak{c} := e^{-\nu\alpha_0},$$

which, together with (2.66) and choosing M sufficiently large, yields

$$\mathbb{E}_u^{\mathcal{B}} \int_0^M \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t + \tilde{\mathfrak{a}}_M \left(\mathfrak{P}_M^{\mathcal{B}} \mathcal{V}(u) - \mathcal{V}(u)\right) \le \frac{\epsilon M}{4}, \tag{2.67}$$

where $\tilde{\mathfrak{a}}_M := (1 - \mathfrak{c}^M)^{-1} \mathcal{C}_9$. Define $\mathfrak{I}(jM) := \int_0^{jM} \mathrm{dist}_V(u_t, K)^2 \mathrm{d}t$ and

$$\tilde{\Gamma}_j := \mathfrak{I}((j+1)M) - \mathfrak{I}(jM) + \tilde{\mathfrak{a}}_M \left(\mathcal{V}(u_{(j+1)M}) - \mathcal{V}(u_{jM})\right).$$

An application of the Markov property combined with (2.67) yields

$$\mathbb{E}_u^{\mathcal{B}}(\tilde{\Gamma}_j | \mathcal{F}_{jM}) \le \frac{\epsilon M}{4}.$$

Consequently, it follows that

$$\mathfrak{I}(kM) + \tilde{\mathfrak{a}}_M(\mathcal{V}(u_{kM}) - \mathcal{V}(u)) \le \frac{k\epsilon M}{4} + \textstyle\sum_{j=0}^{k-1} \tilde{d}_j, \tag{2.68}$$

where $\tilde{d}_j := \tilde{\Gamma}_j - \mathbb{E}_u^{\mathcal{B}}(\tilde{\Gamma}_j | \mathcal{F}_{jM})$ is a sequence of martingale differences.

Repeating the derivation of (2.49), for any $q \ge 2$, one has

$$\sup_{j \ge 0} \mathbb{E}_u^{\mathcal{B}} |\tilde{d}_j|^q \lesssim_{q,\mathcal{R},\epsilon,\nu,h} \mathcal{V}(u)^q.$$

Together with the pathwise estimate (2.68) and the argument carried out in Step VI in the proof of Proposition 2.2, this completes the proof. □

3 Mixing for finite-energy solutions

The goal of this section is to establish polynomial mixing for the finite-energy solutions of the stochastic NS equation (1.1). The proof is based on the classical coupling framework; see Section 3.1.3 in [KS12] and [Shi08]. For the reader's convenience, we briefly describe the main ingredients of this framework below.

Let $\{\boldsymbol{u}_t\}_{t\geq 0}$ be a process valued in $H\times H$, and let $\mathbb{P}_{\boldsymbol{u}}$ be a probability measure on the path space $C([0,\infty);H\times H)$ parameterized by $\boldsymbol{u}\in H\times H$. Given a Markov family $(u_t,\mathbb{P}_u)$, the pair $(\boldsymbol{u}_t,\mathbb{P}_{\boldsymbol{u}})$ is said to be a coupling of two copies of $(u_t,\mathbb{P}_u)$, if under $\mathbb{P}_{\boldsymbol{u}}$, the distributions of $\{\pi_1\boldsymbol{u}_t\}_{t\geq 0}$ and $\{\pi_2\boldsymbol{u}_t\}_{t\geq 0}$ coincide with that of $\{u_t\}_{t\geq 0}$ under $\mathbb{P}_{\pi_1\boldsymbol{u}}$ and $\mathbb{P}_{\pi_2\boldsymbol{u}}$, respectively. Here, for $i=1,2$, π_i denotes the projection onto the i-th component. As presented in [NZ24b], the uniqueness of the invariant measure and the polynomial convergence estimate (1.17) hold, provided that the following result is true.

Proposition 3.1. *Under the assumption of Theorem 1.4, for any $q\geq 2$, there exists a time $T>0$, a closed ball $B\subset H$, a coupling $(\boldsymbol{u}_t,\mathbb{P}_{\boldsymbol{u}})$ of $(u_t,\mathbb{P}_u)$, and a stopping time σ such that $(\boldsymbol{u}_{kT},\mathbb{P}_{\boldsymbol{u}})$ is a Markov process satisfying the following properties.*

1. ***Recurrence.*** *There are constants $\delta, C_{\mathrm{rec}}>0$ such that*

$$\mathbb{E}_{\boldsymbol{u}}e^{\delta\tau_B}\leq C_{\mathrm{rec}}\big(1+\|u\|^2+\|u'\|^2\big) \tag{3.1}$$

for any $\boldsymbol{u}=(u,u')\in H\times H$, where

$$\tau_B:=\inf\{k\geq 0:\boldsymbol{u}_{kT}\in B\times B\}. \tag{3.2}$$

2. ***Polynomial squeezing.*** *There is a constant $C_{\mathrm{sq}}>0$ such that*

$$\|u_t-u_t'\|^2\leq C_{\mathrm{sq}}e^{-\frac{\nu\alpha_0}{2}t}\|u-u'\|^2,\qquad 0\leq t\leq\sigma \tag{3.3}$$

for any $\boldsymbol{u}=(u,u')\in B\times B$, where

$$u_t:=\pi_1\boldsymbol{u}_t,\qquad u_t':=\pi_2\boldsymbol{u}_t.$$

Moreover, the stopping time σ satisfies the following properties.

(a) For any $\boldsymbol{u}=(u,u')\in B\times B$,

$$\mathbb{P}_{\boldsymbol{u}}\{\sigma=\infty\}\geq\tfrac{1}{2}. \tag{3.4}$$

(b) There is a constant $C_q>0$ such that

$$\mathbb{E}_{\boldsymbol{u}}[\mathbb{I}_{\{\sigma<\infty\}}\sigma^q]\leq C_q, \tag{3.5}$$

and

$$\mathbb{E}_{\boldsymbol{u}}[\mathbb{I}_{\{\sigma<\infty\}}(\|u_\sigma\|^{2q}+\|u_\sigma'\|^{2q})]\leq C_q \tag{3.6}$$

for any $\boldsymbol{u}\in B\times B$.

The proof of this proposition is carried out in Sections 3.3 and 3.4. To prepare for the proof, we establish necessary groundwork in the next two subsections.

3.1 Construction of the coupling

In this subsection, we construct a coupling $(\boldsymbol{u}_t,\mathbb{P}_{\boldsymbol{u}})$, following the approach developed in [NZ24a, NZ24b], which is briefly described below.

Introduce parameters $a>0$, $N\geq 1$, and $T>0$, whose values will be specified later. For any $u,u'\in H$, we denote by $\{u(t)\}_{t\geq 0}$ and $\{u'(t)\}_{t\geq 0}$ the solutions of (2.2) issued from u and u', respectively. Let $\{v(t)\}_{t\geq 0}$ satisfy the auxiliary equation (2.6). Define

$$\lambda_T(u,u'):=\mathcal{D}(\{v(t)\}_{t\in[0,T]}),\qquad \lambda_T'(u'):=\mathcal{D}(\{u'(t)\}_{t\in[0,T]}). \tag{3.7}$$

By Theorem 1.2.28 in [KS12], these measures have a maximal coupling $(\mathcal{V}_T(u,u'),\mathcal{V}'_T(u,u'))$ on a probability space $(\tilde\Omega,\tilde{\mathcal{F}},\tilde{\mathbb{P}})$. In particular,

$$\tilde{\mathbb{P}}\{\mathcal{V}_T(u,u')\neq\mathcal{V}'_T(u,u')\}=\|\lambda_T(u,u')-\lambda'_T(u')\|_{\mathrm{var}}. \tag{3.8}$$

Let $\{\tilde v(t)\}_{t\in[0,T]}$ and $\{\tilde u'(t)\}_{t\in[0,T]}$ denote the flows of $\mathcal{V}_T(u,u')$ and $\mathcal{V}'_T(u,u')$, respectively. Then, $\{\tilde v(t)\}_{t\in[0,T]}$ satisfies

$$\partial_t\tilde v-\nu L\tilde v+B(\tilde v)+a\mathrm{P}_N\tilde v=h+\tilde\eta,\qquad \tilde v|_{t=0}=u', \tag{3.9}$$

where $\tilde\eta$ has the same distribution as that of $\eta+a\mathrm{P}_Nu$. Define $\{\tilde u(t)\}_{t\in[0,T]}$ as the solution of

$$\partial_t\tilde u-\nu L\tilde u+B(\tilde u)+a\mathrm{P}_N\tilde u=h+\tilde\eta,\qquad \tilde u|_{t=0}=u. \tag{3.10}$$

As presented in Section 2 of [NZ24b], by the uniqueness-in-law of the above stochastic PDE, one has

$$\mathcal{D}(\{\tilde u(t)\}_{t\in[0,T]})=\mathcal{D}(\{u(t)\}_{t\in[0,T]}).$$

For any $u,u'\in H$, $\omega\in\tilde\Omega$, and $t\in[0,T]$, we define

$$\mathcal{R}_t(u,u',\omega):=\tilde u(t),\qquad \mathcal{R}'_t(u,u',\omega):=\tilde u'(t),\qquad \mathcal{R}''_t(u,u',\omega):=\tilde v(t).$$

Let $\{(\Omega^k,\mathcal{F}^k,\mathbb{P}^k)\}_{k\geq0}$ be independent copies of $(\tilde\Omega,\tilde{\mathcal{F}},\tilde{\mathbb{P}})$ and let $(\Omega,\mathcal{F},\mathbb{P})$ denote their product. For $\omega=(\omega^0,\omega^1,\dots)\in\Omega$ and $t=kT+s$ with $s\in(0,T]$, $k\geq0$, we recursively define

$$\tilde u_t:=\mathcal{R}_s\big(\tilde u_{kT},\tilde u'_{kT},\omega^k\big),\quad \tilde u'_t:=\mathcal{R}'_s\big(\tilde u_{kT},\tilde u'_{kT},\omega^k\big),\quad \tilde v_t:=\mathcal{R}''_s\big(\tilde u_{kT},\tilde u'_{kT},\omega^k\big). \tag{3.11}$$

Let $\boldsymbol{u}_t:=(\tilde u_t,\tilde u'_t)$ and denote the resulting family by $(\boldsymbol{u}_t,\mathbb{P}_{\boldsymbol{u}})$. It is clear that this family is a coupling of two copies of $(u_t,\mathbb{P}_u)$ and that $(\boldsymbol{u}_{kT},\mathbb{P}_{\boldsymbol{u}})_{k\geq0}$ is a time-homogeneous Markov chain. In the following subsections, we shall choose a, N, and T so that $(\boldsymbol{u}_t,\mathbb{P}_{\boldsymbol{u}})$ satisfies the recurrence and polynomial squeezing properties stated in Proposition 3.1.

3.2 Girsanov estimate

In this subsection, we establish an estimate on the total variational distance between the distributions $\mathcal{D}(\{v(t)\}_{t\geq0})$ and $\mathcal{D}(\{u'(t)\}_{t\geq0})$. The proof is based on Propositions 2.1 and 2.7 combined with the Girsanov theorem.

Lemma 3.2. *For any $\mathcal{R}>0$, there is an integer $N_1\geq1$ and a constant $a>0$ depending on $\mathcal{R},\nu,h$ such that for any $q\geq2$, $T,\rho>0$, and $u,u'\in H$,*

$$\|\lambda_T(u,u')-\lambda'_T(u')\|_{\mathrm{var}}\leq\mathcal{C}_q\frac{1+\|u\|^{8q}}{(1+\rho)^q}+\frac{\sqrt{6}a}{b_N^*\sqrt{\nu\alpha_0}}e^{\frac{\mathcal{C}_1\rho}{2}}\|u-u'\|, \tag{3.12}$$

provided that the conditions (1.14) and (1.15) are satisfied with $N\geq N_1$. Here, the probability distributions $\lambda_T(u,u')$, $\lambda'_T(u')$ are defined in (3.7),

$$b_N^*:=\min_{1\leq j\leq N}|b_j|>0,$$

α_0 is the constant in the Poincaré inequality, $\mathcal{C}_1$ is the constant given in (2.7), and $\mathcal{C}_q$ depends on $q,\mathcal{R},\nu,h$.

Proof. Let $\{u(t)\}_{t\geq0}$ be the solution of (2.2) issued from u, and let $\{v(t)\}_{t\geq0}$ satisfy the auxiliary equation (2.6). The proof is divided into the following three steps.

Step I. Fix $\mathcal{R}>0$. Applying Proposition 2.7 with $\epsilon=\frac{\nu\alpha_0}{2\mathcal{C}_1}$, there is a finite set $K\subset C_0^\infty\cap H$ such that the probabilistic growth estimate (2.58) holds. Let N_1,a be the constants indicated

in Proposition 2.1, which corresponds to the finite set K and thus depends only on $\mathcal{R}, \nu, h$. Suppose that the noise coefficients $\{b_j\}_{j\geq 1}$ satisfy (1.14) and (1.15) with $N \geq N_1$. Define

$$\Lambda_u(t) := \int_0^t \mathrm{dist}_V(u(s), K)^2 \mathrm{d}s - \frac{\nu\alpha_0}{2\mathcal{C}_1} t, \qquad \tau_\rho^u := \inf\{t \geq 0 | \Lambda_u(t) \geq \rho\}. \tag{3.13}$$

Introduce $\{v^\rho(t)\}_{t\geq 0}$ as the solution of the truncated auxiliary equation

$$\partial_t v^\rho - \nu L v^\rho + B(v^\rho) + a\mathbb{I}_{\{t<\tau_\rho^u\}} \mathrm{P}_N(v^\rho - u) = h + \eta, \qquad v^\rho|_{t=0} = u'. \tag{3.14}$$

In particular, one has

$$v^\rho(t) = v(t), \qquad 0 \leq t \leq \tau_\rho^u. \tag{3.15}$$

Therefore, for any $t \leq \tau_\rho^u$, an application of (2.7) yields

$$\begin{aligned} \|u(t) - v^\rho(t)\|^2 &\leq \|u - u'\|^2 \exp\left(-\nu\alpha_0 t + \mathcal{C}_1 \int_0^t \mathrm{dist}_V(u(s), K)^2 \mathrm{d}s\right) \\ &\leq e^{\mathcal{C}_1 \rho} e^{-\frac{\nu\alpha_0}{2} t} \|u - u'\|^2. \end{aligned} \tag{3.16}$$

On the other hand, since $\{v^\rho(t)\}_{t\geq 0}$ coincides with $\{v(t)\}_{t\geq 0}$ on $[0, T]$ whenever $\tau_\rho^u > T$, by applying Proposition 2.7, its distribution $\lambda_T^\rho(u, u') := \mathcal{D}(\{v^\rho(t)\}_{t\in[0,T]})$ satisfies

$$\begin{aligned} \|\lambda_T^\rho(u, u') - \lambda_T(u, u')\|_{\mathrm{var}} &\leq \mathbb{P}\{\tau_\rho^u \leq T\} \\ &\leq \mathbb{P}\Big\{\sup_{t\geq 0} \Lambda_u(t) \geq \rho\Big\} \leq C_{q,\mathcal{R},\nu,h} \frac{1 + \|u\|^{8q}}{(1+\rho)^q}. \end{aligned} \tag{3.17}$$

Step Ⅱ. It remains to estimate the total variational distance between $\lambda_T^\rho(u, u')$ and $\lambda_T'(u')$. Such an estimate is obtained through the Girsanov theorem. Without loss of generality, we assume that the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ has the following specific structure. The sample space $\Omega := C_0([0, \infty); H)$ is the space of continuous functions valued in H and vanishing at $t = 0$, endowed with the topology of uniform convergence on compact intervals. The probability $\mathbb{P}$ is the distribution of the Wiener process

$$W(t) := \textstyle\sum_{j\geq 1} b_j e_j \beta_j(t), \tag{3.18}$$

and $\mathcal{F}$ is the completion of the Borel σ-algebra of Ω. Define a measurable transform

$$\mathbf{\Phi}^{u,u'} : \omega_t \mapsto \omega_t - a\int_0^t \mathbb{I}_{\{s<\tau_\rho^u\}} \mathrm{P}_N(v^\rho(s) - u(s))\mathrm{d}s, \qquad \Omega \to \Omega.$$

By the pathwise uniqueness of the stochastic NS equation (2.2), one has

$$\mathbb{P}\big\{u'(t, \mathbf{\Phi}^{u,u'}(\omega)) = v^\rho(t, \omega) \text{ for any } t \geq 0\big\} = 1.$$

This yields

$$\|\lambda_T^\rho(u, u') - \lambda_T'(u')\|_{\mathrm{var}} \leq \|\mathbb{P} - \mathbf{\Phi}_*^{u,u'}\mathbb{P}\|_{\mathrm{var}}, \tag{3.19}$$

where $\mathbf{\Phi}_*^{u,u'}\mathbb{P}$ denotes the push-forward measure under the transform $\mathbf{\Phi}^{u,u'}$. To bound the right-hand side of (3.19), we introduce the following decomposition

$$\Omega = \mathrm{P}_N\Omega \oplus \mathrm{Q}_N\Omega,$$

where P_N and $\mathrm{Q}_N := I - \mathrm{P}_N$ are understood as natural projections

$$\mathrm{P}_N : \Omega \to C_0([0, \infty); \mathrm{P}_N H), \qquad \mathrm{Q}_N : \Omega \to C_0([0, \infty); \mathrm{Q}_N H).$$

Accordingly, we write $\omega = (\omega^{(1)}, \omega^{(2)})$. Thus, the transformation $\mathbf{\Phi}^{u,u'}$ can be expressed as

$$\mathbf{\Phi}^{u,u'}(\omega^{(1)}, \omega^{(2)}) = (\mathbf{\Psi}^{u,u'}(\omega^{(1)}, \omega^{(2)}), \omega^{(2)}),$$

where

$$\mathbf{\Psi}^{u,u'} : \omega_t \mapsto \omega_t^{(1)} - a\int_0^t \mathbb{I}_{\{s<\tau_\rho^u\}} \mathrm{P}_N(v^\rho(s,\omega) - u(s,\omega))\mathrm{d}s, \qquad \Omega \to C_0([0,\infty); \mathrm{P}_N H).$$

Let $\mathbb{P}_N := (\mathrm{P}_N)_*\mathbb{P}$ and $\mathbb{P}_N^\perp := (\mathrm{Q}_N)_*\mathbb{P}$. An application of Lemma 3.3.13 in [KS12] implies

$$\|\mathbb{P} - \mathbf{\Phi}_*^{u,u'}\mathbb{P}\|_{\mathrm{var}} \le \int_{\mathrm{Q}_N\Omega} \|\mathbf{\Psi}_*^{u,u'}(\mathbb{P}_N, \omega^{(2)}) - \mathbb{P}_N\|_{\mathrm{var}} \mathbb{P}_N^\perp(\mathrm{d}\omega^{(2)}). \tag{3.20}$$

Furthermore, by the Girsanov theorem, for each $\omega^{(2)}$, one has

$$\begin{aligned}
&\|\mathbf{\Psi}_*^{u,u'}(\mathbb{P}_N, \omega^{(2)}) - \mathbb{P}_N\|_{\mathrm{var}} \\
&\le \frac{1}{2}\left(\left(\mathbb{E}_N \exp\left(6a^2(b_N^*)^{-2}\int_0^{\tau_\rho^u} \|\mathrm{P}_N(v^\rho(t,\omega) - u(t,\omega))\|^2 \mathrm{d}t\right)\right)^{\frac{1}{2}} - 1\right)^{\frac{1}{2}},
\end{aligned} \tag{3.21}$$

if the following Novikov condition

$$\mathbb{E}_N \exp\left(c\int_0^{\tau_\rho^u} \|\mathrm{P}_N(v^\rho(t,\omega) - u(t,\omega))\|^2 \mathrm{d}t\right) < \infty$$

is satisfied for any $c > 0$ and $\omega^{(2)}$. Here, $\mathbb{E}_N$ is the expectation with respect to $\mathbb{P}_N$.

Step III. By applying (3.16), it follows that

$$\int_0^{\tau_\rho^u} \|\mathrm{P}_N(v^\rho(t) - u(t))\|^2 \mathrm{d}t \le \int_0^{\tau_\rho^u} \|v^\rho(t) - u(t)\|^2 \mathrm{d}t \le \frac{2}{\nu\alpha_0} e^{\mathcal{C}_1\rho}\|u - u'\|^2,$$

which verifies the Novikov condition

$$\mathbb{E}_N \exp\left(c\int_0^{\tau_\rho^u} \|\mathrm{P}_N(v^\rho(t,\omega) - u(t,\omega))\|^2 \mathrm{d}t\right) \le \exp\left(\frac{2c}{\nu\alpha_0} e^{\mathcal{C}_1\rho}\|u - u'\|^2\right). \tag{3.22}$$

Combining the above estimate and (3.19)–(3.21), one has

$$\|\lambda_T^\rho(u,u') - \lambda_T'(u')\|_{\mathrm{var}} \le 1 \wedge \left(\frac{1}{2}\left(\exp\left(\frac{6a^2}{\nu\alpha_0(b_N^*)^2} e^{\mathcal{C}_1\rho}\|u - u'\|^2\right) - 1\right)^{\frac{1}{2}}\right).$$

This, together with (3.17) and the estimate

$$1 \wedge \left(\frac{1}{2}\sqrt{e^s - 1}\right) \le \sqrt{s}, \qquad \forall s \ge 0,$$

implies (3.12) and thus completes the proof. □

3.3 Recurrence

In this subsection, we verify the recurrence property stated in Proposition 3.1, by following the argument in Section 5.1 of [NZ24b]. We begin with the following irreducibility property of the Markov family $(u_t, \mathbb{P}_u)$ associated with the stochastic NS equation (2.2) in the finite-energy setting.

Lemma 3.3. *There is an integer $N_2 \geq 1$ depending only on ν, h such that if (1.15) is satisfied for some $N \geq N_2$, then there exists a function $\mathfrak{u} \in H^1$ with $\|\mathfrak{u}\| \leq 1$ depending on ν, h, N and satisfying the following assertion. For any $R, d > 0$, there is $T_0 > 0$ such that*

$$\inf_{u_0 \in B_H(R)} P_T(u_0, B_H(\mathfrak{u}, d)) \geq p > 0 \tag{3.23}$$

for any $T \geq T_0$. Here, T_0 depends on R, d, ν and p depends on $T, d, \nu, h, N, \{b_j\}_{j\geq 1}$.

The proof follows that of Lemma 5.1 in [NZ24b], with the Poincaré inequality providing the dissipation supplied there by the damping term. We therefore omit the details. Moreover, combining Lemmas 3.2 and 3.3 with Proposition 2.7, one can obtain the irreducibility of the coupling $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$. The argument follows that of Proposition 4.2 in [NZ24a] for the complex Ginzburg–Landau equation and is likewise omitted.

Lemma 3.4. *For any $\mathcal{R} > 0$, there is an integer $N_{\mathrm{rec}} \geq 1$ depending on $\mathcal{R}, \nu, h$ such that if (1.14) and (1.15) are satisfied for some $N \geq N_{\mathrm{rec}}$, then there exists a function $\mathfrak{u} \in H^1$ with $\|\mathfrak{u}\| \leq 1$ depending on ν, h, N and satisfying the following assertion. For any $R, d > 0$, there is $T_0 > 0$ such that*

$$\inf_{\boldsymbol{u} \in B_H(R) \times B_H(R)} \mathbb{P}_{\boldsymbol{u}}\{\boldsymbol{u}_T \in B_H(\mathfrak{u}, d) \times B_H(\mathfrak{u}, d)\} \geq p > 0 \tag{3.24}$$

for every $T \geq T_0$. Here, T_0 depends on $\mathcal{R}, R, d, \nu, h, N, \{b_j\}_{j\geq 1}$, while p additionally depends on T.

Finally, by (A.4), the function

$$F(u, u') := 1 + \|u\|^2 + \|u'\|^2$$

is a Lyapunov function for the coupling process $(\boldsymbol{u}_{kT}, \mathbb{P}_{\boldsymbol{u}})$. Combining this fact with (3.24), we obtain the following recurrence property, as explained in Section 3 in [Shi08].

Proposition 3.5. *Fix arbitrary $\mathcal{R} > 0$ and let N_{rec} be the integer given in Lemma 3.4. Suppose that (1.14) and (1.15) are satisfied for some $N \geq N_{\mathrm{rec}}$. Let $\mathfrak{u}$ be the corresponding function given in Lemma 3.4. Then, for any $d > 0$, there is $T_{\mathrm{rec}} > 0$ such that for any $T \geq T_{\mathrm{rec}}$ and $\boldsymbol{u} = (u, u') \in H \times H$,*

$$\mathbb{E}_{\boldsymbol{u}} e^{\delta_T \tau_d} \leq C_T\big(1 + \|u\|^2 + \|u'\|^2\big), \tag{3.25}$$

for some constants δ_T, C_T. Here

$$\tau_d := \inf\left\{k \geq 0 | \boldsymbol{u}_{kT} \in \overline{B}_H(\mathfrak{u}, d) \times \overline{B}_H(\mathfrak{u}, d)\right\},$$

and the constant T_{rec} depends on $\mathcal{R}, d, \nu, h, N, \{b_j\}_{j\geq 1}$, while δ_T and C_T additionally depend on T.

3.4 Polynomial squeezing and proof of Theorem 1.4

This subsection is devoted to the verification of the polynomial squeezing property stated in Proposition 3.1 for the coupling process $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$. To simplify the presentation, we denote by u_t, u'_t, v_t the processes constructed in (3.11). Let

$$\sigma := \sigma_1 \wedge \tau, \qquad \sigma_1 := \inf\{t \geq 0 | u'_t \neq v_t\}, \qquad \tau := \tau_\rho^u \wedge \tau_\rho^{u'}, \tag{3.26}$$

where $\tau_\rho^u, \tau_\rho^{u'}$ are defined in (3.13). Introduce the random events

$$\mathcal{Q}'_k := \{kT \leq \sigma < (k+1)T,\ \sigma_1 \geq \tau\}, \quad \mathcal{Q}''_k := \{kT \leq \sigma < (k+1)T,\ \sigma_1 < \tau\}, \quad k \geq 0.$$

Here, we recall that $T > 0$ is the time step used in the construction (3.11).

Lemma 3.6. *For any $\mathcal{R} > 0$, there is an integer $N_{\rm sq} \geq 1$ such that if (1.14) and (1.15) are satisfied for some $N \geq N_{\rm sq}$, then the following assertion holds. For any $q \geq 2$, there exist constants $\rho > 0$, $d \in (0,1]$, and $T_{\rm sq} > 0$ depending on $q, \mathcal{R}, \nu, h, N, \{b_j\}_{j\geq 1}$ such that*

$$\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k') \vee \mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k'') \leq \frac{q-1}{16q}(k+1)^{-4q} \tag{3.27}$$

for any $k \geq 0$, $T \geq T_{\rm sq}$, and $\boldsymbol{u} = (u, u')$ satisfying

$$\boldsymbol{u} \in \overline{B}_H(0,2) \times \overline{B}_H(0,2), \qquad \|u - u'\| \leq 2d. \tag{3.28}$$

Proof. Fix arbitrary $\mathcal{R} > 0$. Let $N_{\rm sq} = N_1$, where N_1 is given in Lemma 3.2. Suppose that (1.14) and (1.15) are satisfied for some $N \geq N_{\rm sq}$.

Step I. We begin with the estimate for the random event $\mathcal{Q}_k'$. Applying Proposition 2.7 with q replaced by $4q$ and $\epsilon = \frac{\nu\alpha_0}{2\mathcal{C}_1}$, where $\mathcal{C}_1$ is the constant in (2.7) and α_0 is the constant in the Poincaré inequality, it follows

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k') &\leq \mathbb{P}_u\{kT \leq \tau_\rho^u < (k+1)T\} + \mathbb{P}_{u'}\{kT \leq \tau_\rho^{u'} < (k+1)T\} \\ &\leq \mathbb{P}_u\Big\{\sup_{t\geq kT} \Lambda_u(t) \geq \rho\Big\} + \mathbb{P}_{u'}\Big\{\sup_{t\geq kT} \Lambda_{u'}(t) \geq \rho\Big\} \leq \frac{C_{q,\mathcal{R},\nu,h}}{(1+kT+\rho)^{4q}}.\end{aligned} \tag{3.29}$$

Step II. Next, we turn to the random event $\mathcal{Q}_k''$. For $k = 0$, by applying Lemma 3.2 combined with (3.8), one has

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_0'') &\leq \mathbb{P}_u\{0 \leq \sigma_1 < T\} = \|\lambda_T(u,u') - \lambda_T'(u')\|_{\rm var} \\ &\leq \frac{\mathcal{C}_q}{(1+\rho)^{4q}} + \frac{\sqrt{6}a}{b_N^*\sqrt{\nu\alpha_0}} e^{\frac{\mathcal{C}_1\rho}{2}} \|u - u'\|,\end{aligned} \tag{3.30}$$

where $\mathcal{C}_q, a$ are given in Lemma 3.2. The case $k \geq 1$ is estimated by applying the Markov property, which gives

$$\begin{aligned}\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k'') &= \mathbb{P}(\mathcal{Q}_k'' \cap \{\sigma \geq kT\}) = \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma\geq kT\}}\mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\mathcal{Q}_k''}\middle|\mathcal{F}_{kT}\right)\right) \\ &\leq \mathbb{E}_{\boldsymbol{u}}\left(\mathbb{I}_{\{\sigma\geq kT\}}\mathbb{P}_{\boldsymbol{u}_{kT}}\{0 \leq \sigma_1 < T\}\right),\end{aligned} \tag{3.31}$$

where, by applying Lemma 3.2 with q, ρ replaced by $4q, \rho + \frac{\nu\alpha_0}{4\mathcal{C}_1}kT$, respectively, one has

$$\mathbb{P}_{\boldsymbol{u}_{kT}}\{0 \leq \sigma_1 < T\} \leq \mathcal{C}_q \frac{1 + \|u_{kT}\|^{32q}}{(1+\rho+\frac{\nu\alpha_0}{4\mathcal{C}_1}kT)^{4q}} + \frac{\sqrt{6}a}{b_N^*\sqrt{\nu\alpha_0}} e^{\frac{\mathcal{C}_1\rho}{2}} e^{\frac{\nu\alpha_0}{8}kT} \|u_{kT} - u_{kT}'\|. \tag{3.32}$$

On the other hand, by (3.15), (3.16), and the definition (3.26) of σ, it follows that

$$v_t^\rho = v_t = u_t', \qquad \|u_t - v_t^\rho\| \leq e^{\frac{\mathcal{C}_1\rho}{2}} e^{-\frac{\nu\alpha_0}{4}t} \|u - u'\| \tag{3.33}$$

for any $t \leq \sigma$. Here, v_t^ρ is the solution of the truncated auxiliary equation (3.14). Thus, by combining (3.31)–(3.33) with the moment estimate (A.4), it follows that

$$\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k'') \leq \frac{\mathcal{C}_q C_{q,\nu,h,\mathfrak{B}_0}}{(1+\rho+\frac{\nu\alpha_0}{4\mathcal{C}_1}kT)^{4q}} + \frac{\sqrt{6}a}{b_N^*\sqrt{\nu\alpha_0}} e^{\mathcal{C}_1\rho} e^{-\frac{\nu\alpha_0}{8}kT} \|u - u'\|. \tag{3.34}$$

Summarizing the estimates (3.30) and (3.34) gives

$$\mathbb{P}_{\boldsymbol{u}}(\mathcal{Q}_k'') \leq \frac{C_{q,\mathcal{R},\nu,h}}{(1+\rho+kT)^{4q}} + \frac{C_{\mathcal{R},\nu,h}}{b_N^*} e^{\mathcal{C}_1\rho} e^{-\frac{\nu\alpha_0}{8}kT} d. \tag{3.35}$$

Step III. Now, we choose $\rho, T_{\rm sq}$ sufficiently large so that

$$\frac{C_{q,\mathcal{R},\nu,h}}{\rho^{4q} \wedge T_{\rm sq}^{4q}} < \frac{q-1}{32q},$$

and then take $d \ll 1$ sufficiently small such that

$$\frac{C_{\mathcal{R},\nu,h}}{b_N^*} e^{\mathcal{C}_1 \rho} e^{-\frac{\nu \alpha_0}{8} kT} d \leq \frac{q-1}{32q} (1+k)^{-4q}.$$

This, together with (3.29) and (3.35), yields the desired estimate (3.27). □

Now we are in a position to prove Proposition 3.1.

Proof of Proposition 3.1. Fix arbitrary $q \geq 2$ and $\mathcal{R} > 0$. We choose the parameter a as given in Lemma 3.2 and $N_0 := N_{\mathrm{rec}} \vee N_{\mathrm{sq}}$. Suppose that (1.14) and (1.15) are satisfied with $N \geq N_0$ and set $T := T_{\mathrm{sq}} \vee T_{\mathrm{rec}}$ as the time step corresponding to N, d with the parameter d given in Lemma 3.6. Let $(\boldsymbol{u}_t, \mathbb{P}_{\boldsymbol{u}})$ be the coupling process constructed in Section 3.1, which corresponds to these parameters a, N_0, T. Based on these parameters, Proposition 3.5 and Lemma 3.6 are now applicable.

The closed ball in Proposition 3.1 is defined as $B := \overline{B}_H(\mathfrak{u}, d)$, and the recurrence property is obtained by applying Proposition 3.5. We turn to the verification of the polynomial squeezing properties (3.3)–(3.6). Since $\|\mathfrak{u}\| \leq 1$ and $d \leq 1$, every $\boldsymbol{u} \in B \times B$ satisfies (3.28). This, together with (3.33), gives (3.3). As for (3.4), we apply Lemma 3.6, which gives

$$\mathbb{P}_{\boldsymbol{u}}\{\sigma = \infty\} \geq 1 - \textstyle\sum_{k=0}^{\infty} \mathbb{P}_{\boldsymbol{u}}\{\sigma \in [kT, (k+1)T)\} \geq \frac{1}{2} > 0.$$

To prove (3.5), we derive from Lemma 3.6 that

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{u}} \left(\mathbb{I}_{\{\sigma < \infty\}} \sigma^q \right) &\leq \textstyle\sum_{k=0}^{\infty} \mathbb{E}_{\boldsymbol{u}} \left(\mathbb{I}_{\{\sigma \in [kT,(k+1)T)\}} \sigma^q \right) \\
&\lesssim_{q,T} \textstyle\sum_{k=0}^{\infty} (k+1)^q \mathbb{P}_{\boldsymbol{u}}\{\sigma \in [kT, (k+1)T)\} \\
&\lesssim_{q,T} \textstyle\sum_{k=0}^{\infty} (k+1)^{-3q} \leq C_q.
\end{aligned}$$

Finally, to establish (3.6), we need the following auxiliary estimate

$$\sup_{k \geq 0} \mathbb{E}_u \sup_{t \in [kT,(k+1)T)} \|u_t\|^{4q} \leq C_{q,T}, \tag{3.36}$$

which follows from an application of (A.4), (A.6), and the Markov property. This, together with Lemma 3.6 and the Cauchy–Schwarz inequality, implies that

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{u}} \left(\mathbb{I}_{\{\sigma < \infty\}} \left(\|u_\sigma\|^{2q} + \|u'_\sigma\|^{2q} \right) \right) \\
&\leq \textstyle\sum_{k=0}^{\infty} \mathbb{E}_{\boldsymbol{u}} \left(\mathbb{I}_{\{\sigma \in [kT,(k+1)T)\}} \sup_{t \in [kT,(k+1)T)} \left(\|u_t\|^{2q} + \|u'_t\|^{2q} \right) \right) \\
&\lesssim_{q,T} \textstyle\sum_{k=0}^{\infty} \left(\mathbb{P}_{\boldsymbol{u}}\{\sigma \in [kT, (k+1)T)\} \right)^{\frac{1}{2}} \leq C_{q,T}.
\end{aligned}$$

This completes the proof of Proposition 3.1. □

Finally, we prove Theorem 1.4.

Proof of Theorem 1.4. The uniqueness of the invariant measure and the second statement are direct consequences of Proposition 3.1; see Section 2 in [NZ24b]. We turn to the third statement. Fix arbitrary $\lambda \in \mathcal{P}(H)$ and function $f \in \mathrm{Lip}(H)$ satisfying $\|f\|_{\mathrm{Lip}(H)} \leq 1$. An application of (1.17) gives

$$\begin{aligned}
|\langle f, \mathcal{D}(u(t;\lambda)) \rangle - \langle f, \mu \rangle| &= |\langle f, \mathfrak{P}_t^* \lambda \rangle - \langle f, \mu \rangle| = |\langle \mathfrak{P}_t f, \lambda \rangle - \langle f, \mu \rangle| \\
&\leq \int_{B_H(R)} |\mathfrak{P}_t f(u) - \langle f, \mu \rangle| \lambda(\mathrm{d}u) + 2\lambda(B_H(R)^c) \\
&\leq C_q (1+R^2)(1+t)^{-q} + 2\lambda(B_H(R)^c).
\end{aligned}$$

Since $\lim_{R\to\infty}\lambda(B_H(R)^c)=0$, by taking supremum in f, sending $t\to\infty$, and then letting $R\to\infty$, the convergence (1.18) holds. Now it remains to prove (1.16). By the Portmanteau theorem, (1.17), and (A.4) with $q=1$, one has

$$\int_H \|u\|^2 \mathrm{d}\mu \le \liminf_{t\to\infty} \mathbb{E}\|u(t;0)\|^2 \le C_{\nu,h}(1+\mathfrak{B}_0),$$

which, together with taking $t\to\infty$, implies

$$\int_H \|u\|^2 \mathrm{d}\mu < \infty. \tag{3.37}$$

To prove the support property of μ, we apply the stationarity of the invariant measure μ again combined with (3.37) and the energy equality (A.1), which gives

$$\nu\int_H \|\nabla u\|^2 \mu(\mathrm{d}u) = \nu\mathbb{E}\int_0^1 \|\nabla u(t;\mu)\|^2 \mathrm{d}t \lesssim_\nu \|h\|^2 + \mathfrak{B}_0 < \infty.$$

Therefore, any stationary process $\{u_{\mathrm{inv}}(t)\}_{t\ge 0}$ has V-valued initial data. Together with the fact $\mathfrak{B}_1<\infty$ and standard regularity result for the stochastic NS equation (2.2), this implies

$$u_{\mathrm{inv}} \in L^2_{\mathrm{loc}}(0,\infty; H^2\cap V).$$

By stationarity again, it follows

$$\mu(H^2\cap V) = \mathbb{E}\int_0^1 \mathbb{I}_{\{u_{\mathrm{inv}}(t)\in H^2\cap V\}}\mathrm{d}t = 1,$$

which, combined with the Sobolev embedding, implies $\mu(H\cap L^\infty)=1$. □

4 Decay estimates for infinite-energy solutions of the homogeneous NS equation

In this section, our goal is to establish decay estimates for infinite-energy solutions of the homogeneous NS equation (1.10). The main result is presented below.

Proposition 4.1. *Let $w(t)$ be an infinite-energy solution of the homogeneous NS equation (1.10) issued from the initial data $w_0\in H_b\cap L^\infty$. Then, for any $R>0$, there are constants $\mathscr{C}_1,\alpha_1$ depending on R,ν such that for any w_0 satisfying $\|w_0\|_{L^2_b}\le R$,*

$$\int_0^\infty \|w(t)\|_\infty^2 \mathrm{d}t \le \mathscr{C}_1\big(1+\log(1+\|w_0\|_\infty)\big), \tag{4.1}$$

and

$$\|w(t)\|_{L^2_b} + \|w(t)\|_\infty \le \mathscr{C}_1 e^{-\alpha_1 t}, \qquad \forall t\ge 1. \tag{4.2}$$

We divide the proof into two steps, each treated in a separate subsection below. Throughout the paper, we need the following weight function

$$\theta_{\varepsilon,\mathfrak{z}}(x) := \frac{1}{\sqrt{1+\varepsilon^2(x-\mathfrak{z})^2}}, \tag{4.3}$$

which was introduced in [Zel07]. The following properties play an important role. Although proofs of these properties can be found in [Zel03, Zel07, AZ14], we collect their proofs in Section B for the reader's convenience.

Proposition 4.2. *The following statements hold for the weight function $\theta_{\varepsilon,\mathfrak{z}}$.*

1. For any $\varepsilon\in(0,1)$ and $p\in(1,\infty)$, it follows

$$\|f\|_{L^p_b} \lesssim_p \sup_{\mathfrak{z}\in\mathbb{R}} \|\theta_{\varepsilon,\mathfrak{z}} f\|_p \lesssim_p \varepsilon^{-\frac{1}{p}}\|f\|_{L^p_b}. \tag{4.4}$$

2. *For any $\varepsilon \in (0,1)$, it follows*

$$|\partial_x \theta_{\varepsilon,\mathfrak{z}}(x)| \le \varepsilon \theta^2_{\varepsilon,\mathfrak{z}}(x) \tag{4.5}$$

3. *For any $\varepsilon \in (0,1)$ and integer $j \ge 1$, it follows*

$$\|\partial_x^j \log \theta_{\varepsilon,\mathfrak{z}}\|_\infty \lesssim_j \varepsilon^j. \tag{4.6}$$

4.1 Estimate on unit time interval

We begin with an estimate of $w(t)$ on the unit time interval. The main result is given below.

Proposition 4.3. *Let $w(t)$ be an infinite-energy solution of the homogeneous NS equation (1.10) issued from the initial data $w_0 \in H_b \cap L^\infty$. Then, for any $R > 0$, there is a constant $\mathscr{C}_2$ depending on R, ν such that for any w_0 satisfying $\|w_0\|_{L^2_b} \le R$,*

$$\int_0^1 \|w(t)\|^2_\infty \mathrm{d}t \le \mathscr{C}_2\big(1 + \log(1 + \|w_0\|_\infty)\big). \tag{4.7}$$

Proof. **Step I.** First, we establish an estimate around $t = 0$ through the Duhamel formulation

$$w(t) = S(t)w_0 - \int_0^t S(t-s)\Pi \operatorname{div}(w \otimes w)(s)\mathrm{d}s, \tag{4.8}$$

and the estimates for the Stokes semigroup in uniformly local spaces. For any $T > 0$, define

$$\|w\|_{\mathcal{Y}_T} := \sup_{0<t\le T} (\|w(t)\|_{L^4_b} + t^{\frac{1}{4}}\|w(t)\|_\infty).$$

By applying (C.2) combined with the Gagliardo-–Nirenberg inequality

$$\|f\|_\infty \lesssim \|f\|^{\frac{1}{2}}_{L^4_b}\|f\|^{\frac{1}{2}}_{W^{1,4}_b} \tag{4.9}$$

and the Poincaré inequality, one has

$$\|S(t)w_0\|_{L^4_b} \lesssim_\nu \|w_0\|_{L^4_b}, \qquad \|S(t)w_0\|_\infty \lesssim_\nu t^{-\frac{1}{4}}\|w_0\|_{L^4_b},$$

which implies

$$\|S(\cdot)w_0\|_{\mathcal{Y}_T} \lesssim_\nu \|w_0\|_{L^4_b}. \tag{4.10}$$

Similarly, an application of (C.3) gives

$$\|S(t-s)\Pi \operatorname{div}(w\otimes w)(s)\|_{L^4_b} \lesssim_\nu \frac{1}{(t-s)^{\frac{1}{2}}}\|(w\otimes w)(s)\|_{L^4_b} \lesssim_\nu \frac{1}{s^{\frac{1}{4}}(t-s)^{\frac{1}{2}}}\|w\|^2_{\mathcal{Y}_T}, \tag{4.11}$$

and

$$\|\nabla S(t-s)\Pi \operatorname{div}(w\otimes w)(s)\|_{L^4_b} \lesssim_\nu \frac{1}{t-s}\|(w\otimes w)(s)\|_{L^4_b} \lesssim_\nu \frac{1}{s^{\frac{1}{4}}(t-s)}\|w\|^2_{\mathcal{Y}_T}.$$

Together with the Gagliardo–Nirenberg inequality (4.9), this yields

$$\|S(t-s)\Pi \operatorname{div}(w\otimes w)(s)\|_\infty \lesssim_\nu \frac{1}{s^{\frac{1}{4}}(t-s)^{\frac{3}{4}}}\|w\|^2_{\mathcal{Y}_T}.$$

As a result, by combining this and (4.11), one has

$$\begin{aligned}\left\|\int_0^t S(t-s)\Pi \operatorname{div}(w\otimes w)(s)\mathrm{d}s\right\|_{L^4_b} &\lesssim_\nu \|w\|^2_{\mathcal{Y}_T}\int_0^t s^{-\frac{1}{4}}(t-s)^{-\frac{1}{2}}\mathrm{d}s \\ &= t^{\frac{1}{4}}B\left(\tfrac{3}{4},\tfrac{1}{2}\right)\|w\|^2_{\mathcal{Y}_T},\end{aligned}$$

and

$$t^{\frac{1}{4}}\left\|\int_0^t S(t-s)\Pi\,\mathrm{div}(w\otimes w)(s)\mathrm{d}s\right\|_\infty \lesssim_\nu t^{\frac{1}{4}}\|w\|_{\mathcal{Y}_T}^2\int_0^t s^{-\frac{1}{4}}(t-s)^{-\frac{3}{4}}\mathrm{d}s$$
$$= t^{\frac{1}{4}}B\left(\tfrac{3}{4},\tfrac{1}{4}\right)\|w\|_{\mathcal{Y}_T}^2,$$

where $B(\cdot,\cdot)$ denotes the Beta function. Consequently, it follows that

$$\left\|\int_0^\cdot S(\cdot-s)\Pi\,\mathrm{div}(w\otimes w)(s)\mathrm{d}s\right\|_{\mathcal{Y}_T} \lesssim_\nu T^{\frac{1}{4}}\|w\|_{\mathcal{Y}_T}^2,$$

which, combined with (4.10), leads to

$$\|w\|_{\mathcal{Y}_T} \le C_\nu\|w_0\|_{L_b^4} + C_\nu T^{\frac{1}{4}}\|w\|_{\mathcal{Y}_T}^2.$$

This, together with a bootstrap argument, yields

$$\|w\|_{\mathcal{Y}_T} \le 2C_\nu\|w_0\|_{L_b^4}, \tag{4.12}$$

provided that $T^{1/4}\|w_0\|_{L_b^4} \le \frac{1}{8C_\nu^2}$.

Step II. Let

$$T_0 := 1\wedge\frac{1}{4096C_\nu^8(1+\|w_0\|_\infty^2)(1+R^2)}.$$

By the Hölder inequality, it follows that

$$T_0^{\frac{1}{4}}\|w_0\|_{L_b^4} \le \frac{1}{8C_\nu^2},$$

which, together with (4.12) and the Hölder inequality again, gives

$$\int_0^{T_0}\|w(t)\|_\infty^2\mathrm{d}t \le 8C_\nu^2\|w_0\|_{L_b^4}^2T_0^{\frac{1}{2}} \le \frac{1}{8C_\nu^2}. \tag{4.13}$$

We turn to the estimate of $w(t)$ on $[T_0,1]$. By applying Theorem 4.1 and Corollary 6.4 in [AZ14], one has

$$\|w(t)\|_{W_b^{2,2}} \le C_Rt^{-1}, \qquad \forall t\in(0,1],$$

and

$$\|w(t)\|_{L_b^2} \le C_R, \qquad \forall t\ge 0.$$

Together with the Gagliardo—Nirenberg inequality

$$\|f\|_\infty \lesssim \|f\|_{L_b^2}^{\frac{1}{2}}\|f\|_{W_b^{2,2}}^{\frac{1}{2}}, \tag{4.14}$$

this implies

$$\int_{T_0}^1\|w(t)\|_\infty^2\mathrm{d}t \lesssim_R \log T_0^{-1} \lesssim_R C_\nu + \log(1+R^2) + \log(1+\|w_0\|_\infty^2)$$

By combining this and (4.13), we obtain (4.7) as desired. □

4.2 Long-time estimate

In this subsection, we aim to establish the following exponential decay estimate for $w(t)$ on $[1,\infty)$. The proof follows the idea of Theorem 6.1 in [AZ14] with a refined treatment of weighted energy estimates. This yields exponential decay to 0, when the cross-sectional flux vanishes, sharpening the dissipative bound stated in that theorem.

Proposition 4.4. *Let $w(t)$ be an infinite-energy solution of the homogeneous NS equation (1.10) issued from the initial data $w_0 \in H_b \cap L^\infty$. Then, for any $R>0$, there are constants $\mathscr{C}_1, \alpha_1$ depending on R, ν such that for any w_0 satisfying $\|w_0\|_{L^2_b} \le R$,*

$$\|w(t)\|_{L^2_b} + \|w(t)\|_\infty \le \mathscr{C}_1 e^{-\alpha_1 t} \tag{4.15}$$

for any $t \ge 1$.

Proof. **Step I.** Let $w(t)$ be an infinite-energy solution of (1.10) issued from w_0. Introduce an auxiliary function $v_{\varepsilon,\mathfrak{z}}(t)$ which satisfies the following backward generalized Stokes equation

$$\begin{cases} -\partial_t v_{\varepsilon,\mathfrak{z}} + \nabla \mathfrak{p}_{\varepsilon,\mathfrak{z}} = \nu \Delta v_{\varepsilon,\mathfrak{z}}, \\ \operatorname{div} v_{\varepsilon,\mathfrak{z}} = 2\theta_{\varepsilon,\mathfrak{z}} \partial_x \theta_{\varepsilon,\mathfrak{z}} w_1, \\ v_{\varepsilon,\mathfrak{z}}|_{t=T} = \left(0, 2\theta_{\varepsilon,\mathfrak{z}} \partial_x \theta_{\varepsilon,\mathfrak{z}} \mathcal{J} w(T)\right), \qquad v_{\varepsilon,\mathfrak{z}}|_{\partial D} = 0, \end{cases} \tag{4.16}$$

where $\mathcal{J} w := \int_{-1}^{y} w_1(x,y')\mathrm{d}y'$ denotes the stream function of the velocity field w. According to Corollary 3.4 in [AZ14], the above equation is well-posed and there exists a constant $\mathscr{C}_3$ depending only on ν such that

$$\|\theta_{\varepsilon,\mathfrak{z}}^{-2} v_{\varepsilon,\mathfrak{z}}\|_{C([0,T];L^2)} + \|\theta_{\varepsilon,\mathfrak{z}}^{-2} v_{\varepsilon,\mathfrak{z}}\|_{C([0,T];L^3)} \le \mathscr{C}_3 \varepsilon \|\theta_{\varepsilon,\mathfrak{z}} w\|_{C([0,T];L^2)}. \tag{4.17}$$

By testing (1.10) against $\theta_{\varepsilon,\mathfrak{z}}^2 w - v_{\varepsilon,\mathfrak{z}}$ and using the fact

$$\operatorname{div}(\theta_{\varepsilon,\mathfrak{z}}^2 w - v_{\varepsilon,\mathfrak{z}}) = 0$$

and $(4.16)_1$, it follows that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\|\theta_{\varepsilon,\mathfrak{z}} w\|^2 - \langle w, v_{\varepsilon,\mathfrak{z}}\rangle\right) + \nu \langle \nabla w, \nabla(\theta_{\varepsilon,\mathfrak{z}}^2 w)\rangle = -\langle (w\cdot\nabla)w, \theta_{\varepsilon,\mathfrak{z}}^2 w - v_{\varepsilon,\mathfrak{z}}\rangle. \tag{4.18}$$

The above weighted energy identity can be rigorously justified by an approximation argument; see Corollary 3.5 in [AZ14]. We estimate the terms in (4.18) in the following way. First, by applying (4.5), the fact $\theta_{\varepsilon,\mathfrak{z}} \le 1$, and the Poincaré inequality, one has

$$\begin{aligned} \langle \nabla w, \nabla(\theta_{\varepsilon,\mathfrak{z}}^2 w)\rangle &= \|\theta_{\varepsilon,\mathfrak{z}} \nabla w\|^2 + 2\langle \partial_x w, \theta_{\varepsilon,\mathfrak{z}} \partial_x \theta_{\varepsilon,\mathfrak{z}} w\rangle \\ &\ge \|\theta_{\varepsilon,\mathfrak{z}} \nabla w\|^2 - 2\varepsilon \|\theta_{\varepsilon,\mathfrak{z}} \nabla w\| \|\theta_{\varepsilon,\mathfrak{z}} w\| \ge (1 - \sqrt{\alpha_0}\varepsilon)\|\theta_{\varepsilon,\mathfrak{z}} \nabla w\|^2. \end{aligned} \tag{4.19}$$

Next, by integrating by parts combined with an application of (4.5) and the Ladyzhenskaya inequality, it follows

$$\begin{aligned} |\langle (w\cdot\nabla)w, \theta_{\varepsilon,\mathfrak{z}}^2 w\rangle| &= |\langle |w|^2, w_1 \theta_{\varepsilon,\mathfrak{z}} \partial_x \theta_{\varepsilon,\mathfrak{z}}\rangle| \\ &\le \varepsilon \|\theta_{\varepsilon,\mathfrak{z}} w\| \|\theta_{\varepsilon,\mathfrak{z}} w\|_4^2 \lesssim \varepsilon \|\theta_{\varepsilon,\mathfrak{z}} w\|^2 \|\theta_{\varepsilon,\mathfrak{z}} \nabla w\|. \end{aligned} \tag{4.20}$$

On the other hand, by using the fact $\theta_{\varepsilon,\mathfrak{z}} \le 1$ and the Sobolev embedding $H^1 \hookrightarrow L^6$, one has

$$|\langle (w\cdot\nabla)w, v_{\varepsilon,\mathfrak{z}}\rangle| \le \|\theta_{\varepsilon,\mathfrak{z}} w\|_6 \|\theta_{\varepsilon,\mathfrak{z}} \nabla w\| \|\theta_{\varepsilon,\mathfrak{z}}^{-2} v_{\varepsilon,\mathfrak{z}}\|_3 \lesssim \|\theta_{\varepsilon,\mathfrak{z}} \nabla w\|^2 \|\theta_{\varepsilon,\mathfrak{z}}^{-2} v_{\varepsilon,\mathfrak{z}}\|_3, \tag{4.21}$$

and

$$|\langle w, v_{\varepsilon,\mathfrak{z}}\rangle| \le \|\theta_{\varepsilon,\mathfrak{z}} w\| \|\theta_{\varepsilon,\mathfrak{z}}^{-2} v_{\varepsilon,\mathfrak{z}}\|. \tag{4.22}$$

Combining (4.17)–(4.22) and choosing $\varepsilon>0$ such that $1-\sqrt{\alpha_0}\varepsilon>\frac{1}{2}$, it follows that for any $T>0$,

$$\|\theta_{\varepsilon,\mathfrak{z}}w(T)\|^2+\nu\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla w\|^2\mathrm{d}t\lesssim\|\theta_{\varepsilon,\mathfrak{z}}w(T)\|\|\theta_{\varepsilon,\mathfrak{z}}^{-2}v_{\varepsilon,\mathfrak{z}}(T)\|+\|\theta_{\varepsilon,\mathfrak{z}}w_0\|\|\theta_{\varepsilon,\mathfrak{z}}^{-2}v_{\varepsilon,\mathfrak{z}}(0)\|$$
$$+\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2+\varepsilon\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}w\|^2\|\theta_{\varepsilon,\mathfrak{z}}\nabla w\|\mathrm{d}t+\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla w\|^2\|\theta_{\varepsilon,\mathfrak{z}}^{-2}v_{\varepsilon,\mathfrak{z}}\|_3\mathrm{d}t,$$

which, together with $(4.16)_3$, (4.17), and (4.5), implies that

$$\|\theta_{\varepsilon,\mathfrak{z}}w(T)\|^2+\nu\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla w\|^2\mathrm{d}t\le\mathscr{C}_4\varepsilon\|\theta_{\varepsilon,\mathfrak{z}}w(T)\|^2+\mathscr{C}_4\varepsilon\mathscr{M}_T\|\theta_{\varepsilon,\mathfrak{z}}w_0\|$$
$$+\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2+\mathscr{C}_4\varepsilon\mathscr{M}_T\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla w\|^2\mathrm{d}t,$$

where $\mathscr{C}_4$ depends only on ν and

$$\mathscr{M}_T:=\|\theta_{\varepsilon,\mathfrak{z}}w\|_{C([0,T];L^2)}.$$

Consequently, by shrinking $\varepsilon>0$ so that $\mathscr{C}_4\varepsilon<\frac{1}{4}$, one gets

$$\|\theta_{\varepsilon,\mathfrak{z}}w(T)\|^2+(\nu-\mathscr{C}_4\varepsilon\mathscr{M}_T)\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla w\|^2\mathrm{d}t\le\mathscr{C}_4\varepsilon\mathscr{M}_T\|\theta_{\varepsilon,\mathfrak{z}}w_0\|+\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2.\quad(4.23)$$

Step Ⅱ. We close the above weighted energy estimate via a standard bootstrap argument. Fix $R>0$ and initial data $w_0\in H_b\cap L^\infty$ satisfying $\|w_0\|_{L_b^2}\le R$. Choose $\varepsilon>0$ sufficiently small so that

$$1-\sqrt{\alpha_0}\varepsilon>\frac{1}{2},\qquad\mathscr{C}_4\varepsilon<\frac{1}{4},\qquad2\mathscr{C}_4C\varepsilon^{\frac{1}{2}}R<\frac{\nu}{2},\quad(4.24)$$

where C denotes the constant in (4.4). As a result, one has

$$2\mathscr{C}_4\varepsilon\|\theta_{\varepsilon,\mathfrak{z}}w_0\|\le2\mathscr{C}_4C'\varepsilon^{\frac{1}{2}}\|w_0\|_{L_b^2}<\frac{\nu}{2}.\quad(4.25)$$

Suppose that at time T, it follows

$$\mathscr{M}_T\le2\|\theta_{\varepsilon,\mathfrak{z}}w_0\|,$$

which, together with (4.23)–(4.25), leads to

$$\|\theta_{\varepsilon,\mathfrak{z}}w(T)\|^2+\frac{\nu}{2}\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla w\|^2\mathrm{d}t\le\frac{3}{2}\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2.\quad(4.26)$$

Applying the above estimate to $T_*:=\inf\{t\ge0|\mathscr{M}_T=2\|\theta_{\varepsilon,\mathfrak{z}}w_0\|\}$ gives

$$\mathscr{M}_{T_*}^2=\sup_{t\le T_*}\|\theta_{\varepsilon,\mathfrak{z}}w(t)\|^2\le\frac{3}{2}\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2<4\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2.$$

This contradiction indicates that $T_*=\infty$ and (4.26) holds for any $T>0$. In particular, one has

$$\sup_{t\le T}\|\theta_{\varepsilon,\mathfrak{z}}w(t)\|^2+\frac{\nu}{2}\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla w\|^2\mathrm{d}t\le3\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2,\qquad\forall T\ge0.\quad(4.27)$$

Step Ⅲ. In this step, we establish the exponential decay of $\|w(t)\|_{L_b^2}$. First, for any $s\ge0$, by shrinking $\varepsilon>0$ so that

$$2\mathscr{C}_4\varepsilon\|\theta_{\varepsilon,\mathfrak{z}}w(s)\|\le2\sqrt{3}\mathscr{C}_4C'\varepsilon^{\frac{1}{2}}\|w_0\|_{L_b^2}\le2\sqrt{3}\mathscr{C}_4C'\varepsilon^{\frac{1}{2}}R<\frac{\nu}{2},\quad(4.28)$$

and repeating the bootstrap argument in Step **II**, it follows that

$$\sup_{s\le t\le s+T}\|\theta_{\varepsilon,\mathfrak{z}}w(t)\|^2+\frac{\nu}{2}\int_s^{s+T}\|\theta_{\varepsilon,\mathfrak{z}}\nabla w\|^2\mathrm{d}t\le 3\|\theta_{\varepsilon,\mathfrak{z}}w(s)\|^2,\qquad \forall T\ge 0. \tag{4.29}$$

In particular, one has

$$\|\theta_{\varepsilon,\mathfrak{z}}w(s+T)\|^2\le 3\|\theta_{\varepsilon,\mathfrak{z}}w(t)\|^2,\qquad \forall t\in[s,s+T].$$

Integrating the above inequality with respect to t yields

$$T\|\theta_{\varepsilon,\mathfrak{z}}w(s+T)\|^2\le 3\int_s^{s+T}\|\theta_{\varepsilon,\mathfrak{z}}w(t)\|^2\mathrm{d}t,$$

which, combined with the Poincaré inequality and (4.29), leads to

$$T\|\theta_{\varepsilon,\mathfrak{z}}w(s+T)\|^2\le 3\alpha_0^{-1}\int_s^{s+T}\|\theta_{\varepsilon,\mathfrak{z}}\nabla w(t)\|^2\mathrm{d}t\le\frac{18}{\nu\alpha_0}\|\theta_{\varepsilon,\mathfrak{z}}w(s)\|^2.$$

By choosing $T_\nu:=\max\{1,36\nu^{-1}\alpha_0^{-1}\}$, it follows

$$\|\theta_{\varepsilon,\mathfrak{z}}w(s+T_\nu)\|^2\le\frac{1}{2}\|\theta_{\varepsilon,\mathfrak{z}}w(s)\|^2.$$

Iterating the above inequality gives

$$\|\theta_{\varepsilon,\mathfrak{z}}w(nT_\nu)\|^2\le\frac{1}{2^n}\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2,\qquad \forall n\ge 0.$$

Consequently, for any $t\ge 0$, by applying the above inequality and (4.29), one gets

$$\|\theta_{\varepsilon,\mathfrak{z}}w(t)\|^2\le 3\|\theta_{\varepsilon,\mathfrak{z}}w(\lfloor tT_\nu^{-1}\rfloor T_\nu)\|^2\le 3e^{-\lfloor tT_\nu^{-1}\rfloor\log 2}\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2\le 6e^{-(T_\nu^{-1}\log 2)t}\|\theta_{\varepsilon,\mathfrak{z}}w_0\|^2.$$

Since ε depends only on R,ν, see (4.24) and (4.28), by combining the above estimate and (4.4), it follows

$$\|w(t)\|_{L^2_b}\lesssim_{R,\nu} e^{-\alpha_1 t},\qquad \forall t\ge 0. \tag{4.30}$$

Step IV. Finally, we estimate $\|w(t)\|_\infty$. For any $t\ge 1$, by applying Corollary 6.4 in [AZ14], there is an increasing function $Q:\mathbb{R}_+\to\mathbb{R}_+$ such that

$$\|w(t)\|_{W^{2,2}_b}\le 2Q(\|w(t-2^{-1})\|_{L^2_b}),$$

which, combined with (4.30), yields

$$\|w(t)\|_{W^{2,2}_b}\le C_{R,\nu}. \tag{4.31}$$

Together with (4.30) and the Gagliardo–Nirenberg inequality (4.14), this implies

$$\|w(t)\|_\infty\lesssim_{R,\nu} e^{-\frac{\alpha_1}{2}t}$$

for any $t\ge 1$, and thus completes the proof. □

Now we turn to the proof of Proposition 4.1.

Proof of Proposition 4.1. Notice that estimate (4.2) is obtained in Proposition 4.4. As for estimate (4.1), by applying Propositions 4.3 and 4.4, one has

$$\begin{aligned}\int_0^\infty\|w(t)\|_\infty^2\mathrm{d}t&=\int_0^1\|w(t)\|_\infty^2\mathrm{d}t+\int_1^\infty\|w(t)\|_\infty^2\mathrm{d}t\\&\le C_{R,\nu}\big(1+\log(1+\|w_0\|_\infty)\big)+C_{R,\nu}.\end{aligned}$$

This completes the proof. □

5 Mixing for infinite-energy solutions

The goal of this section is to establish Theorem 1.5. To this end, we introduce the following decomposition

$$u(t) = w(t) + u_{\rm rem}(t),$$

where $w(t)$ is an infinite-energy solution of the homogeneous NS equation (1.10) issued from u_0 and $u_{\rm rem}(t)$ solves the white-forced NS-type equation (1.23) with zero initial data.

5.1 Probabilistic estimates for the remainder

In this subsection, we establish probabilistic estimates for the finite-energy remainder $u_{\rm rem}$ that will be used in the proof of Theorem 1.5.

Proposition 5.1. *Suppose that $u_0 \in H_b \cap L^\infty$ and $\mathfrak{B}_0 < \infty$. Then, the following statements hold.*

1. *For any $q \in [1,\infty)$, it follows that*

$$\sup_{t\geq 0} \mathbb{E}\|u_{\rm rem}(t)\|^{2q} \lesssim_{q,\nu,h,\mathfrak{B}_0} \exp\left(\frac{2q}{\nu}\int_0^\infty \|w(s)\|_\infty^2 \mathrm{d}s\right). \tag{5.1}$$

2. *For any $q \in [1,\infty)$ and $t, T \geq 0$, it follows that*

$$\mathbb{E}\left(\sup_{t\leq s\leq t+T}\|u_{\rm rem}(s)\|^2 + \nu\int_t^{t+T}\|\nabla u_{\rm rem}(s)\|^2\mathrm{d}s\right)^q \lesssim_{q,\nu,h,\mathfrak{B}_0} (1+T)^q \exp\left(\frac{2q}{\nu}\int_0^\infty \|w(s)\|_\infty^2 \mathrm{d}s\right). \tag{5.2}$$

Proof. **Step I.** Write

$$\Theta(t) := \exp\left(-\frac{2}{\nu}\int_0^t \|w(s)\|_\infty^2 \mathrm{d}s\right).$$

By applying the Itô formula, one has

$$\begin{aligned}\mathrm{d}\|u_{\rm rem}(t)\|^2\Theta(t) = &-\frac{2}{\nu}\Theta(t)\|u_{\rm rem}(t)\|^2\|w(t)\|_\infty^2\mathrm{d}t \\ &+ 2\Theta(t)\langle u_{\rm rem}, -(u_{\rm rem}\cdot\nabla)w + \nu\Delta u_{\rm rem} + h\rangle\mathrm{d}t \\ &+ 2\Theta(t)\textstyle\sum_{j\geq 1} b_j\langle u_{\rm rem}, e_j\rangle\mathrm{d}\beta_j + \mathfrak{B}_0\Theta(t)\mathrm{d}t.\end{aligned} \tag{5.3}$$

Together with the Itô formula again, it follows that

$$\begin{aligned}\mathrm{d}\left(\|u_{\rm rem}(t)\|^2\Theta(t)\right)^q = &-\frac{2q}{\nu}\Theta^q(t)\|u_{\rm rem}(t)\|^{2q}\|w(t)\|_\infty^2\mathrm{d}t \\ &+ 2q\Theta^q(t)\|u_{\rm rem}(t)\|^{2(q-1)}\langle u_{\rm rem}, -(u_{\rm rem}\cdot\nabla)w + \nu\Delta u_{\rm rem} + h\rangle\mathrm{d}t \\ &+ 2q\Theta^q(t)\|u_{\rm rem}(t)\|^{2(q-1)}\textstyle\sum_{j\geq 1} b_j\langle u_{\rm rem}, e_j\rangle\mathrm{d}\beta_j \\ &+ \mathfrak{B}_0 q\Theta^q(t)\|u_{\rm rem}(t)\|^{2(q-1)}\mathrm{d}t \\ &+ 2q(q-1)\Theta^q(t)\|u_{\rm rem}(t)\|^{2(q-2)}\textstyle\sum_{j\geq 1} b_j^2\langle u_{\rm rem}, e_j\rangle^2\mathrm{d}t,\end{aligned} \tag{5.4}$$

where, by the cancellation property of the convection term combined with the Young and Poincaré inequalities, one has

$$\begin{aligned}
&2q\Theta^q(t)\|u_{\rm rem}\|^{2(q-1)}\langle u_{\rm rem}, -(u_{\rm rem}\cdot\nabla)w+\nu\Delta u_{\rm rem}+h\rangle\\
&\quad= 2q\Theta^q(t)\|u_{\rm rem}\|^{2(q-1)}\left(\langle (u_{\rm rem}\cdot\nabla)u_{\rm rem}, w\rangle-\nu\|\nabla u_{\rm rem}\|^2+\langle u_{\rm rem}, h\rangle\right)\\
&\quad\le 2q\Theta^q(t)\|u_{\rm rem}\|^{2(q-1)}\left(\frac{1}{2\nu}\|u_{\rm rem}\|^2\|w\|_\infty^2-\frac{\nu}{4}\|\nabla u_{\rm rem}\|^2+\nu^{-1}\alpha_0^{-1}\|h\|^2\right)\\
&\quad\le \frac{q}{\nu}\Theta^q(t)\|u_{\rm rem}\|^{2q}\|w\|_\infty^2-\frac{q\nu}{2}\Theta^q(t)\|u_{\rm rem}\|^{2(q-1)}\|\nabla u_{\rm rem}\|^2\\
&\qquad+\frac{2q}{\nu\alpha_0}\Theta^q(t)\|u_{\rm rem}\|^{2(q-1)}\|h\|^2\\
&\quad\le \frac{q}{\nu}\Theta^q(t)\|u_{\rm rem}\|^{2q}\|w\|_\infty^2-\frac{q\nu}{4}\Theta^q(t)\|u_{\rm rem}\|^{2(q-1)}\|\nabla u_{\rm rem}\|^2\\
&\qquad+C_{q,\nu}\Theta^q(t)\|h\|^{2q},
\end{aligned}\tag{5.5}$$

and

$$\begin{aligned}
&\mathfrak{B}_0 q\Theta^q(t)\|u_{\rm rem}\|^{2(q-1)}+2q(q-1)\Theta^q(t)\|u_{\rm rem}\|^{2(q-2)}\textstyle\sum_{j\ge1}b_j^2\langle u_{\rm rem}, e_j\rangle^2\\
&\quad\le C_q\mathfrak{B}_0\Theta^q(t)\|u_{\rm rem}\|^{2(q-1)}\le\frac{q\nu}{8}\Theta^q(t)\|u_{\rm rem}\|^{2(q-1)}\|\nabla u_{\rm rem}\|^2+C_{q,\nu}\mathfrak{B}_0^q\Theta^q(t).
\end{aligned}\tag{5.6}$$

By taking expectation on both sides of (5.4) and applying (5.5), (5.6), and the Gronwall inequality, one obtains

$$\Theta^q(t)\mathbb{E}\|u_{\rm rem}(t)\|^{2q}\le C_{q,\nu,h,\mathfrak{B}_0},\qquad\forall t\ge0,\tag{5.7}$$

which ensures (5.1).

Step II. Integrating on both sides of (5.3) from t to $t+T$ and applying (5.5) with $q=1$, one has

$$\begin{aligned}
&\|u_{\rm rem}(t+T)\|^2\Theta(t+T)-\|u_{\rm rem}(t)\|^2\Theta(t)+\frac{\nu}{2}\int_t^{t+T}\Theta(s)\|\nabla u_{\rm rem}\|^2{\rm d}s\\
&\qquad\le C_{\nu,h,\mathfrak{B}_0}\int_t^{t+T}\Theta(s){\rm d}s+2\int_t^{t+T}\Theta(s)\textstyle\sum_{j\ge1}b_j\langle u_{\rm rem}, e_j\rangle{\rm d}\beta_j,
\end{aligned}\tag{5.8}$$

where, by the Burkholder–Davis–Gundy inequality, it follows that

$$\mathbb{E}\sup_{t'\in[0,T]}\left|\int_t^{t+t'}\Theta(s)\textstyle\sum_{j\ge1}b_j\langle u_{\rm rem}, e_j\rangle{\rm d}\beta_j\right|^q\lesssim_q\mathfrak{B}_0^{\frac{q}{2}}\mathbb{E}\left(\int_t^{t+T}\Theta^2(s)\|u_{\rm rem}\|^2{\rm d}s\right)^{\frac{q}{2}}.$$

Consequently, by combining this with (5.7), (5.8), and the Hölder inequality, one gets

$$\begin{aligned}
&\mathbb{E}\sup_{t\le s\le t+T}\|u_{\rm rem}(s)\|^{2q}\Theta^q(s)+\mathbb{E}\left(\nu\int_t^{t+T}\Theta(s)\|\nabla u_{\rm rem}\|^2{\rm d}s\right)^q\\
&\qquad\lesssim_{q,\nu,h,\mathfrak{B}_0}T^q+\Theta^q(t)\mathbb{E}\|u_{\rm rem}(t)\|^{2q}+\mathbb{E}\left(\int_t^{t+T}\Theta^2(s)\|u_{\rm rem}\|^2{\rm d}s\right)^{\frac{q}{2}}\\
&\qquad\lesssim_{q,\nu,h,\mathfrak{B}_0}T^q+1.
\end{aligned}$$

Since $\Theta(t)\ge\exp\left(-2\nu^{-1}\int_0^\infty\|w(s)\|_\infty^2{\rm d}s\right)$, this implies (5.2). □

5.2 *The asymptotic behavior of the remainder*

In this subsection, we show that the remainder $u_{\rm rem}$ can be asymptotically approximated by a family of finite-energy solutions of (1.1), as $t \to \infty$. To formulate the result, we define $\{u_{\rm app}(t;T)\}_{t\geq T}$ as the finite-energy solution of (1.1) satisfying the initial condition

$$u_{\rm app}(T;T) = u_{\rm rem}(T). \tag{5.9}$$

Proposition 5.2. *Suppose that $u_0 \in H_b \cap L^\infty$ and $\mathfrak{B}_0 < \infty$. Then, for any $R > 0$, there is a constant $\alpha_2 > 0$ depending on ν, R such that for any u_0 satisfying $\|u_0\|_{L^2_b} \leq R$, $q \in [1,\infty)$, and $T \geq 4$,*

$$\mathbb{E} d_{\rm loc}(u_{\rm app}(\mathfrak{T};T), u_{\rm rem}(\mathfrak{T})) \lesssim_{q,R,\nu,h,\mathfrak{B}_0} \frac{(1+\|u_0\|_\infty)^{\alpha_2 q}}{(1+\mathfrak{T})^{\frac{q}{2}}}, \tag{5.10}$$

where $\mathfrak{T} := T + \sqrt{T}$.

Proof. **Step I.** Let $\mathfrak{g}(t) := u_{\rm rem}(t) - u_{\rm app}(t;T)$ for $t \geq T$. Then, $\mathfrak{g}$ satisfies

$$\begin{cases} \partial_t \mathfrak{g} - \nu L\mathfrak{g} + B(\mathfrak{g}, u_{\rm rem}) + B(u_{\rm app}, \mathfrak{g}) + B(w, u_{\rm rem}) + B(u_{\rm rem}, w) = 0, \\ \mathfrak{g}(T) = 0, \end{cases}$$

where the operators L and $B(\cdot,\cdot)$ are defined in (2.1). By testing the above equation against $\mathfrak{g}$ and applying the cancellation property of the convection term and the Ladyzhenskaya inequality, it follows

$$\begin{aligned} \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\mathfrak{g}(t)\|^2 + \nu\|\nabla\mathfrak{g}\|^2 &= -\langle B(\mathfrak{g}, u_{\rm rem}) + B(w, u_{\rm rem}) + B(u_{\rm rem}, w), \mathfrak{g}\rangle \\ &\lesssim \|\mathfrak{g}\|\|\nabla\mathfrak{g}\|\|\nabla u_{\rm rem}\| + \|w\|_\infty\|\nabla\mathfrak{g}\|\|u_{\rm rem}\|, \end{aligned}$$

which, together with Cauchy–Schwarz inequality, implies

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\mathfrak{g}(t)\|^2 + \nu\|\nabla\mathfrak{g}\|^2 \lesssim_\nu \|\mathfrak{g}\|^2\|\nabla u_{\rm rem}\|^2 + \|w\|_\infty^2\|u_{\rm rem}\|^2.$$

An application of the Gronwall inequality yields

$$\|\mathfrak{g}(\mathfrak{T})\|^2 \lesssim_\nu \int_T^{\mathfrak{T}} \|w(t)\|_\infty^2\|u_{\rm rem}(t)\|^2 \exp\left(C_\nu \int_t^{\mathfrak{T}} \|\nabla u_{\rm rem}(s)\|^2 \mathrm{d}s\right) \mathrm{d}t. \tag{5.11}$$

Step II. Define

$$\mathcal{I}_T := \int_T^{\mathfrak{T}} \|\nabla u_{\rm rem}(t)\|^2 \mathrm{d}t, \qquad \mathcal{A}_T := \{\mathcal{I}_T \leq C_\nu^{-1}\alpha_1 T\},$$

where α_1 is the constant given in Proposition 4.1 and C_ν is given in (5.11). Notice that on the event $\mathcal{A}_T$, one has

$$\|\mathfrak{g}(\mathfrak{T})\|^2 \lesssim_{R,\nu} Te^{-\alpha_1 T} \lesssim_{R,\nu} e^{-\frac{\alpha_1}{2}T}. \tag{5.12}$$

On the other hand, an application of (5.2) and (4.1) implies

$$\begin{aligned} \mathbb{E}\mathcal{I}_T^q &\lesssim_{q,\nu,h,\mathfrak{B}_0} (1+\sqrt{T})^q \exp\left(\frac{2q}{\nu}\int_0^\infty \|w(s)\|_\infty^2 \mathrm{d}s\right) \\ &\lesssim_{q,R,\nu,h,\mathfrak{B}_0} (1+\sqrt{T})^q (1+\|u_0\|_\infty)^{\alpha_2 q}, \end{aligned}$$

where α_2 is a constant depending on R, ν. Together with the Chebyshev inequality, this yields

$$\mathbb{P}(\mathcal{A}_T^c) \lesssim_{R,\nu} \frac{\mathbb{E}\mathcal{I}_T^q}{T^q} \lesssim_{q,R,\nu,h,\mathfrak{B}_0} \frac{(1+\|u_0\|_\infty)^{\alpha_2 q}}{(1+T)^{\frac{q}{2}}}.$$

Consequently, by combining the above estimate with (5.12), it follows

$$\mathbb{E}(1\wedge\|\mathfrak{g}(\mathfrak{T})\|)\le\mathbb{E}(\|\mathfrak{g}(\mathfrak{T})\|\mathbb{I}_{\mathcal{A}_T})+\mathbb{P}(\mathcal{A}_T^c)\lesssim_{q,R,\nu,h,\mathfrak{B}_0} e^{-\frac{\alpha_1}{4}T}+\frac{(1+\|u_0\|_\infty)^{\alpha_2 q}}{(1+T)^{\frac{q}{2}}}$$

$$\lesssim_{q,R,\nu,h,\mathfrak{B}_0}\frac{(1+\|u_0\|_\infty)^{\alpha_2 q}}{(1+\mathfrak{T})^{\frac{q}{2}}}.$$

Since by the definition (1.8),

$$d_{\mathrm{loc}}(u,v)\le 1\wedge\|u-v\|,\qquad \forall u,v\in H,$$

this implies (5.10). □

5.3 *Proof of Theorem 1.5*

To prove Theorem 1.5, we need the following quantitative smoothing estimate for the infinite-energy solution $\{u(t)\}_{t\ge0}$.

Lemma 5.3. *Suppose that $u_0\in H_b$ and $\mathfrak{B}_1<\infty$. Then, for any $\mathscr{L},R>0$, there is a constant $\mathscr{C}_5$ depending on $\mathscr{L},R,\nu,h$ such that for any u_0 satisfying $\|u_0\|_{L_b^2}\le R$,*

$$\|u(t)\|_{L_b^2}\le\mathscr{C}_5,\qquad \|u(t)\|_\infty\le\mathscr{C}_5 t^{-\frac12},\qquad \forall t\in(0,1] \tag{5.13}$$

on the random event

$$\mathcal{A}_{t,\mathscr{L}}:=\Big\{\sup_{s\in[0,t]}\|\mathcal{Z}_s\|_{H^{\frac32}}\le\mathscr{L}\Big\},\qquad \mathcal{Z}_t:=\textstyle\sum_{j\ge1}b_j\int_0^t e^{\nu L(t-s)}e_j\mathrm{d}\beta_j. \tag{5.14}$$

Moreover, it follows that

$$\mathbb{P}(\mathcal{A}_{t,\mathscr{L}}^c)\lesssim\exp\left(-\frac{\alpha_3\mathscr{L}^2}{\mathfrak{B}_1\sqrt{t}}\right), \tag{5.15}$$

where the constant α_3 depends only on ν.

We postpone the proof of this lemma to the end of Appendix D and turn to the proof of Theorem 1.5.

Proof of Theorem 1.5. Let $(u_t^H,\mathbb{P}_u^H)$ and $(u_t^X,\mathbb{P}_u^X)$ be the Markov families associated with the finite-energy and infinite-energy solutions, respectively. We denote the corresponding transition operators by $\mathfrak{P}_t^H$ and $\mathfrak{P}_t^X$; see (2.3) for the definition of $\mathfrak{P}_t^H$ and $\mathfrak{P}_t^X$ is defined in the same way. Let $\mathfrak{P}_t^{H*}$ and $\mathfrak{P}_t^{X*}$ be the dual operators of $\mathfrak{P}_t^H$ and $\mathfrak{P}_t^X$, respectively. The proof is divided into the following three steps.

Step I. Suppose that

$$u_0\in H_b\cap L^\infty,\qquad \|u_0\|_{L_b^2}\le R,\qquad R>0.$$

Fix a function $f\in\mathrm{Lip}(X)$ satisfying

$$\|f\|_{\mathrm{Lip}(X)}\le1. \tag{5.16}$$

Since $d_{\mathrm{loc}}(u,u')\le\|u-u'\|$, then $f|_H\in\mathrm{Lip}(H)$ and the above condition also holds for $f|_H$. For simplicity, we do not distinguish f and its restriction $f|_H$ on H. Let u_{app} be the

finite-energy solution constructed in Proposition 5.2 and $\mathfrak{T} := T + \sqrt{T}$. By the Markov property, the relation (5.9), Theorem 1.4, and Propositions 4.1 and 5.1, it follows

$$\begin{aligned}|\mathbb{E}f(u_{\mathrm{app}}(\mathfrak{T};T)) - \langle f,\mu\rangle| = |\mathbb{E}\mathfrak{P}^H_{\sqrt{T}}f(u_{\mathrm{rem}}(T)) - \langle f,\mu\rangle| &\lesssim_q \frac{1+\mathbb{E}\|u_{\mathrm{rem}}(T)\|^2}{(1+\sqrt{T})^q}\\ &\lesssim_{q,\nu,h,\{b_j\}_{j\geq 1}} \frac{\exp\left(\frac{2}{\nu}\int_0^\infty \|w(s)\|_\infty^2 \mathrm{d}s\right)}{(1+\sqrt{T})^q}\\ &\lesssim_{q,R,\nu,h,\{b_j\}_{j\geq 1}} \frac{(1+\|u_0\|_\infty)^{\alpha_4}}{(1+T)^{\frac{q}{2}}}\end{aligned}$$

for any $q \geq 1$ and $T \geq 4$, where $\alpha_4 > 0$ is a constant depending only on R, ν. Together with Proposition 5.2, the decay estimate (4.2) in Proposition 4.1, and the fact

$$d_{\mathrm{loc}}(u,u') \leq \sum_{j\geq 1} \frac{1 \wedge \sqrt{2j}\|u-u'\|_{L^2_b}}{2^j} \leq 2(1 \wedge \|u-u'\|_{L^2_b}),$$

this implies

$$\begin{aligned}|\mathbb{E}f(u(\mathfrak{T})) - \langle f,\mu\rangle| &\lesssim_{q,R,\nu,h,\{b_j\}_{j\geq 1}} e^{-\alpha_1\mathfrak{T}} + \frac{(1+\|u_0\|_\infty)^{\alpha_2 q}}{(1+\mathfrak{T})^{\frac{q}{2}}} + \frac{(1+\|u_0\|_\infty)^{\alpha_4}}{(1+\mathfrak{T})^{\frac{q}{2}}}\\ &\lesssim_{q,R,\nu,h,\{b_j\}_{j\geq 1}} \frac{(1+\|u_0\|_\infty)^{\alpha_5 q}}{(1+\mathfrak{T})^{\frac{q}{2}}} \qquad (5.17)\end{aligned}$$

for any $q \geq 1$ and $\mathfrak{T} \geq 6$. Here, the constant $\alpha_5 := \alpha_2 \vee \alpha_4$ depends only on R, ν.

Step II. In this step, we establish (1.19). Suppose that

$$u_0 \in H_b, \qquad \|u_0\|_{L^2_b} \leq R, \qquad R > 0.$$

Define $t_0 := t^{-c}$ with $c > 0$ to be determined in (5.18). Let $\mathcal{A}_{t,\mathscr{L}}$ be the random event introduced in (5.14). Since $t_0 \leq 1$, an application of Lemma 5.3 gives

$$\|u(t_0)\|_{L^2_b} \leq C_{R,\nu,h}, \qquad \|u(t_0)\|_\infty \leq C_{R,\nu,h} t_0^{-\frac{1}{2}}$$

on the event $\mathcal{A}_{t_0,1}$. Consequently, by applying (5.17) to $u(t)$ with $t \geq 8$ combined with the Markov property, one has

$$\begin{aligned}\left|\mathbb{E}\left(\mathbb{I}_{\mathcal{A}_{t_0,1}}\left(f(u(t)) - \langle f,\mu\rangle\right)\right)\right| &\leq \mathbb{E}\left(\mathbb{I}_{\mathcal{A}_{t_0,1}}|\mathfrak{P}^X_{t-t_0}f(u(t_0)) - \langle f,\mu\rangle|\right)\\ &\lesssim_{q,R,\nu,h,\{b_j\}_{j\geq 1}} \frac{1+\mathbb{E}(\mathbb{I}_{\mathcal{A}_{t_0,1}}\|u(t_0)\|_\infty^{\alpha_6 q})}{(1+t-t_0)^{\frac{q}{2}}}\\ &\lesssim_{q,R,\nu,h,\{b_j\}_{j\geq 1}} \frac{1+t^{\frac{c\alpha_6 q}{2}}}{(1+t-t_0)^{\frac{q}{2}}}\end{aligned}$$

for some constant α_6 depending on R, ν, h. Together with choosing

$$c = \frac{1}{2\alpha_6} \qquad (5.18)$$

and using the fact $t - t_0 \geq \frac{t}{2}$, this yields

$$\left|\mathbb{E}\left(\mathbb{I}_{\mathcal{A}_{t_0,1}}\left(f(u(t)) - \langle f,\mu\rangle\right)\right)\right| \lesssim_{q,R,\nu,h,\{b_j\}_{j\geq 1}} \frac{1}{(1+t)^{\frac{q}{4}}}.$$

Combining this with (5.15), one gets

$$|\mathbb{E}f(u(t)) - \langle f,\mu\rangle| \lesssim_{q,R,\nu,h,\{b_j\}_{j\geq 1}} \frac{1}{(1+t)^{\frac{q}{4}}}$$

for any $t \geq 8$. The case $t < 8$ is bounded by the trivial bound

$$|\mathbb{E}f(u(t)) - \langle f, \mu \rangle| \leq 2.$$

Since q is arbitrary, by replacing q by $4q$, the convergence estimate (1.19) holds.

Step III. It remains to establish (1.20), which ensures the uniqueness of the invariant measure. Let $\lambda \in \mathcal{P}(X)$. For any function $f \in \mathrm{Lip}(X)$ satisfying (5.16), by applying (1.19), it follows

$$\begin{aligned}
|\langle f, \mathcal{D}(u(t;\lambda))\rangle - \langle f, \mu\rangle| &= |\langle f, \mathfrak{P}_t^{X*}\lambda\rangle - \langle f, \mu\rangle| = |\langle \mathfrak{P}_t^X f, \lambda\rangle - \langle f, \mu\rangle| \\
&\leq \int_{B_{H_b}(R)} |\mathfrak{P}_t^X f(u) - \langle f, \mu\rangle| \lambda(\mathrm{d}u) + 2\lambda(B_{H_b}(R)^c) \\
&\leq C_{q,R}(1+t)^{-q} + 2\lambda(B_{H_b}(R)^c).
\end{aligned}$$

Since $\lim_{R\to\infty} \lambda(B_{H_b}(R)^c) = 0$, by taking supremum in f, sending $t \to \infty$, and then letting $R \to \infty$, the convergence (1.20) holds. □

A A priori energy estimates

This appendix collects basic energy estimates used throughout the paper.

Proposition A.1. *Let $\{u(t)\}_{t\geq 0}$ be the solution of (2.2) issued from $u \in H$. Suppose that $\mathfrak{B}_0 := \sum_{j\geq 1} b_j^2 < \infty$ and $h \in H$. Then, the following estimates hold.*

1. ***Energy equality.*** *For any $t \geq 0$, it follows*

$$\mathbb{E}\|u(t)\|^2 + 2\nu\mathbb{E}\int_0^t \|\nabla u(s)\|^2 \mathrm{d}s = 2\mathbb{E}\int_0^t \langle u(s), h\rangle \mathrm{d}s + \|u\|^2 + \mathfrak{B}_0 t. \tag{A.1}$$

2. ***Moment estimates.*** *Let $\kappa_0 := \frac{\nu\alpha_0}{16(1+\mathfrak{B}_0)}$ and*

$$\mathcal{E}(t) := \sup_{s\leq t} \|u(s)\|^2 + \nu\int_0^t \|\nabla u(s)\|^2 \mathrm{d}s, \tag{A.2}$$

where α_0 is the constant in the Poincaré inequality. Then, for any $0 < \kappa \leq \kappa_0$, $q \geq 1$, and $t \geq 0$, it follows

$$\mathbb{E}e^{\kappa\|u(t)\|^2} \leq e^{-\kappa t}e^{\kappa\|u\|^2} + C_{\nu,h} + \mathfrak{B}_0, \tag{A.3}$$

$$\mathbb{E}\|u(t)\|^{2q} \leq e^{-q\nu\alpha_0 t}\|u\|^{2q} + C_{q,\nu,h}(1+\mathfrak{B}_0)^q, \tag{A.4}$$

$$\mathbb{E}e^{\kappa\mathcal{E}(t)} \leq 2e^{\kappa(C_{\nu,h}+\mathfrak{B}_0)t}e^{2\kappa\|u\|^2}, \tag{A.5}$$

$$\mathbb{E}\mathcal{E}^q(t) \lesssim_{q,\nu,h} (1+\mathfrak{B}_0)^q(t+\|u\|^2)^q. \tag{A.6}$$

3. ***Fractional regularity estimate.*** *For any $T > 0$ and $\gamma \in (0, \frac{1}{2})$, it follows*

$$\mathbb{E}\|u(\cdot)\|_{W^{\gamma,2}(0,T;V^*)} \lesssim_{\gamma,T,\nu,h,\mathfrak{B}_0} 1 + \|u\|^2. \tag{A.7}$$

Proof. The proof of estimates (A.1) and (A.3)–(A.6) can be found in Section 2.4 in [KS12] and is thus omitted. We turn to the proof of (A.7). Notice that

$$\begin{aligned}
\mathbb{E}\|u(\cdot)\|_{W^{\gamma,2}(0,T;V^*)} \leq{}& \mathbb{E}\left\|\int_0^\cdot (\nu Lu(t) - B(u(t)) + h)\mathrm{d}t\right\|_{H^1(0,T;V^*)} \\
&+ \mathbb{E}\left\|\int_0^\cdot \eta \mathrm{d}t\right\|_{W^{\gamma,2}(0,T;V^*)} + \sqrt{T}\|u\|,
\end{aligned}$$

where, by applying (A.6) with $q = 1$, one has

$$\begin{aligned}\mathbb{E}\left\|\int_0^{\cdot}(\nu Lu - B(u) + h)\mathrm{d}t\right\|_{H^1(0,T;V^*)} &\lesssim_{T,\nu} \|h\| + \mathbb{E}\|u(\cdot)\|_{L^2(0,T;V)} \\ &\quad + \mathbb{E}\left(\|u(\cdot)\|_{L^\infty(0,T;H)}\|u(\cdot)\|_{L^2(0,T;V)}\right) \\ &\lesssim_{T,\nu,h,\mathfrak{B}_0} 1 + \|u\|^2.\end{aligned}$$

As for the moment estimate of the noise η, we use Lemma 2.1 in [FG95] to obtain

$$\mathbb{E}\left\|\int_0^{\cdot}\eta\mathrm{d}t\right\|_{W^{\gamma,2}(0,T;V^*)} \lesssim_\gamma (1+T)\sqrt{\mathfrak{B}_0}.$$

Combining the above estimates, we obtain (A.7). □

Next, we establish the continuity of the finite-energy solutions with respect to the initial data and noise coefficients.

Proposition A.2. *Suppose that $\{u(t)\}_{t\geq 0}$ and $\{u'(t)\}_{t\geq 0}$ are the solutions of (2.2) issued from u, u' with the noise coefficients given by*

$$\mathcal{B} := \{b_j\}_{j\geq 1}, \qquad \mathcal{B}' := \{b_j'\}_{j\geq 1},$$

respectively. Then, for any $R, \mathcal{R}, T > 0$, $u, u' \in B_H(R)$, and $\mathcal{B}, \mathcal{B}' \in B_{\ell^2}(\mathcal{R})$,

$$\mathbb{E}\int_0^T \|\nabla(u(t) - u'(t))\|^2\mathrm{d}t \lesssim_{\mathcal{R},R,T,\nu,h} d + d^{c_0}, \tag{A.8}$$

where $d := \|u - u'\| + \|\mathcal{B} - \mathcal{B}'\|_{\ell^2}$, and c_0 is a constant depending on $\mathcal{R}, \nu$.

Proof. Notice that if $d \geq 1$, then (A.8) is a direct consequence of (A.6). Thus, in what follows, we shall assume $d < 1$. The difference $w(t) := u(t) - u'(t)$ satisfies

$$\partial_t w - \nu Lw + B(w, u(t)) + B(u'(t), w) = \partial_t \textstyle\sum_{j\geq 1}(b_j - b_j')e_j\beta_j(t).$$

Applying the Itô formula, it follows

$$\begin{aligned}&\mathrm{d}\|w(t)\|^2 + 2\left(\nu\|\nabla w\|^2 + \langle B(w, u(t)), w\rangle\right)\mathrm{d}t \\ &\qquad = 2\textstyle\sum_{j\geq 1}(b_j - b_j')\langle e_j, w\rangle\mathrm{d}\beta_j + \|\mathcal{B} - \mathcal{B}'\|_{\ell^2}^2\mathrm{d}t.\end{aligned}$$

Introduce $\mathfrak{J}(t) := \exp\left(-2c\int_0^t\|\nabla u\|^2\mathrm{d}s\right)$ with c to be chosen later. By the Itô product rule, one has

$$\begin{aligned}&\mathrm{d}(\mathfrak{J}(t)\|w(t)\|^2) + 2\mathfrak{J}\left(\nu\|\nabla w\|^2 + \langle B(w, u(t)), w\rangle + c\|\nabla u\|^2\|w\|^2\right)\mathrm{d}t \\ &\qquad = 2\mathfrak{J}\textstyle\sum_{j\geq 1}(b_j - b_j')\langle e_j, w\rangle\mathrm{d}\beta_j + \mathfrak{J}\|\mathcal{B} - \mathcal{B}'\|_{\ell^2}^2\mathrm{d}t.\end{aligned}$$

Together with taking expectation on both sides and using the Ladyzhenskaya inequality, one gets

$$\begin{aligned}&2\mathbb{E}\int_0^T \mathfrak{J}(t)\left(\nu\|\nabla w\|^2 + c\|\nabla u\|^2\|w\|^2\right)\mathrm{d}t \\ &\qquad \leq \|u - u'\|^2 + \|\mathcal{B} - \mathcal{B}'\|_{\ell^2}^2 T + 2\mathbb{E}\int_0^T \mathfrak{J}(t)|\langle B(w, u(t)), w\rangle|\mathrm{d}t \\ &\qquad \leq \|u - u'\|^2 + \|\mathcal{B} - \mathcal{B}'\|_{\ell^2}^2 T + \mathbb{E}\int_0^T \mathfrak{J}(t)\left(\nu\|\nabla w\|^2 + C_\nu\|\nabla u\|^2\|w\|^2\right)\mathrm{d}t.\end{aligned}$$

Consequently, by taking $2c = C_\nu$, it follows that

$$\mathbb{E}\int_0^T \mathfrak{J}(t)\|\nabla w\|^2 \mathrm{d}t \lesssim_{T,\nu} d^2.$$

Let

$$\mathcal{A}_{M,T} := \left\{ \int_0^T \|\nabla u(t)\|^2 \mathrm{d}t \le M \right\}, \qquad M > 0.$$

Notice that on the event $\mathcal{A}_{M,T}$, it follows that

$$\mathbb{E}\left(\mathbb{I}_{\mathcal{A}_{M,T}} \int_0^T \|\nabla w\|^2 \mathrm{d}t \right) \lesssim_{T,\nu} e^{C_\nu M} d^2. \tag{A.9}$$

On the other hand, by applying (A.5), one has

$$\mathbb{P}(\mathcal{A}_{M,T}^c) \le e^{-\kappa_0 \nu M} \mathbb{E} e^{\kappa_0 \mathcal{E}(T)} \lesssim_{R,\mathcal{R},T,\nu,h} e^{-\kappa_0 \nu M}.$$

Together with (A.6), this implies

$$\mathbb{E}\left(\mathbb{I}_{\mathcal{A}_{M,T}^c} \int_0^T \|\nabla w\|^2 \mathrm{d}t \right) \le \left(\mathbb{P}(\mathcal{A}_{M,T}^c)\right)^{\frac{1}{2}} \left(\mathbb{E}\left(\int_0^T \|\nabla w\|^2 \mathrm{d}t \right)^2 \right)^{\frac{1}{2}} \lesssim_{\mathcal{R},R,T,\nu,h} e^{-\frac{\kappa_0 \nu M}{2}}.$$

Choosing $M := C_\nu^{-1} \log d^{-1}$ and combining the above estimate with (A.9), one gets

$$\mathbb{E}\int_0^T \|\nabla w\|^2 \mathrm{d}t \lesssim_{\mathcal{R},R,T,\nu,h} d + d^{\frac{\kappa_0 \nu C_\nu^{-1}}{2}} \lesssim_{\mathcal{R},R,T,\nu,h} d + d^{\frac{\nu^2 \alpha_0 C_\nu^{-1}}{32(1+\mathcal{R}^2)}}.$$

This completes the proof. □

B Proof of Proposition 4.2

In this section, we present the proof of Proposition 4.2. To establish the first property, notice that

$$0 < \theta_{\varepsilon,\mathfrak{z}}(x) \le 1, \qquad \inf_{|x-\mathfrak{z}|\le 1} \theta_{\varepsilon,\mathfrak{z}}(x) \ge \frac{1}{\sqrt{2}}.$$

Consequently, for any $\mathfrak{z} \in \mathbb{R}$, it follows

$$\|f\|_{L^p(D_\mathfrak{z})} \le 2^{\frac{p}{2}} \|\theta_{\varepsilon,\mathfrak{z}} f\|_p,$$

which, together with taking supremum in $\mathfrak{z}$ on both sides, gives

$$\|f\|_{L_b^p} \lesssim_p \sup_{\mathfrak{z}\in\mathbb{R}} \|\theta_{\varepsilon,\mathfrak{z}} f\|_p. \tag{B.1}$$

On the other hand, for any $\mathfrak{z} \in \mathbb{R}$, one has

$$\begin{aligned}
\|\theta_{\varepsilon,\mathfrak{z}} f\|_p^p &= \textstyle\sum_{l\in\mathbb{Z}} \displaystyle\int_{D_{\mathfrak{z}+l}} \theta_{\varepsilon,\mathfrak{z}}^p |f|^p \mathrm{d}x\mathrm{d}y \\
&\le \textstyle\sum_{l\ge 0} \dfrac{1}{(1+\varepsilon^2 l^2)^{\frac{p}{2}}} \displaystyle\int_{D_{\mathfrak{z}+l}} |f|^p \mathrm{d}x\mathrm{d}y + \sum_{l<0} \dfrac{1}{(1+\varepsilon^2 (l+1)^2)^{\frac{p}{2}}} \displaystyle\int_{D_{\mathfrak{z}+l}} |f|^p \mathrm{d}x\mathrm{d}y \\
&\le 2\left(1 + \int_0^\infty \theta_{\varepsilon,0}^p \mathrm{d}x\mathrm{d}y\right) \|f\|_{L_b^p}^p \lesssim_p \varepsilon^{-1} \|f\|_{L_b^p}^p.
\end{aligned}$$

Together with (B.1), this ensures (4.4).

The second property follows from direct calculation. Indeed, it follows that

$$\partial_x \theta_{\varepsilon,\mathfrak{z}}(x) = -\frac{\varepsilon^2(x-\mathfrak{z})}{(1+\varepsilon^2(x-\mathfrak{z})^2)^{\frac{3}{2}}}, \tag{B.2}$$

which ensures (4.5). We turn to the third property. Notice that

$$\partial_x \log \theta_{\varepsilon,\mathfrak{z}} = \theta_{\varepsilon,\mathfrak{z}}^{-1} \partial_x \theta_{\varepsilon,\mathfrak{z}} = -\frac{\varepsilon^2(x-\mathfrak{z})}{1+\varepsilon^2(x-\mathfrak{z})^2}. \tag{B.3}$$

This yields

$$\|\partial_x \log \theta_{\varepsilon,\mathfrak{z}}\|_\infty \leq \varepsilon.$$

Since

$$\partial_x \log \theta_{\varepsilon,\mathfrak{z}}(x) = -\varepsilon \mathfrak{f}(\varepsilon(x-\mathfrak{z})), \qquad \mathfrak{f}(x) = \frac{x}{1+x^2},$$

and

$$\|\partial_x^j \mathfrak{f}\|_\infty \leq C_j,$$

by taking derivatives on both sides of (B.3), one obtains (4.6).

C Estimate for the Stokes semigroup in the uniformly local spaces

In this section, we establish estimates for the linear Stokes equation

$$\begin{cases} \partial_t u + \nabla p = \nu \Delta u, \\ \operatorname{div} u = 0, \\ u|_{t=0} = u_0, \qquad u|_{\partial D} = 0 \end{cases} \tag{C.1}$$

in the uniformly local spaces $L^p_{b,\mathrm{div}}$. These estimates are used in Section 4.

Proposition C.1. *Let $S(t) := e^{\nu t L}$ denote the Stokes semigroup. For any $p \in (1,\infty)$ and $t \in (0,1]$, it follows that*

$$\|S(t)u\|_{L^p_b} + t^{\frac{1}{2}} \|\nabla S(t)u\|_{L^p_b} \lesssim_{p,\nu} \|u\|_{L^p_b} \tag{C.2}$$

for any $u \in L^p_{b,\mathrm{div}}$ and

$$t^{\frac{1}{2}} \|S(t)\Pi \operatorname{div} f\|_{L^p_b} + t\|\nabla S(t)\Pi \operatorname{div} f\|_{L^p_b} \lesssim_{p,\nu} \|f\|_{L^p_b} \tag{C.3}$$

for any $f \in L^p_b$.

Proof. **Step I.** To begin with, we recall that the Stokes operator

$$L : \mathcal{D}(L) \to L^p_{\mathrm{div}}, \qquad \mathcal{D}(L) := L^p_{\mathrm{div}} \cap W^{2,p} \cap W^{1,p}_0$$

generates an analytic semigroup; see [AW05]. We claim that

$$\mathcal{J} : u \mapsto \mathcal{J}u(x,y) := \int_{-1}^{y} u_1(x,y')\mathrm{d}y', \qquad L^p_{\mathrm{div}} \to W^{1,p}_0$$

is an isomorphism. Indeed, for any $u \in L^p_{\mathrm{div}}$, by the divergence-free and no-penetration conditions, one has

$$\nabla^\perp \mathcal{J} u = u, \qquad \nabla^\perp := (\partial_y, -\partial_x). \tag{C.4}$$

Consequently, $\mathcal{J}u \in W^{1,p}$ and $\mathcal{J}$ is bounded from L^p_{div} to $W^{1,p}$. On the other hand, notice that

$$\partial_x \mathcal{J}u(x,1) = \int_{-1}^{1} \partial_x u_1(x,y')\mathrm{d}y' = -\int_{-1}^{1} \partial_y u_2(x,y')\mathrm{d}y' = 0,$$

which implies that $\mathcal{J}u(x,1) = \int_{-1}^{1} u_1(x,y')\mathrm{d}y' \equiv C$ for some constant C. Together with the fact that

$$\left\| \int_{-1}^{1} u_1(\cdot,y')\mathrm{d}y' \right\|_{L^p(\mathbb{R})} \lesssim \|u\|_p,$$

this implies that $\mathcal{J}u(x,1) \equiv 0$ and thus $\mathcal{J}u \in W^{1,p}_0$. Moreover, the relation (C.4) ensures that $\mathcal{J}$ is both one-to-one and onto. Therefore, $\mathcal{J}$ is an isomorphism as claimed.

Define the Stokes operator on the space of stream functions

$$\mathfrak{L} : \mathcal{D}(\mathfrak{L}) \to W^{1,p}_0, \qquad \psi \mapsto \mathcal{J}L\nabla^{\perp}\psi,$$

where

$$\mathcal{D}(\mathfrak{L}) := \{\psi \in W^{3,p} | \psi|_{\partial D} = \partial_y \psi|_{\partial D} = 0\}.$$

Since $\mathfrak{L} = \mathcal{J}L\nabla^{\perp} = \mathcal{J}L\mathcal{J}^{-1}$ is similar to L, it is also sectorial and thus $\nu\mathfrak{L}$ generates an analytic semigroup $\mathcal{S}(t) := e^{\nu t\mathfrak{L}}$. Moreover, notice that $\psi(t) := \mathcal{S}(t)\psi_0$ satisfies

$$\nabla^{\perp}\psi(t) = \nabla^{\perp}\mathcal{J}S(t)\mathcal{J}^{-1}\psi_0 = S(t)\mathcal{J}^{-1}\psi_0, \tag{C.5}$$

which, by taking curl on both sides of (C.1), yields the following equation

$$\begin{cases} \partial_t \Delta\psi = \nu\Delta^2\psi, \\ \psi|_{\partial D} = \partial_y\psi|_{\partial D} = 0, \\ \psi|_{t=0} = \psi_0. \end{cases} \tag{C.6}$$

Consequently, one has the following representation formula

$$\mathfrak{L} = \Delta_D^{-1}\Delta^2, \tag{C.7}$$

where Δ_D denotes the classical Dirichlet–Laplacian and Δ^2 is realized as the bi-Laplacian with the boundary conditions $(\text{C.6})_2$.

Step II. In view of (4.4), it suffices to establish uniform-in-$\mathfrak{z}$ estimate for $S(t)$ in the weighted Lebesgue space $L^p(D;\theta^p_{\varepsilon,\mathfrak{z}}\mathrm{d}x\mathrm{d}y)$. To avoid the additional complications arising from the interaction between the divergence-free condition and weighted estimates, we consider the Stokes semigroup $\mathcal{S}(t)$ in the stream-function formulation.

Fix arbitrary $\mathfrak{z} \in \mathbb{R}$ and $\varepsilon \in (0,1)$. Notice that $\phi_{\varepsilon,\mathfrak{z}}(t) := \theta_{\varepsilon,\mathfrak{z}}\mathcal{S}(t)\psi_0$ solves the equation

$$\begin{cases} \partial_t\phi_{\varepsilon,\mathfrak{z}} = \nu\theta_{\varepsilon,\mathfrak{z}}\mathfrak{L}\theta^{-1}_{\varepsilon,\mathfrak{z}}\phi_{\varepsilon,\mathfrak{z}} =: \nu\mathfrak{L}_{\varepsilon,\mathfrak{z}}\phi_{\varepsilon,\mathfrak{z}}, \\ \phi_{\varepsilon,\mathfrak{z}}|_{t=0} = \theta_{\varepsilon,\mathfrak{z}}\psi_0. \end{cases} \tag{C.8}$$

This reduces the weighted estimates of $\mathcal{S}(t)$ to the estimate of $\mathcal{S}_{\varepsilon,\mathfrak{z}}(t) := \exp(\nu t\mathfrak{L}_{\varepsilon,\mathfrak{z}})$ in the usual Sobolev space $W^{1,p}_0$. We claim that there exist constants C_p, ε independent of $\mathfrak{z} \in \mathbb{R}$ such that

$$\|(\mathfrak{L}_{\varepsilon,\mathfrak{z}} - \mathfrak{L})\psi\|_{1,p} \le C_p\varepsilon\left(\|\psi\|_{1,p} + \|\mathfrak{L}\psi\|_{1,p}\right) \tag{C.9}$$

for any $\psi \in \mathcal{D}(\mathfrak{L})$. To see this, note that from (C.7), one has

$$\mathfrak{L}_{\varepsilon,\mathfrak{z}} = \left(\theta_{\varepsilon,\mathfrak{z}}\Delta_D^{-1}\theta^{-1}_{\varepsilon,\mathfrak{z}}\right)\left(\theta_{\varepsilon,\mathfrak{z}}\Delta^2\theta^{-1}_{\varepsilon,\mathfrak{z}}\right),$$

which implies

$$\begin{aligned}\mathfrak{L}_{\varepsilon,\mathfrak{z}}-\mathfrak{L}&=\left(\theta_{\varepsilon,\mathfrak{z}}\Delta_D^{-1}\theta_{\varepsilon,\mathfrak{z}}^{-1}\right)\left[\theta_{\varepsilon,\mathfrak{z}}\Delta^2\theta_{\varepsilon,\mathfrak{z}}^{-1}-\left(\theta_{\varepsilon,\mathfrak{z}}\Delta_D\theta_{\varepsilon,\mathfrak{z}}^{-1}\right)\Delta_D^{-1}\Delta^2\right]\\&=\left(\theta_{\varepsilon,\mathfrak{z}}\Delta_D^{-1}\theta_{\varepsilon,\mathfrak{z}}^{-1}\right)\left[\theta_{\varepsilon,\mathfrak{z}}\Delta^2\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta^2-\left(\theta_{\varepsilon,\mathfrak{z}}\Delta_D\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta_D\right)\mathfrak{L}\right].\end{aligned}\tag{C.10}$$

Consequently, it remains to estimate $\theta_{\varepsilon,\mathfrak{z}}\Delta_D\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta_D$ and $\theta_{\varepsilon,\mathfrak{z}}\Delta^2\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta^2$. Let $\psi\in W_0^{1,p}$. By direct calculation, it follows

$$\begin{aligned}(\theta_{\varepsilon,\mathfrak{z}}\Delta_D\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta_D)\psi&=2\theta_{\varepsilon,\mathfrak{z}}\partial_x\theta_{\varepsilon,\mathfrak{z}}^{-1}\partial_x\psi+\theta_{\varepsilon,\mathfrak{z}}\partial_x^2\theta_{\varepsilon,\mathfrak{z}}^{-1}\psi\\&=2\partial_x(\log\theta_{\varepsilon,\mathfrak{z}}^{-1})\partial_x\psi+\partial_x(\theta_{\varepsilon,\mathfrak{z}}\partial_x\theta_{\varepsilon,\mathfrak{z}}^{-1})\psi-\partial_x\theta_{\varepsilon,\mathfrak{z}}\partial_x\theta_{\varepsilon,\mathfrak{z}}^{-1}\psi\\&=2\partial_x(\log\theta_{\varepsilon,\mathfrak{z}}^{-1})\partial_x\psi+\partial_x^2(\log\theta_{\varepsilon,\mathfrak{z}}^{-1})\psi+(\partial_x\theta_{\varepsilon,\mathfrak{z}}\theta_{\varepsilon,\mathfrak{z}}^{-1})^2\psi\\&=(\partial_x\log\theta_{\varepsilon,\mathfrak{z}})^2\psi-2\partial_x(\log\theta_{\varepsilon,\mathfrak{z}})\partial_x\psi-\partial_x^2(\log\theta_{\varepsilon,\mathfrak{z}})\psi.\end{aligned}$$

Together with (4.6), this implies that for any $\varphi\in W_0^{1,p'}$,

$$\begin{aligned}\langle(\theta_{\varepsilon,\mathfrak{z}}\Delta_D\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta_D)\psi,\varphi\rangle&=\langle(\partial_x\log\theta_{\varepsilon,\mathfrak{z}})^2\psi-2\partial_x(\log\theta_{\varepsilon,\mathfrak{z}})\partial_x\psi-\partial_x^2(\log\theta_{\varepsilon,\mathfrak{z}})\psi,\varphi\rangle\\&\lesssim\varepsilon\|\psi\|_{1,p}\|\varphi\|_{1,p'},\end{aligned}$$

and thus that

$$\|(\theta_{\varepsilon,\mathfrak{z}}\Delta_D\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta_D)\psi\|_{-1,p}\lesssim\varepsilon\|\psi\|_{1,p}.\tag{C.11}$$

Similarly, for any $\psi\in\mathcal{D}(\mathfrak{L})$, it follows that

$$\begin{aligned}\theta_{\varepsilon,\mathfrak{z}}\Delta^2\theta_{\varepsilon,\mathfrak{z}}^{-1}\psi&=(\theta_{\varepsilon,\mathfrak{z}}\Delta\theta_{\varepsilon,\mathfrak{z}}^{-1})^2\psi\\&=\left[(\partial_x\log\theta_{\varepsilon,\mathfrak{z}})^2-2\partial_x(\log\theta_{\varepsilon,\mathfrak{z}})\partial_x-\partial_x^2(\log\theta_{\varepsilon,\mathfrak{z}})+\Delta\right]^2\psi,\end{aligned}$$

which, combined with (4.6), leads to

$$\|(\theta_{\varepsilon,\mathfrak{z}}\Delta^2\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta^2)\psi\|_{-1,p}\lesssim\varepsilon\|\psi\|_{3,p}.\tag{C.12}$$

Moreover, by combining (C.11) with the Neumann series

$$\begin{aligned}\left(\theta_{\varepsilon,\mathfrak{z}}\Delta_D^{-1}\theta_{\varepsilon,\mathfrak{z}}^{-1}\right)&=\Delta_D^{-1}(I+(\theta_{\varepsilon,\mathfrak{z}}\Delta_D\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta_D)\Delta_D^{-1})^{-1}\\&=\textstyle\sum_{j\geq0}\Delta_D^{-1}\left(-(\theta_{\varepsilon,\mathfrak{z}}\Delta_D\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta_D)\Delta_D^{-1}\right)^j,\end{aligned}$$

one gets

$$\left\|\left(\theta_{\varepsilon,\mathfrak{z}}\Delta_D^{-1}\theta_{\varepsilon,\mathfrak{z}}^{-1}\right)\right\|_{\mathcal{L}(W^{-1,p};W_0^{1,p})}\lesssim_p\textstyle\sum_{j\geq0}\varepsilon^j\|\Delta_D^{-1}\|_{\mathcal{L}(W^{-1,p};W_0^{1,p})}^j<\infty,\tag{C.13}$$

provided that $\varepsilon\ll1$ is sufficiently small.

Now by combining (C.10)–(C.13), it follows that for any $\psi\in\mathcal{D}(\mathfrak{L})$,

$$\begin{aligned}\|(\mathfrak{L}_{\varepsilon,\mathfrak{z}}-\mathfrak{L})\psi\|_{1,p}&=\left\|\left(\theta_{\varepsilon,\mathfrak{z}}\Delta_D^{-1}\theta_{\varepsilon,\mathfrak{z}}^{-1}\right)\left[\theta_{\varepsilon,\mathfrak{z}}\Delta^2\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta^2-\left(\theta_{\varepsilon,\mathfrak{z}}\Delta_D\theta_{\varepsilon,\mathfrak{z}}^{-1}-\Delta_D\right)\mathfrak{L}\right]\psi\right\|_{1,p}\\&\lesssim_p\varepsilon\|\psi\|_{3,p}+\varepsilon\|\mathfrak{L}\psi\|_{1,p}\lesssim_p\varepsilon(\|\psi\|_{1,p}+\|\mathfrak{L}\psi\|_{1,p})\end{aligned}$$

as claimed. Here in the last inequality, we used the fact that the graph norm of $\mathfrak{L}$ is equivalent to $W^{3,p}$-norm on $\mathcal{D}(\mathfrak{L})$.

Step Ⅲ. In this step, we aim to show that $\mathcal{S}_{\varepsilon,\mathfrak{z}}(t)$ is an analytic semigroup. For $\vartheta\in(\frac{\pi}{2},\pi)$, define

$$\Sigma_\vartheta:=\{z\in\mathbb{C}|z\neq0,\ |\arg z|<\vartheta\}.$$

Since $\mathcal{S}(t)$ is an analytic semigroup generated by $\nu\mathfrak{L}$, there is $z_0 \geq 0$ and $\vartheta_0 \in (\frac{\pi}{2}, \pi)$ such that

$$z_0 + \Sigma_{\vartheta_0} \subset \rho(\mathfrak{L}),$$

and

$$\|(z-\mathfrak{L})^{-1}\|_{\mathcal{L}(W_0^{1,p})} \lesssim_p \frac{1}{|z-z_0|}, \qquad \forall z \in z_0 + \Sigma_{\vartheta_0}, \tag{C.14}$$

where $\rho(\mathfrak{L})$ is the resolvent set of $\mathfrak{L}$. Fix arbitrary $z \in z_0 + 1 + \Sigma_{\vartheta_0}$ and $\psi \in W_0^{1,p}$. By combining (C.9), (C.14), and the relation

$$(z-\mathfrak{L}_{\varepsilon,\mathfrak{z}})^{-1} = (z-\mathfrak{L})^{-1}(I + (\mathfrak{L}-\mathfrak{L}_{\varepsilon,\mathfrak{z}})(z-\mathfrak{L})^{-1})^{-1},$$

one has

$$\|(z-\mathfrak{L}_{\varepsilon,\mathfrak{z}})^{-1}\psi\|_{1,p} \lesssim_p \frac{1}{|z-z_0|}\|(I + (\mathfrak{L}-\mathfrak{L}_{\varepsilon,\mathfrak{z}})(z-\mathfrak{L})^{-1})^{-1}\psi\|_{1,p},$$

where

$$\begin{aligned}
\|(\mathfrak{L}-\mathfrak{L}_{\varepsilon,\mathfrak{z}})(z-\mathfrak{L})^{-1}\psi\|_{1,p} &\lesssim_p \varepsilon\left(\|(z-\mathfrak{L})^{-1}\psi\|_{1,p} + \|\mathfrak{L}(z-\mathfrak{L})^{-1}\psi\|_{1,p}\right)\\
&\lesssim_p \varepsilon\left(\frac{1}{|z-z_0|} + 1 + \frac{|z|}{|z-z_0|}\right)\|\psi\|_{1,p}\\
&\lesssim_p \varepsilon\|\psi\|_{1,p}.
\end{aligned}$$

Consequently, by shrinking ε if necessary, it follows that

$$\|(z-\mathfrak{L}_{\varepsilon,\mathfrak{z}})^{-1}\|_{\mathcal{L}(W_0^{1,p})} \lesssim_p \frac{1}{|z-z_0|} \lesssim_p \frac{1}{|z-z_0-1|}. \tag{C.15}$$

This implies that $\mathcal{S}_{\varepsilon,\mathfrak{z}}(t)$ is an analytic semigroup in $W_0^{1,p}$.

Step IV. Since the sector $z_0 + 1 + \Sigma_{\vartheta_0}$ and the constant in the resolvent estimate (C.15) are independent of the parameter $\mathfrak{z}$, by the analyticity of $\mathcal{S}_{\varepsilon,\mathfrak{z}}(t)$, one has

$$\|\theta_{\varepsilon,\mathfrak{z}}\mathcal{S}(t)\psi_0\|_{1,p} + t\|\theta_{\varepsilon,\mathfrak{z}}\mathfrak{L}\mathcal{S}(t)\psi_0\|_{1,p} \leq C_{p,\nu}\|\theta_{\varepsilon,\mathfrak{z}}\psi_0\|_{1,p}, \qquad \forall t \in (0,1], \tag{C.16}$$

where the constant $C_{p,\nu}$ does not depend on $\mathfrak{z}$; see Theorem 2.2 in [Sch04] and Property (10.5) in [DPS87]. On the other hand, for any $\psi \in \mathcal{D}(\mathfrak{L})$, by applying the standard regularity theory of the Stokes operator combined with (C.9), one has

$$\begin{aligned}
\|\theta_{\varepsilon,\mathfrak{z}}\psi\|_{3,p} &\lesssim_p \|\theta_{\varepsilon,\mathfrak{z}}\psi\|_{1,p} + \|\mathfrak{L}(\theta_{\varepsilon,\mathfrak{z}}\psi)\|_{1,p}\\
&\lesssim_p \|\theta_{\varepsilon,\mathfrak{z}}\psi\|_{1,p} + \|\mathfrak{L}_{\varepsilon,\mathfrak{z}}(\theta_{\varepsilon,\mathfrak{z}}\psi)\|_{1,p} + \|(\mathfrak{L}_{\varepsilon,\mathfrak{z}}-\mathfrak{L})(\theta_{\varepsilon,\mathfrak{z}}\psi)\|_{1,p},\\
&\lesssim_p \|\theta_{\varepsilon,\mathfrak{z}}\psi\|_{1,p} + \|\theta_{\varepsilon,\mathfrak{z}}\mathfrak{L}\psi\|_{1,p} + \varepsilon(\|\theta_{\varepsilon,\mathfrak{z}}\psi\|_{1,p} + \|\mathfrak{L}(\theta_{\varepsilon,\mathfrak{z}}\psi)\|_{1,p}),
\end{aligned}$$

which, together with shrinking ε if necessary, yields

$$\|\theta_{\varepsilon,\mathfrak{z}}\psi\|_{3,p} \lesssim_p \|\theta_{\varepsilon,\mathfrak{z}}\psi\|_{1,p} + \|\theta_{\varepsilon,\mathfrak{z}}\mathfrak{L}\psi\|_{1,p}.$$

As a result, for any $t \in (0,1]$, one has

$$t\|\theta_{\varepsilon,\mathfrak{z}}\mathcal{S}(t)\psi_0\|_{3,p} \lesssim_p t\|\theta_{\varepsilon,\mathfrak{z}}\mathcal{S}(t)\psi_0\|_{1,p} + t\|\theta_{\varepsilon,\mathfrak{z}}\mathfrak{L}\mathcal{S}(t)\psi_0\|_{1,p} \leq C_{p,\nu}\|\theta_{\varepsilon,\mathfrak{z}}\psi_0\|_{1,p}.$$

This, combined with (C.16) and the standard Sobolev interpolation inequality, gives

$$t^{\frac{1}{2}}\|\nabla(\theta_{\varepsilon,\mathfrak{z}}\mathcal{S}(t)\psi_0)\|_{1,p} \leq C_{p,\nu}\|\theta_{\varepsilon,\mathfrak{z}}\psi_0\|_{1,p}. \tag{C.17}$$

Since

$$\nabla(\theta_{\varepsilon,\mathfrak{z}}\psi) = \theta_{\varepsilon,\mathfrak{z}}\nabla\psi + (\partial_x\theta_{\varepsilon,\mathfrak{z}}\psi, 0) = \theta_{\varepsilon,\mathfrak{z}}\nabla\psi + (\partial_x \log\theta_{\varepsilon,\mathfrak{z}}(\theta_{\varepsilon,\mathfrak{z}}\psi), 0),$$

by applying (C.17) and (4.6), it follows that

$$\|\theta_{\varepsilon,\mathfrak{z}}\mathcal{S}(t)\psi_0\|_{1,p} + t^{\frac{1}{2}}\|\theta_{\varepsilon,\mathfrak{z}}\nabla\mathcal{S}(t)\psi_0\|_{1,p} \le C_{p,\nu}\|\theta_{\varepsilon,\mathfrak{z}}\psi_0\|_{1,p}, \qquad \forall t\in(0,1]. \tag{C.18}$$

Now, we are in a position to prove (C.2). Let $u_0 \in L^p_{b,\mathrm{div}}$, $\psi_0 = \mathcal{J}u_0$, $u(t) := S(t)u_0$, and $\psi(t) := \mathcal{S}(t)\psi_0$. From (C.5), it follows that $\nabla^\perp\psi(t) = u(t)$, which implies

$$\theta_{\varepsilon,\mathfrak{z}}u(t) = \theta_{\varepsilon,\mathfrak{z}}\nabla^\perp\psi(t) = \nabla^\perp(\theta_{\varepsilon,\mathfrak{z}}\psi(t)) + (0, \partial_x(\log\theta_{\varepsilon,\mathfrak{z}})\theta_{\varepsilon,\mathfrak{z}}\psi(t)).$$

Together with (4.6), this implies

$$\|\theta_{\varepsilon,\mathfrak{z}}u(t)\|_p \lesssim \|\theta_{\varepsilon,\mathfrak{z}}\psi(t)\|_{1,p}. \tag{C.19}$$

Similarly, it follows

$$\|\theta_{\varepsilon,\mathfrak{z}}\nabla u(t)\|_p \lesssim \|\theta_{\varepsilon,\mathfrak{z}}\nabla\psi(t)\|_{1,p}. \tag{C.20}$$

Plugging (C.19) and (C.20) into (C.18) and applying the Poincaré inequality, one gets

$$\|\theta_{\varepsilon,\mathfrak{z}}S(t)u_0\|_p + t^{\frac{1}{2}}\|\theta_{\varepsilon,\mathfrak{z}}\nabla S(t)u_0\|_p \le C_{p,\nu}\|\theta_{\varepsilon,\mathfrak{z}}\psi_0\|_{1,p} \le C_{p,\nu}\|\theta_{\varepsilon,\mathfrak{z}}u_0\|_p \tag{C.21}$$

for any $t\in(0,1]$, which, together with (4.4), leads to the desired estimate (C.2).

Step V. We turn to the proof of (C.3). To this end, we note that the stability estimate (C.9) relies only on (4.6), which is satisfied for the inverse weight $\theta^{-1}_{\varepsilon,\mathfrak{z}}$. As a result, one has

$$\|\theta^{-1}_{\varepsilon,\mathfrak{z}}S(t)u_0\|_p + t^{\frac{1}{2}}\|\theta^{-1}_{\varepsilon,\mathfrak{z}}\nabla S(t)u_0\|_p \le C_{p,\nu}\|\theta^{-1}_{\varepsilon,\mathfrak{z}}u_0\|_p, \qquad \forall t\in(0,1]. \tag{C.22}$$

We also need the following weighted estimate for the Leray projection Π, whose proof is postponed to the end of this section.

Lemma C.2. *For any $p\in(1,\infty)$ and integers $k\ge 0$, $l\in\mathbb{Z}$, there exists a sufficiently small $\varepsilon = \varepsilon(l,k,p) > 0$ such that*

$$\|\theta^l_{\varepsilon,\mathfrak{z}}\Pi f\|_{k,p} \lesssim_{k,p} \|\theta^l_{\varepsilon,\mathfrak{z}}f\|_{k,p} \tag{C.23}$$

for any $\mathfrak{z}\in\mathbb{R}$ and f satisfying $\theta^l_{\varepsilon,\mathfrak{z}}f\in W^{k,p}$.

Let $f,g\in C^\infty_0$. Notice that

$$\langle S(t)\Pi\,\mathrm{div}\, f, g\rangle = -\langle f, \nabla S(t)\Pi g\rangle \le \|\theta_{\varepsilon,\mathfrak{z}}f\|_p\|\theta^{-1}_{\varepsilon,\mathfrak{z}}\nabla S(t)\Pi g\|_{p'},$$

where, by applying (C.22) and (C.23) with p replaced by p' and $k=0$, $l=-1$, one has

$$\|\theta^{-1}_{\varepsilon,\mathfrak{z}}\nabla S(t)\Pi g\|_{p'} \lesssim_{p,\nu} t^{-\frac{1}{2}}\|\theta^{-1}_{\varepsilon,\mathfrak{z}}\Pi g\|_{p'} \lesssim_{p,\nu} t^{-\frac{1}{2}}\|\theta^{-1}_{\varepsilon,\mathfrak{z}}g\|_{p'}.$$

Consequently, it follows that

$$\langle S(t)\Pi\,\mathrm{div}\, f, g\rangle \lesssim_{p,\nu} t^{-\frac{1}{2}}\|\theta_{\varepsilon,\mathfrak{z}}f\|_p\|\theta^{-1}_{\varepsilon,\mathfrak{z}}g\|_{p'},$$

which, by density, implies

$$\sqrt{t}\|\theta_{\varepsilon,\mathfrak{z}}S(t)\Pi\,\mathrm{div}\, f\|_p \lesssim_{p,\nu} \|\theta_{\varepsilon,\mathfrak{z}}f\|_p, \qquad \forall t\in(0,1]. \tag{C.24}$$

On the other hand, note that an application of (C.21) and (C.24) gives

$$\begin{aligned} t\|\theta_{\varepsilon,\mathfrak{z}}\nabla S(t)\Pi\,\mathrm{div}\, f\|_p &= \sqrt{2t}\sqrt{\frac{t}{2}}\|\theta_{\varepsilon,\mathfrak{z}}\nabla S(t/2)S(t/2)\Pi\,\mathrm{div}\, f\|_p \\ &\lesssim_{p,\nu} \sqrt{t}\|\theta_{\varepsilon,\mathfrak{z}}S(t/2)\Pi\,\mathrm{div}\, f\|_p \lesssim_{p,\nu} \|\theta_{\varepsilon,\mathfrak{z}}f\|_p. \end{aligned}$$

This, together with (C.24) and (4.4), gives the estimate (C.3). □

Notice that the squared inverse weight function $\theta_{\varepsilon,\mathfrak{z}}^{-2}$ also satisfies (4.6). As discussed in the derivation of (C.22), this and Lemma C.2 with $k=0$ and $l=-2$ yield the following result.

Corollary C.3. *Let $S(t):=e^{\nu t L}$ denote the Stokes semigroup. For any $p\in(1,\infty)$, there exists a sufficiently small $\varepsilon=\varepsilon(p)>0$ such that*

$$\|\theta_{\varepsilon,\mathfrak{z}}^{-2}S(t)\Pi u\|_p \lesssim_{p,\nu} \|\theta_{\varepsilon,\mathfrak{z}}^{-2}u\|_p \tag{C.25}$$

for any $t\in[0,1]$, $\mathfrak{z}\in\mathbb{R}$, and u satisfying $\theta_{\varepsilon,\mathfrak{z}}^{-2}u\in L^p$.

Finally, we present the proof of Lemma C.2.

Proof of Lemma C.2. According to Proposition 2.6 in [AZ14], it suffices to show that $\theta_{\varepsilon,\mathfrak{z}}^{l}$ with $l\in\mathbb{Z}$ is a weight function of exponential growth rate. Indeed, by applying (4.6), one has

$$\|\partial_x \log\theta_{\varepsilon,\mathfrak{z}}^{l}\|_\infty \lesssim |l|\varepsilon,$$

which, together with the mean value theorem, shows that

$$\log\theta_{\varepsilon,\mathfrak{z}}^{l}(x)-\log\theta_{\varepsilon,\mathfrak{z}}^{l}(x')\lesssim |l|\varepsilon|x-x'|,$$

and thus that

$$\theta_{\varepsilon,\mathfrak{z}}^{l}(x)\le e^{C|l|\varepsilon|x-x'|}\theta_{\varepsilon,\mathfrak{z}}^{l}(x').$$

Therefore, $\theta_{\varepsilon,\mathfrak{z}}^{l}$ has an exponential growth rate $C|l|\varepsilon$, and estimate (C.23) holds. □

D Global well-posedness in the uniformly local spaces

The goal of this section is to establish Theorem 1.2. Before proceeding, we show that the phase space X is a standard measurable space, which provides the necessary measure-theoretic settings for Markovian dynamics. Indeed,

$$X=\cup_{m\ge1}\cap_{k\in\mathbb{Z}}\left\{u\in H_{\mathrm{loc}}\middle|\|u\|_{L^2(D_k)}\le m\right\},$$

where the space H_{loc} is defined in (1.27) and $D_k:=(k,k+1)\times(-1,1)$. Notice that each set on the right-hand side is a closed subset of the Polish space $(H_{\mathrm{loc}},d_{\mathrm{loc}})$, so X is a Borel subset of H_{loc} and thus standard measurable.

Next, we prove the unique solvability of the stochastic NS equation (1.1) in the uniformly local spaces.

Proposition D.1. *Under the assumption of Theorem 1.2, there is a unique infinite-energy solution of (1.1), which exists globally in time.*

Proof. The proof follows the idea in [AZ14] with slight modifications. Introduce the following decomposition

$$u(t)=z(t)+u_{\mathrm{det}}(t),$$

where $\{z(t)\}_{t\ge0}$ is the solution of the stochastic Stokes equation

$$\begin{cases}\partial_t z+\nabla\mathfrak{p}_1-\nu\Delta z=h+\eta,\\ \operatorname{div} z=0,\\ z|_{t=0}=0,\qquad z|_{\partial D}=0,\end{cases}\tag{D.1}$$

and $\{u_{\mathrm{det}}(t)\}_{t\ge0}$ satisfies the following PDE with random parameter

$$\begin{cases}\partial_t u_{\mathrm{det}}+(u_{\mathrm{det}}\cdot\nabla)u_{\mathrm{det}}+(u_{\mathrm{det}}\cdot\nabla)z+(z\cdot\nabla)u_{\mathrm{det}}+\nabla\mathfrak{p}_2-\nu\Delta u_{\mathrm{det}}=-(z\cdot\nabla)z,\\ \operatorname{div} u_{\mathrm{det}}=0,\\ u_{\mathrm{det}}|_{t=0}=u_0,\qquad u_{\mathrm{det}}|_{\partial D}=0.\end{cases}\tag{D.2}$$

Since $\mathfrak{B}_1 < \infty$, one has

$$z \in C([0,\infty); V) \cap L^2_{\mathrm{loc}}(0,\infty; H^2 \cap V) \tag{D.3}$$

$\mathbb{P}$-almost surely. Consequently, it remains to prove the unique solvability of (D.2) for any given trajectory $z(\cdot)$ satisfying (D.3). For simplicity, in what follows, we shall omit the subscripts in $u_{\mathrm{det}}, \mathfrak{p}_2$. This will not lead to any confusion.

Step I. In this step, we establish a priori estimates in the uniformly local space H_b. Notice that such estimates have been established in the proof of Proposition 4.4, provided that $z \equiv 0$; see (4.27). To simplify the presentation, we only mention the necessary modifications required in the general case. Let $\theta_{\varepsilon,\mathfrak{z}}$ be the weight function defined in (4.3). It suffices to estimate

$$\mathcal{I}_1 := -\langle (u\cdot\nabla)z + (z\cdot\nabla)u + (z\cdot\nabla)z, \theta^2_{\varepsilon,\mathfrak{z}} u\rangle,$$

and

$$\mathcal{I}_2 := \langle (u\cdot\nabla)z + (z\cdot\nabla)u + (z\cdot\nabla)z, v_{\varepsilon,\mathfrak{z}}\rangle.$$

Indeed, by the Hölder and Gagliardo–Nirenberg inequalities combined with (4.5), it follows that

$$\begin{aligned} |\langle (u\cdot\nabla)z, \theta^2_{\varepsilon,\mathfrak{z}} u\rangle| &\le \|\nabla z\|_4 \|\theta_{\varepsilon,\mathfrak{z}} u\|^2_{\frac{8}{3}} \le \|\nabla z\|_4 \|\theta_{\varepsilon,\mathfrak{z}} u\|^{\frac{3}{2}} \|\theta_{\varepsilon,\mathfrak{z}}\nabla u\|^{\frac{1}{2}} \\ &\le \delta \|\theta_{\varepsilon,\mathfrak{z}}\nabla u\|^2 + C_\delta \|\nabla z\|_4^{\frac{4}{3}} \|\theta_{\varepsilon,\mathfrak{z}} u\|^2. \end{aligned} \tag{D.4}$$

Moreover, by integrating by parts and using (4.5) and $\operatorname{div} z = 0$, one has

$$\begin{aligned} |\langle (z\cdot\nabla)u, \theta^2_{\varepsilon,\mathfrak{z}} u\rangle| &= |\langle z\theta_{\varepsilon,\mathfrak{z}}\partial_x\theta_{\varepsilon,\mathfrak{z}}, |u|^2\rangle| \le \|z\|_4 \|\theta_{\varepsilon,\mathfrak{z}} u\|^2_{\frac{8}{3}} \\ &\le \delta \|\theta_{\varepsilon,\mathfrak{z}}\nabla u\|^2 + C_\delta \|z\|_4^{\frac{4}{3}} \|\theta_{\varepsilon,\mathfrak{z}} u\|^2. \end{aligned} \tag{D.5}$$

Similarly, an application of integrating by parts and the Poincaré inequality gives

$$\begin{aligned} |\langle (z\cdot\nabla)z, \theta^2_{\varepsilon,\mathfrak{z}} u\rangle| &= |\langle z\otimes z, \nabla(\theta^2_{\varepsilon,\mathfrak{z}} u)\rangle| \le C\|z\|_4^2 \|\theta_{\varepsilon,\mathfrak{z}}\nabla u\| \\ &\le \delta \|\theta_{\varepsilon,\mathfrak{z}}\nabla u\|^2 + C_\delta \|z\|_4^4. \end{aligned} \tag{D.6}$$

By combining (D.4)–(D.6), one gets

$$\mathcal{I}_1 \le 3\delta \|\theta_{\varepsilon,\mathfrak{z}}\nabla u\|^2 + C_\delta \|\nabla z\|_4^{\frac{4}{3}} \|\theta_{\varepsilon,\mathfrak{z}} u\|^2 + C_\delta \|z\|_4^4. \tag{D.7}$$

We turn to the estimate of $\mathcal{I}_2$. By applying (4.5), the embedding $H^1 \hookrightarrow L^6$, and the Poincaré inequality, it follows

$$\begin{aligned} \mathcal{I}_2 &\le \|\theta_{\varepsilon,\mathfrak{z}} u\|_6 \|\theta_{\varepsilon,\mathfrak{z}}\nabla z\| \|\theta^{-2}_{\varepsilon,\mathfrak{z}} v_{\varepsilon,\mathfrak{z}}\|_3 \\ &\quad + \|\theta_{\varepsilon,\mathfrak{z}} z\|_6 \|\theta_{\varepsilon,\mathfrak{z}}\nabla u\| \|\theta^{-2}_{\varepsilon,\mathfrak{z}} v_{\varepsilon,\mathfrak{z}}\|_3 + \|\theta_{\varepsilon,\mathfrak{z}} z\|_6 \|\theta_{\varepsilon,\mathfrak{z}}\nabla z\| \|\theta^{-2}_{\varepsilon,\mathfrak{z}} v_{\varepsilon,\mathfrak{z}}\|_3 \\ &\le \delta \|\theta_{\varepsilon,\mathfrak{z}}\nabla u\|^2 + C_\delta \|\nabla z\|^2 \|\theta^{-2}_{\varepsilon,\mathfrak{z}} v_{\varepsilon,\mathfrak{z}}\|_3^2 + C\|\nabla z\|^2 \|\theta^{-2}_{\varepsilon,\mathfrak{z}} v_{\varepsilon,\mathfrak{z}}\|_3, \end{aligned} \tag{D.8}$$

where, by applying (4.17), one has

$$\int_0^T \left(\|\nabla z\|^2 \|\theta^{-2}_{\varepsilon,\mathfrak{z}} v_{\varepsilon,\mathfrak{z}}\|_3^2 + \|\nabla z\|^2 \|\theta^{-2}_{\varepsilon,\mathfrak{z}} v_{\varepsilon,\mathfrak{z}}\|_3\right) \mathrm{d}t \lesssim_\nu \left(\varepsilon^2 \mathscr{M}_T^2 + \varepsilon \mathscr{M}_T\right) \int_0^T \|\nabla z\|^2 \mathrm{d}t \tag{D.9}$$

with $\mathscr{M}_T := \|\theta_{\varepsilon,\mathfrak{z}} u\|_{C([0,T];L^2)}$. Now, by combining (D.7)–(D.9) and repeating the derivation (4.23) with sufficiently small $\varepsilon > 0$, one gets

$$\begin{aligned}
&\|\theta_{\varepsilon,\mathfrak{z}} u(T)\|^2 + (\nu - C_\nu \varepsilon \mathscr{M}_T) \int_0^T \|\theta_{\varepsilon,\mathfrak{z}} \nabla u\|^2 \mathrm{d}t \\
&\qquad \le C_\nu \varepsilon \mathscr{M}_T \|\theta_{\varepsilon,\mathfrak{z}} u_0\| + \|\theta_{\varepsilon,\mathfrak{z}} u_0\|^2 + C_\nu \left(\varepsilon^2 \mathscr{M}_T^2 + \varepsilon \mathscr{M}_T\right) \int_0^T \|\nabla z\|^2 \mathrm{d}t \\
&\qquad + C_\nu \int_0^T \|\nabla z\|_4^{\frac{4}{3}} \|\theta_{\varepsilon,\mathfrak{z}} u\|^2 \mathrm{d}t + C_\nu \int_0^T \|z\|_4^4 \mathrm{d}t. \qquad \text{(D.10)}
\end{aligned}$$

Notice that (D.3) ensures

$$\int_0^T \left(\|\nabla z\|_4^{\frac{4}{3}} + \|z\|_4^4 + \|\nabla z\|^2 \right) \mathrm{d}t < \infty.$$

Consequently, by shrinking $\varepsilon > 0$ if necessary, repeating the bootstrap argument in Step **II** in the proof of Proposition 4.4, and using (4.4), it follows

$$\begin{aligned}
\sup_{0\le t\le T} \|u(t)\|_{L^2_b}^2 + \nu \sup_{\mathfrak{z}\in\mathbb{R}} \int_0^T \|\nabla u(t)\|_{L^2(D_\mathfrak{z})}^2 \mathrm{d}t &\le C_\nu \exp\left(C_\nu \int_0^T \|\nabla z\|_4^{\frac{4}{3}} \mathrm{d}t \right) \\
&\times \left(1 + \|u_0\|_{L^2_b}^2 + \int_0^T \|z(t)\|_4^4 dt + \left(\int_0^T \|\nabla z(t)\|^2 dt \right)^2 \right)^2. \qquad \text{(D.11)}
\end{aligned}$$

Step II. Let $u(t), u'(t)$ be the solutions of (D.2) corresponding to the same trajectory $z(\cdot)$, which start from u_0 and u_0', respectively. The goal of this step is to establish an estimate for the difference $\tilde{g}(t) := u(t) - u'(t)$, which satisfies

$$\begin{cases}
\partial_t \tilde{g} - \nu \Delta \tilde{g} + ((u' + z) \cdot \nabla)\tilde{g} + (\tilde{g} \cdot \nabla)(u + z) + \nabla \tilde{\mathfrak{p}} = 0 \\
\operatorname{div} \tilde{g} = 0, \\
\tilde{g}|_{t=0} = u_0 - u_0', \qquad \tilde{g}|_{\partial D} = 0.
\end{cases} \qquad \text{(D.12)}$$

To this end, for $\mathfrak{z} \in \mathbb{R}$, we define a cut-off function

$$\psi_\mathfrak{z} := \psi(x - \mathfrak{z}), \qquad \psi \in C_0^\infty((-1,2)), \qquad 0 \le \psi \le 1, \qquad \psi \equiv 1 \text{ on } [0,1]. \qquad \text{(D.13)}$$

Similar to Step I in the proof of Proposition 4.4, we introduce an auxiliary function $v_\mathfrak{z}$ based on the weight $\psi_\mathfrak{z}$ through the backward Stokes equation

$$\begin{cases}
-\partial_t v_\mathfrak{z} + \nabla \mathfrak{p}_\mathfrak{z} = \nu \Delta v_\mathfrak{z}, \\
\operatorname{div} v_\mathfrak{z} = 2\psi_\mathfrak{z} \partial_x \psi_\mathfrak{z} \tilde{g}_1, \\
v_\mathfrak{z}|_{t=T} = \left(0, 2\psi_\mathfrak{z} \partial_x \psi_\mathfrak{z} \mathcal{J}\tilde{g}(T)\right), \qquad v_\mathfrak{z}|_{\partial D} = 0,
\end{cases} \qquad \text{(D.14)}$$

where $\mathcal{J}\tilde{g} := \int_{-1}^y \tilde{g}_1(x, y')\mathrm{d}y'$. By applying Proposition 2.2 and Theorem 3.2 in [AZ14] with $\phi := \theta_{\varepsilon,\mathfrak{z}}^{-2}$, the Sobolev embedding $H^{\frac{1}{3}} \hookrightarrow L^p$ with $p = 2, 3$, and the estimate

$$\|\theta_{\varepsilon,\mathfrak{z}}^{-2} \psi_\mathfrak{z} \partial_x \psi_\mathfrak{z} \mathcal{J}\tilde{g}\|_{H^1} \lesssim \|\theta_{\varepsilon,\mathfrak{z}} \tilde{g}\|, \qquad \text{(D.15)}$$

one has

$$\sup_{t\in[0,T]} \|\theta_{\varepsilon,\mathfrak{z}}^{-2} v_\mathfrak{z}(t)\| + \sup_{t\in[0,T]} \|\theta_{\varepsilon,\mathfrak{z}}^{-2} v_\mathfrak{z}(t)\|_3 \lesssim_\nu \sup_{t\in[0,T]} \|\theta_{\varepsilon,\mathfrak{z}} \tilde{g}(t)\|. \qquad \text{(D.16)}$$

Testing (D.12) against $\psi_{\mathfrak{z}}^2\tilde{g} - v_{\mathfrak{z}}$ and using $(\text{D.14})_1$ yield

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\|\psi_{\mathfrak{z}}\tilde{g}\|^2 - \langle \tilde{g}, v_{\mathfrak{z}}\rangle\right) + \nu\langle \nabla\tilde{g}, \nabla(\psi_{\mathfrak{z}}^2\tilde{g})\rangle$$
$$= -\langle ((u'+z)\cdot\nabla)\tilde{g} + (\tilde{g}\cdot\nabla)(u+z), \psi_{\mathfrak{z}}^2\tilde{g} - v_{\mathfrak{z}}\rangle.$$

Comparing with (4.18), since the terminal data $v_{\mathfrak{z}}(T)$ is of order $O(1)$, there is a troublesome term $\langle \tilde{g}(T), v_{\mathfrak{z}}(T)\rangle$ arising from integrating both sides over $[0,T]$. To overcome this difficulty, we modify the auxiliary function $v_{\mathfrak{z}}$ by extracting a divergence-free part so that the resulting auxiliary function $\hat{v}_{\mathfrak{z}}$ has terminal data of the gradient type. As a result, the terminal pairing $\langle \tilde{g}(T), v_{\mathfrak{z}}(T)\rangle$ vanishes.

To see this, we define $\tilde{v}_{\mathfrak{z}}(t) := S(T-t)\Pi v_{\mathfrak{z}}(T)$, where Π is the Leray projection and $S(t) := e^{\nu t L}$ is the Stokes semigroup. Let $\hat{v}_{\mathfrak{z}} := v_{\mathfrak{z}} - \tilde{v}_{\mathfrak{z}}$, so $\hat{v}_{\mathfrak{z}}$ satisfies

$$\begin{cases} -\partial_t\hat{v}_{\mathfrak{z}} + \nabla\hat{\mathfrak{p}}_{\mathfrak{z}} = \nu\Delta\hat{v}_{\mathfrak{z}}, \\ \operatorname{div}\hat{v}_{\mathfrak{z}} = 2\psi_{\mathfrak{z}}\partial_x\psi_{\mathfrak{z}}\tilde{g}_1, \\ \hat{v}_{\mathfrak{z}}|_{t=T} = (I-\Pi)\left(0, 2\psi_{\mathfrak{z}}\partial_x\psi_{\mathfrak{z}}\mathcal{J}\tilde{g}(T)\right), \qquad \hat{v}_{\mathfrak{z}}|_{\partial D} = 0. \end{cases} \tag{D.17}$$

By applying Corollary C.3, the Sobolev embedding $H^1 \hookrightarrow L^p$ with $p = 2, 3$, and estimate (D.15), one has

$$\|\theta_{\varepsilon,\mathfrak{z}}^{-2}\tilde{v}_{\mathfrak{z}}(t)\| + \|\theta_{\varepsilon,\mathfrak{z}}^{-2}\tilde{v}_{\mathfrak{z}}(t)\|_3 \lesssim_\nu \|\theta_{\varepsilon,\mathfrak{z}}^{-2}v_{\mathfrak{z}}(T)\| + \|\theta_{\varepsilon,\mathfrak{z}}^{-2}v_{\mathfrak{z}}(T)\|_3 \lesssim_\nu \|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(T)\|$$
$$\lesssim_\nu \sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|.$$

Consequently, estimate (D.16) holds for the new auxiliary function $\hat{v}_{\mathfrak{z}}$, if ε is sufficiently small. Furthermore, based on this auxiliary function, we have the following energy equality

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\|\psi_{\mathfrak{z}}\tilde{g}\|^2 - \langle \tilde{g}, \hat{v}_{\mathfrak{z}}\rangle\right) + \nu\langle \nabla\tilde{g}, \nabla(\psi_{\mathfrak{z}}^2\tilde{g})\rangle$$
$$= -\langle ((u'+z)\cdot\nabla)\tilde{g} + (\tilde{g}\cdot\nabla)(u+z), \psi_{\mathfrak{z}}^2\tilde{g} - \hat{v}_{\mathfrak{z}}\rangle, \tag{D.18}$$

where the terminal pairing $\langle \tilde{g}(T), \hat{v}_{\mathfrak{z}}(T)\rangle$ disappears, since $\tilde{g}(T)$ is divergence-free and $\hat{v}_{\mathfrak{z}}(T)$ is of the gradient type as indicated in $(\text{D.17})_3$.

We turn to the estimate of each term in (D.18). Throughout the rest of the proof, we fix a sufficiently small ε such that (D.16) holds for $\hat{v}_{\mathfrak{z}}$. Since $\psi_{\mathfrak{z}}$ has compact support, the dissipation term satisfies

$$\nu\langle \nabla\tilde{g}, \nabla(\psi_{\mathfrak{z}}^2\tilde{g})\rangle \geq \frac{\nu}{2}\|\nabla(\psi_{\mathfrak{z}}\tilde{g})\|^2 - C_\nu\|\partial_x\psi_{\mathfrak{z}}\tilde{g}\|^2 \geq \frac{\nu}{2}\|\nabla(\psi_{\mathfrak{z}}\tilde{g})\|^2 - C_\nu\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}\|^2. \tag{D.19}$$

Let

$$D_{\mathfrak{z}}^* := (\mathfrak{z}-1, \mathfrak{z}+2)\times(-1,1). \tag{D.20}$$

Applying the Hölder and Ladyzhenskaya inequalities combined with the support property of $\psi_{\mathfrak{z}}$, one has

$$|\langle (\tilde{g}\cdot\nabla)(u+z), \psi_{\mathfrak{z}}^2\tilde{g}\rangle| \leq C\|\psi_{\mathfrak{z}}\tilde{g}\|\|\nabla(\psi_{\mathfrak{z}}\tilde{g})\|\|\nabla(u+z)\|_{L^2(D_{\mathfrak{z}}^*)}$$
$$\leq \frac{\nu}{8}\|\nabla(\psi_{\mathfrak{z}}\tilde{g})\|^2 + C_\nu\|\psi_{\mathfrak{z}}\tilde{g}\|^2\|\nabla(u+z)\|_{L^2(D_{\mathfrak{z}}^*)}^2. \tag{D.21}$$

Let $\mathfrak{M}_T \geq 1$ be a constant depending on $\nu, T, \|u_0\|_{L_b^2}, \|u_0'\|_{L_b^2}, \|z\|_{C([0,T];V)}, \|z\|_{L^2(0,T;H^2)}$ such that

$$\sup_{0\leq t\leq T}\left(\|u(t)\|_{L_b^2}^2 + \|u'(t)\|_{L_b^2}^2\right) + \nu\sup_{\mathfrak{z}\in\mathbb{R}}\int_0^T\left(\|\nabla u(t)\|_{L^2(D_{\mathfrak{z}})}^2 + \|\nabla u'(t)\|_{L^2(D_{\mathfrak{z}})}^2\right)\mathrm{d}t \leq \mathfrak{M}_T,$$

and

$$\|z\|^2_{C([0,T];V)} + \|z\|^2_{L^2(0,T;H^2)} \le \mathfrak{M}_T.$$

By applying the above bounds and the Ladyzhenskaya inequality, integrating by parts, and using the fact $\operatorname{div}(u'+z)=0$, it follows

$$\begin{aligned}
|\langle (u'+z)\cdot\nabla\tilde{g}, \psi_\mathfrak{z}^2\tilde{g}\rangle| &= |\langle (u'+z)_1\psi_\mathfrak{z}\partial_x\psi_\mathfrak{z}, |\tilde{g}|^2\rangle \le \|u'+z\|_{L^4(D_\mathfrak{z}^*)}\|\psi_\mathfrak{z}\tilde{g}\|_4\|\partial_x\psi_\mathfrak{z}\tilde{g}\| \\
&\le C\mathfrak{M}_T^{\frac14}\|\nabla(u'+z)\|^{\frac12}_{L^2(D_\mathfrak{z}^*)}\|\psi_\mathfrak{z}\tilde{g}\|^{\frac12}\|\nabla(\psi_\mathfrak{z}\tilde{g})\|^{\frac12}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}\| \\
&\le \frac{\nu}{8}\|\nabla(\psi_\mathfrak{z}\tilde{g})\|^2 + \mathfrak{M}_T\|\nabla(u'+z)\|^2_{L^2(D_\mathfrak{z}^*)}\|\psi_\mathfrak{z}\tilde{g}\|^2 + C_\nu\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}\|^2. \qquad \text{(D.22)}
\end{aligned}$$

We turn to the terms involving $\hat{v}_\mathfrak{z}$. By (D.16) with $v_\mathfrak{z}$ replaced by $\hat{v}_\mathfrak{z}$, the Gagliardo–Nirenberg inequality, and estimate (4.4), one has

$$\begin{aligned}
|\langle (\tilde{g}\cdot\nabla)(u+z), \hat{v}_\mathfrak{z}\rangle| &\le \|\theta_{\varepsilon,\mathfrak{z}}^{-2}\hat{v}_\mathfrak{z}\|_3\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}\|_6\|\theta_{\varepsilon,\mathfrak{z}}\nabla(u+z)\| \\
&\le C_\nu \sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}\|^{\frac13}\|\theta_{\varepsilon,\mathfrak{z}}\nabla\tilde{g}\|^{\frac23}\|\theta_{\varepsilon,\mathfrak{z}}\nabla(u+z)\| \\
&\le \delta\|\theta_{\varepsilon,\mathfrak{z}}\nabla\tilde{g}\|^2 + \delta\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2\|\theta_{\varepsilon,\mathfrak{z}}\nabla(u+z)\|^2 + C_{\nu,\delta}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}\|^2, \qquad \text{(D.23)}
\end{aligned}$$

and

$$\begin{aligned}
|\langle (u'+z)\cdot\nabla\tilde{g}, \hat{v}_\mathfrak{z}\rangle| &\le \|\theta_{\varepsilon,\mathfrak{z}}^{-2}\hat{v}_\mathfrak{z}\|_3\|\theta_{\varepsilon,\mathfrak{z}}(u'+z)\|^{\frac13}\|\theta_{\varepsilon,\mathfrak{z}}\nabla(u'+z)\|^{\frac23}\|\theta_{\varepsilon,\mathfrak{z}}\nabla\tilde{g}\| \\
&\le C_\nu\mathfrak{M}_T^{\frac16}\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|\|\theta_{\varepsilon,\mathfrak{z}}\nabla(u'+z)\|^{\frac23}\|\theta_{\varepsilon,\mathfrak{z}}\nabla\tilde{g}\| \\
&\le \delta\|\theta_{\varepsilon,\mathfrak{z}}\nabla\tilde{g}\|^2 + \delta\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2\|\theta_{\varepsilon,\mathfrak{z}}\nabla(u'+z)\|^2 \\
&\quad + C_{\nu,\delta}\mathfrak{M}_T\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2. \qquad \text{(D.24)}
\end{aligned}$$

As for the initial pairing, by (D.16) with $v_\mathfrak{z}$ replaced by $\hat{v}_\mathfrak{z}$, we have

$$\begin{aligned}
|\langle \tilde{g}(0), \hat{v}_\mathfrak{z}(0)\rangle| &\le C_\nu\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|\|\theta_{\varepsilon,\mathfrak{z}}(u_0-u_0')\| \\
&\le C_{\nu,\delta}\|\theta_{\varepsilon,\mathfrak{z}}(u_0-u_0')\|^2 + \delta\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2. \qquad \text{(D.25)}
\end{aligned}$$

Now, by combining (D.18)–(D.25) and using the fact that $\psi_\mathfrak{z}\equiv 1$ on $D_\mathfrak{z}$, it follows

$$\begin{aligned}
\frac12\|\psi_\mathfrak{z}\tilde{g}(T)\|^2 + \frac{\nu}{4}\int_0^T\|\nabla\tilde{g}\|^2_{L^2(D_\mathfrak{z})}\mathrm{d}t &\le C_{\nu,\delta}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(0)\|^2 + 2\delta\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla\tilde{g}\|^2\mathrm{d}t \\
&\quad + (\delta C_\nu\mathfrak{M}_T + C_{\nu,\delta}T\mathfrak{M}_T)\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2 \\
&\quad + (C_\nu+\mathfrak{M}_T)\int_0^T\|\psi_\mathfrak{z}\tilde{g}\|^2\left(\|\nabla(u'+z)\|^2_{L^2(D_\mathfrak{z}^*)} + \|\nabla(u+z)\|^2_{L^2(D_\mathfrak{z}^*)}\right)\mathrm{d}t,
\end{aligned}$$

which, together with taking $T\le 1$ and applying the Gronwall inequality, yields

$$\begin{aligned}
\|\psi_\mathfrak{z}\tilde{g}(T)\|^2 + \nu\int_0^T\|\nabla\tilde{g}\|^2_{L^2(D_\mathfrak{z})}\mathrm{d}t &\le C_{\nu,\delta,\mathfrak{M}_1}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(0)\|^2 + \delta C_{\nu,\mathfrak{M}_1}\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla\tilde{g}\|^2\mathrm{d}t \\
&\quad + (\delta C_{\nu,\mathfrak{M}_1} + C_{\nu,\delta,\mathfrak{M}_1}T)\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2.
\end{aligned}$$

To close the estimate, we use a convolution-type argument. Multiplying on both sides with $\theta^2_{\varepsilon,\tilde{\mathfrak{z}}}(\mathfrak{z}) = (1+\varepsilon^2(\mathfrak{z}-\tilde{\mathfrak{z}})^2)^{-1}$, integrating the resulting estimate with respect to $\mathfrak{z}\in\mathbb{R}$, and using Proposition 2.2 in [AZ14] combined with the fact

$$\int_{\mathbb{R}} \frac{1}{(1+\varepsilon^2(x'-x)^2)(1+\varepsilon^2x^2)}\mathrm{d}x = \frac{2\pi}{\varepsilon(4+\varepsilon^2x'^2)}, \tag{D.26}$$

one gets

$$\begin{aligned}\|\theta_{\varepsilon,\tilde{\mathfrak{z}}}\tilde{g}(T)\|^2 + \nu\int_0^T \|\theta_{\varepsilon,\tilde{\mathfrak{z}}}\nabla\tilde{g}\|^2\mathrm{d}t \le C_{\nu,\delta,\mathfrak{M}_1}\|\theta_{\varepsilon,\tilde{\mathfrak{z}}}\tilde{g}(0)\|^2 + \delta C_{\nu,\mathfrak{M}_1}\int_0^T \|\theta_{\varepsilon,\tilde{\mathfrak{z}}}\nabla\tilde{g}\|^2\mathrm{d}t\\ + (\delta C_{\nu,\mathfrak{M}_1} + C_{\nu,\delta,\mathfrak{M}_1}T)\int_{\mathbb{R}}\theta^2_{\varepsilon,\tilde{\mathfrak{z}}}(\mathfrak{z})\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2\mathrm{d}\mathfrak{z}.\end{aligned}$$

Choose δ, T sufficiently small such that

$$\delta C_{\nu,\mathfrak{M}_1} \le \frac{\nu}{2}, \qquad \delta C_{\nu,\mathfrak{M}_1} + C_{\nu,\delta,\mathfrak{M}_1}T \le \frac{\varepsilon}{16\pi},$$

so one has

$$\sup_{t\in[0,T]}\|\theta_{\varepsilon,\tilde{\mathfrak{z}}}\tilde{g}(t)\|^2 \le C_{\nu,\delta,\mathfrak{M}_1}\|\theta_{\varepsilon,\tilde{\mathfrak{z}}}\tilde{g}(0)\|^2 + \frac{\varepsilon}{16\pi}\int_{\mathbb{R}}\theta^2_{\varepsilon,\tilde{\mathfrak{z}}}(\mathfrak{z})\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2\mathrm{d}\mathfrak{z}.$$

Multiplying again on both sides of the above inequality with $\theta^2_{\varepsilon,\hat{\mathfrak{z}}}(\tilde{\mathfrak{z}})$, integrating the resulting estimate with respect to $\tilde{\mathfrak{z}}\in\mathbb{R}$, and using (D.26) yield

$$\int_{\mathbb{R}}\theta^2_{\varepsilon,\hat{\mathfrak{z}}}(\mathfrak{z})\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2\mathrm{d}\mathfrak{z} \le C_{\nu,\delta,\mathfrak{M}_1}\|\theta_{\varepsilon,\hat{\mathfrak{z}}}\tilde{g}(0)\|^2 + \frac{1}{8}\int_{\mathbb{R}}\theta^2_{\varepsilon,\hat{\mathfrak{z}}}(\mathfrak{z})\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2\mathrm{d}\mathfrak{z}.$$

Consequently, since $\hat{\mathfrak{z}}$ is arbitrary, one has

$$\sup_{t\in[0,T]}\|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(t)\|^2 + \nu\int_0^T\|\theta_{\varepsilon,\mathfrak{z}}\nabla\tilde{g}\|^2\mathrm{d}t \lesssim_{\nu,\mathfrak{M}_1} \|\theta_{\varepsilon,\mathfrak{z}}\tilde{g}(0)\|^2. \tag{D.27}$$

Since T does not depend on the initial data $\tilde{g}(0)$, by iterating the above estimate and using (4.4), it follows

$$\sup_{t\in[0,T]}\|u(t)-u'(t)\|^2_{L^2_b} + \nu\sup_{\mathfrak{z}\in\mathbb{R}}\int_0^T\|\nabla(u-u')\|^2_{L^2(D_{\mathfrak{z}})}\mathrm{d}t \le C_T\|u_0-u_0'\|^2_{L^2_b}, \tag{D.28}$$

where C_T depends on ν, T, u_0, u_0' and the trajectory $z(\cdot)$.

Step III. The uniqueness now follows from (D.28). The existence part requires additional treatment, since in general one can only approximate $u\in H_b$ by elements in H under the weak metric d_{loc} defined in (1.8).

Let $u_0, u_0'\in B_{H_b}(R)$. Notice that for any integer $k\ge 1$ and $\mathfrak{z}\in\mathbb{R}$, by applying (D.27), one has

$$\begin{aligned}\sup_{t\in[0,T]}\|u(t)-u'(t)\|^2_{L^2(D_{\mathfrak{z}})} &\le C_T\|u_0-u_0'\|^2_{L^2(I_{k^2,\mathfrak{z}})} + C_T\int_{I^c_{k^2,\mathfrak{z}}}\frac{|u_0(x)-u_0'(x)|^2}{1+\varepsilon^2(x-\mathfrak{z})^2}\mathrm{d}x\\ &\lesssim C_T\|u_0-u_0'\|^2_{L^2(I_{k^2,\mathfrak{z}})} + C_T\|u_0-u_0'\|^2_{L^2_b}\varepsilon^{-2}k^{-2},\end{aligned}$$

where $I_{k^2,\mathfrak{z}} := (\mathfrak{z}-k^2, \mathfrak{z}+k^2)\times(-1,1)$. This implies

$$\|u(t)-u'(t)\|_{L^2(I_k)} \lesssim C_T\sqrt{k}\|u_0-u_0'\|_{L^2(I_{2k^2})} + C_TR\varepsilon^{-1}k^{-\frac{1}{2}}$$

with $I_k := (-k,k)\times(-1,1)$. Consequently, for any integer $k\geq 1$ and $t\in[0,T]$, one has

$$d_{\mathrm{loc}}(u(t),u'(t)) \leq \left(\sum_{l\leq k} 2^{-l}\right)\|u(t)-u'(t)\|_{L^2(I_k)} + \sum_{l>k}\frac{1\wedge\|u(t)-u'(t)\|_{L^2(I_l)}}{2^l}$$

$$\lesssim C_T(\sqrt{k}\|u_0-u_0'\|_{L^2(I_{2k^2})} + R\varepsilon^{-1}k^{-\frac{1}{2}}) + C_T\sum_{l>k}\frac{\sqrt{l}(1\wedge\|u_0-u_0'\|_{L^2(I_{2l^2})})}{2^l}. \quad \text{(D.29)}$$

Now for any given $u_0\in H_b$, choose $\{u_{0,n}\}_{n\geq 1}\subset H$ such that

$$\sup_{n\geq 1}\|u_{0,n}\|_{L^2_b}<\infty, \qquad \lim_{n\to\infty} d_{\mathrm{loc}}(u_0,u_{0,n})=0.$$

Denote by $u_n(\cdot)$ the finite-energy solution issued from $u_{0,n}$. By choosing k sufficiently large in (D.29) and then sending $n,m\to\infty$, one obtains

$$\lim_{n,m\to\infty}\sup_{t\in[0,T]} d_{\mathrm{loc}}(u_n(t),u_m(t))=0.$$

This shows that $\{u_n(\cdot)\}_{n\geq 1}$ is a Cauchy sequence in $C([0,T];L^2_{\mathrm{loc}})$, and thus converges to some function u. Moreover, the a priori estimate (D.11) shows that

$$\sup_{n\geq 1}\sup_{0\leq t\leq T}\|u_n(t)\|^2_{L^2_b}\leq C_T$$

for some positive constant C_T. Since every closed H_b-ball is complete under d_{loc}, we conclude that $u\in C([0,T];X)$.

Finally, notice that the convergence under the metric of $C([0,T];L^2_{\mathrm{loc}})$ is sufficient to pass to the limit for the nonlinear term. This, together with the weak convergence in $L^2(0,T;L^2_{\mathrm{loc}})$ for the gradient term, ensures the existence of an infinite-energy solution of (D.2). □

Next, we study the smoothing effect of the stochastic NS equation (1.1) in the infinite-energy setting.

Proposition D.2. *Let $\{u(t)\}_{\geq 0}$ be the unique global infinite-energy solution of (1.1). Then, for any $\delta,T>0$, $\sup_{t\in[\delta,T]}\|u(t)\|_\infty<\infty$ holds $\mathbb{P}$-almost surely.*

Proof. The proof is based on Lemma 5.3, whose proof is postponed to the end of this section.
Step I. To begin with, we show that

$$\mathbb{P}_u^X\Big\{\sup_{t\in[\delta,1]}\|u_t\|_\infty<\infty\Big\}=1 \quad \text{(D.30)}$$

for every $u\in H_b$ and $\delta\in(0,1)$. Let $R:=\|u\|_{L^2_b}$. By applying (5.13), it follows that

$$\sup_{t\in[\delta,1]}\|u(t)\|_\infty\leq \mathscr{C}_5\delta^{-\frac{1}{2}}$$

on the random event $\mathcal{A}_{1,\mathscr{L}}$, which implies

$$\mathbb{P}_u^X\Big\{\sup_{t\in[\delta,1]}\|u_t\|_\infty=\infty\Big\}\leq \mathbb{P}_u^X\Big\{\sup_{t\in[\delta,1]}\|u_t\|_\infty>\mathscr{C}_5\delta^{-\frac{1}{2}}\Big\}\leq \mathbb{P}(\mathcal{A}^c_{1,\mathscr{L}})\to 0,$$

as $\mathscr{L}\to\infty$. Therefore, (D.30) holds.
Step II. Without loss of generality, we assume $0<\delta<\frac{1}{2}$. Introduce the following covering

$$[\delta,T]\subset\cup_{j=0}^N[j/2+\delta,j/2+1].$$

By the Markov property, it follows

$$\mathbb{P}_u^X\Big\{\sup_{t\in[\delta,1]}\|u_{j/2+t}\|_\infty<\infty\Big|\mathcal{F}_{j/2}\Big\}=\mathbb{P}_{u_{j/2}}^X\Big\{\sup_{t\in[\delta,1]}\|u(t)\|_\infty<\infty\Big\}=1.$$

Consequently, by taking the intersection of these events, we obtain

$$\sup_{t\in[\delta,T]}\|u(t)\|_\infty<\infty$$

almost surely. □

We turn to the verification of the decomposition stated in Theorem 1.2.

Proposition D.3. *Let $\{u(t)\}_{\geq 0}$ be the unique global infinite-energy solution of (1.1) issued from $u_0\in H_b\cap L^\infty$. Then,*

$$u(t)=w(t)+u_{\rm rem}(t),$$

where $w(t)$ is an infinite-energy solution of the homogeneous NS equation (1.10) and $u_{\rm rem}$ satisfies the path regularity (1.11).

Proof. It suffices to prove (1.11). To this end, we test the remainder equation (1.23) against $u_{\rm rem}$ and apply the Itô formula to derive

$$\begin{aligned}\mathrm{d}\|u_{\rm rem}(t)\|^2+2\nu\|\nabla u_{\rm rem}\|^2\mathrm{d}t&=2\langle h-(u_{\rm rem}\cdot\nabla)w,u_{\rm rem}\rangle\mathrm{d}t\\&\quad+2\textstyle\sum_{j\geq1}b_j\langle u_{\rm rem},e_j\rangle\mathrm{d}\beta_j+\mathfrak{B}_0\mathrm{d}t,\end{aligned}\tag{D.31}$$

where, by the cancellation property of the convection term, one has

$$|\langle(u_{\rm rem}\cdot\nabla)w,u_{\rm rem}\rangle|\leq\|u_{\rm rem}\|\|\nabla u_{\rm rem}\|\|w\|_\infty.$$

As a result, by taking expectation on both sides of (D.31), it follows

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\|u_{\rm rem}(t)\|^2+\nu\mathbb{E}\|\nabla u_{\rm rem}\|^2\leq\mathfrak{B}_0+C_\nu\|h\|^2+C_\nu\|w\|_\infty^2\mathbb{E}\|u_{\rm rem}\|^2\tag{D.32}$$

On the other hand, by Proposition 4.1, one has

$$\int_0^\infty\|w(t)\|_\infty^2\mathrm{d}t\leq C_{\nu,u_0}.$$

This, together with (D.32) combined with the Gronwall inequality, yields

$$\mathbb{E}\|u_{\rm rem}(T)\|^2\leq T(\mathfrak{B}_0+C_\nu\|h\|^2)\exp\left(C_\nu\int_0^\infty\|w(t)\|_\infty^2\mathrm{d}t\right)\leq C_{\nu,u_0}T(\mathfrak{B}_0+\|h\|^2).$$

Plugging this into (D.32) and integrating the resulting estimate over $[0,T]$ imply

$$\mathbb{E}\|u_{\rm rem}(T)\|^2+\nu\mathbb{E}\int_0^T\|\nabla u_{\rm rem}\|^2\mathrm{d}t\leq C_{\nu,u_0}T(\mathfrak{B}_0+\|h\|^2).\tag{D.33}$$

Finally, by the Burkholder–Davis–Gundy and Poincaré inequalities combined with (D.33), it follows

$$\begin{aligned}\mathbb{E}\sup_{t\in[0,T]}\left|\textstyle\sum_{j\geq1}b_j\int_0^t\langle u_{\rm rem},e_j\rangle\mathrm{d}\beta_j\right|&\leq C\mathbb{E}\left(\textstyle\sum_{j\geq1}b_j^2\int_0^T\langle u_{\rm rem},e_j\rangle^2\mathrm{d}t\right)^{\frac12}\\&\leq C\mathfrak{B}_0^{\frac12}\left(\mathbb{E}\int_0^T\|\nabla u_{\rm rem}\|^2\mathrm{d}t\right)^{\frac12}\leq C_{\nu,u_0}\mathfrak{B}_0^{\frac12}T^{\frac12}(\mathfrak{B}_0+\|h\|^2)^{\frac12}.\end{aligned}$$

Together with integrating both sides in t of (D.31) and taking first supremum in $t\in[0,T]$ and then expectation, this implies

$$\mathbb{E}\sup_{t\in[0,T]}\|u_{\rm rem}(t)\|^2\leq C_{\nu,h,\mathfrak{B}_0,T,u_0}<\infty.$$

Combining this and (D.33), we obtain the path regularity (1.11). □

Now, we prove the last statement in Theorem 1.2.

Proposition D.4. *The infinite-energy solutions of* (1.1) *define a Markov family* $(u_t, \mathbb{P}_u^X)$.

Proof. As presented in Example 1.3.6 in [KS12], it suffices to show that (1.1) admits a solution map, which is Borel measurable in the initial data and noise trajectory. Indeed, since $\mathfrak{B}_1 < \infty$, there is a measurable map

$$\mathbf{\Psi}_1 : C([0,\infty); V) \to C([0,\infty); V) \cap L^2_{\mathrm{loc}}(0,\infty; V \cap H^2)$$

such that

$$\mathbf{\Psi}_1 W = z, \qquad \mathbb{P}\text{-a.s.},$$

where $W(t) := \int_0^t \eta(s)\mathrm{d}s$ is the Wiener process and $\{z(t)\}_{t\geq 0}$ is the solution of the stochastic Stokes equation (D.1). Introduce

$$\mathbf{\Psi}_2 : (u, z) \mapsto u_{\mathrm{det}}(\cdot), \qquad X \times \big(C([0,\infty); V) \cap L^2_{\mathrm{loc}}(0,\infty; V \cap H^2)\big) \to C([0,\infty); X)$$

as the solution map of (D.2). Define

$$\mathbf{\Phi} : (u, W) \mapsto \mathbf{\Psi}_2(u, \mathbf{\Psi}_1 W) + \mathbf{\Psi}_1 W, \qquad X \times C([0,\infty); V) \to C([0,\infty); X).$$

By the unique solvability of the stochastic NS equation (1.1) in the infinite-energy setting, one has

$$\mathbf{\Phi}(u, W) = \mathbf{\Psi}_2(u, \mathbf{\Psi}_1 W) + \mathbf{\Psi}_1 W = u(\cdot), \qquad \mathbb{P}\text{-a.s.},$$

where $\{u(t)\}_{t\geq 0}$ is the infinite-energy solution of (1.1). Moreover, the derivation of (D.28) and (D.29) can be extended to two solutions with different noise trajectories, which ensures the Borel measurability of $\mathbf{\Psi}_2$ and thus the Borel measurability of the solution map $\mathbf{\Phi}$. □

Finally, we present the proof of Lemma 5.3.

Proof of Lemma 5.3. Fix arbitrary $t_0 \in (0, 1]$. Introduce the decomposition

$$u(t) = u_{\mathrm{det}}(t) + z(t),$$

where $\{z(t)\}_{t\geq 0}$ satisfies the stochastic Stokes equation (D.1) and $\{u_{\mathrm{det}}(t)\}_{t\geq 0}$ is the solution of (D.2). Throughout the proof, the analysis is carried out on the random event $\mathcal{A}_{t_0,\mathscr{L}}$. Notice that the Stokes semigroup $S(t) := e^{t\nu L}$ satisfies

$$\left\| \int_0^{\cdot} S(\cdot - s) h \mathrm{d}s \right\|_{L^\infty(0,t;H^{3/2})} \lesssim_\nu t^{\frac{1}{4}} \|h\|,$$

which, together with the Sobolev embeddings $H^{\frac{3}{2}} \hookrightarrow L^\infty$ and $H^{\frac{3}{2}} \hookrightarrow W^{1,4}$, yield

$$\sup_{s\in[0,t_0]} \left(\|z(s)\|_\infty + \|z(s)\|_{H^1} + \|z(s)\|_{W^{1,4}}\right) \leq C_{\nu,h,\mathscr{L}}. \tag{D.34}$$

Therefore, it remains to bound u_{det}. As before, to simplify the notation, we shall drop the subscript in u_{det} and omit the dependence of the constants on $h, \mathscr{L}, R$ in what follows.
Step I. Let $\theta_{\varepsilon,\mathfrak{z}}, \psi_{\mathfrak{z}}$ be the weight functions introduced in (4.3) and (D.13), respectively. Denote by $D^*_{\mathfrak{z}}$ the rectangle introduced in (D.20). By applying (D.11) combined with (D.34), one has

$$\sup_{0\leq t\leq t_0} \|u(t)\|^2_{L^2_b} + \nu \sup_{\mathfrak{z}\in\mathbb{R}} \int_0^{t_0} \|\nabla u(t)\|^2_{L^2(D_{\mathfrak{z}})} \mathrm{d}t \leq C_\nu. \tag{D.35}$$

Testing the equation (D.2) against $-\operatorname{div}(\psi_\mathfrak{z}^2\nabla u)$ gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\psi_\mathfrak{z}\nabla u\|^2+\nu\langle\Pi\Delta u,\operatorname{div}(\psi_\mathfrak{z}^2\nabla u)\rangle=\langle\Pi((u+z)\cdot\nabla)(u+z),\operatorname{div}(\psi_\mathfrak{z}^2\nabla u)\rangle. \tag{D.36}$$

Each term in the above energy equality is estimated in the following way. We begin with the dissipation term. Since

$$\operatorname{div}(\psi_\mathfrak{z}^2\nabla u)=\psi_\mathfrak{z}^2\Delta u+2\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u,$$

it follows

$$\begin{aligned}\langle\Pi\Delta u,\operatorname{div}(\psi_\mathfrak{z}^2\nabla u)\rangle&=\langle\Pi\Delta u,\psi_\mathfrak{z}^2\Delta u\rangle+\langle\Pi\Delta u,2\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u\rangle\\&=\|\psi_\mathfrak{z}\Pi\Delta u\|^2+\langle\Pi\Delta u,[\Pi,\psi_\mathfrak{z}^2]\Delta u\rangle+\langle\Pi\Delta u,2\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u\rangle,\end{aligned} \tag{D.37}$$

where $[\Pi,\psi_\mathfrak{z}^2]:=\Pi\psi_\mathfrak{z}^2-\psi_\mathfrak{z}^2\Pi$. The second term on the right-hand side of (D.37) is estimated by using Lemma 6.3[2] in [AZ14], Lemma C.2, Corollary C.3, the trace and interpolation inequalities combined with integrating by parts, which gives

$$\begin{aligned}|\langle\Pi\Delta u,[\Pi,\psi_\mathfrak{z}^2]\Delta u\rangle|&\le|\langle\nabla u,\nabla\Pi[\Pi,\psi_\mathfrak{z}^2]\Delta u\rangle|+|\langle\partial_n u,\Pi[\Pi,\psi_\mathfrak{z}^2]\Delta u\rangle_{\partial D}|\\&\le C\|\theta_{\varepsilon,\mathfrak{z}}\nabla u\|\|\theta_{\varepsilon,\mathfrak{z}}^{-2}[\Pi,\psi_\mathfrak{z}^2]\Delta u\|_{H^1}+\|\theta_{\varepsilon,\mathfrak{z}}\partial_y u\|_{H^{\frac{3}{4}}}\|\theta_{\varepsilon,\mathfrak{z}}^{-2}[\Pi,\psi_\mathfrak{z}^2]\Delta u\|_{H^1}\\&\le C\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^{\frac{3}{2}}\|\theta_{\varepsilon,\mathfrak{z}}u\|^{\frac{1}{2}}+\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^{\frac{15}{8}}\|\theta_{\varepsilon,\mathfrak{z}}u\|^{\frac{1}{8}}\\&\le\delta\nu^{-1}\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2+C_{\delta,\nu}\|\theta_{\varepsilon,\mathfrak{z}}u\|^2.\end{aligned}$$

Together with (D.37), Lemma 6.3 in [AZ14], and

$$2|\langle\Pi\Delta u,\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u\rangle|\le\frac{1}{4}\|\psi_\mathfrak{z}\Pi\Delta u\|^2+C\|\nabla u\|_{L^2(D_\mathfrak{z}^*)}^2,$$

this implies

$$\begin{aligned}\nu\langle\Pi\Delta u,\operatorname{div}(\psi_\mathfrak{z}^2\nabla u)\rangle&\ge\frac{3\nu}{4}\|\psi_\mathfrak{z}\Pi\Delta u\|^2-\delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2-C_{\delta,\nu}\|\theta_{\varepsilon,\mathfrak{z}}u\|^2-C_\nu\|\nabla u\|_{L^2(D_\mathfrak{z}^*)}^2\\&\ge\frac{3\nu}{8}\left(\|\psi_\mathfrak{z}\Pi\Delta u\|^2+c\|\psi_\mathfrak{z}u\|_{H^2}^2\right)\\&\qquad-\delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2-C_{\delta,\nu}\|\theta_{\varepsilon,\mathfrak{z}}u\|^2-C_\nu\|\nabla u\|_{L^2(D_\mathfrak{z}^*)}^2.\end{aligned} \tag{D.38}$$

We turn to the nonlinear terms. To this end, notice that

$$\langle\Pi f,\operatorname{div}(\psi_\mathfrak{z}^2\nabla u)\rangle=\langle\psi_\mathfrak{z}f,\psi_\mathfrak{z}\Pi\Delta u\rangle+\langle f,[\Pi,\psi_\mathfrak{z}^2]\Delta u\rangle+\langle f,\Pi(2\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u)\rangle,$$

where, by Lemma 6.3 in [AZ14] and Lemma C.2, one has

$$\|\theta_{\varepsilon,\mathfrak{z}}^{-2}\nabla[\Pi,\psi_\mathfrak{z}^2]\Delta u\|+\|\theta_{\varepsilon,\mathfrak{z}}^{-2}\nabla\Pi(2\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u)\|\lesssim\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}. \tag{D.39}$$

We begin with

$$\begin{aligned}\mathcal{I}_3:&=\langle\Pi(u\cdot\nabla)u,\operatorname{div}(\psi_\mathfrak{z}^2\nabla u)\rangle\\&=\langle(u\cdot\nabla)(\psi_\mathfrak{z}u),\psi_\mathfrak{z}\Pi\Delta u\rangle-\langle u_1\psi_\mathfrak{z}'u,\psi_\mathfrak{z}\Pi\Delta u\rangle\\&\qquad+\langle(u\cdot\nabla)u,[\Pi,\psi_\mathfrak{z}^2]\Delta u\rangle+\langle(u\cdot\nabla)u,\Pi(2\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u)\rangle\\&=:\mathcal{I}_{3,1}+\mathcal{I}_{3,2}+\mathcal{I}_{3,3}+\mathcal{I}_{3,4}.\end{aligned} \tag{D.40}$$

[2]This lemma does not rely on the smallness assumption of $(\psi^2)''$, which is not assumed in (D.13).

For $\mathcal{I}_{3,1}$ and $\mathcal{I}_{3,2}$, by the Ladyzhenskaya and Young inequalities combined with the support property of $\psi_\mathfrak{z}$, it follows

$$\begin{aligned}
|\mathcal{I}_{3,1}| &\le C\|u\|_{L^4(D_\mathfrak{z}^*)}\|\nabla(\psi_\mathfrak{z} u)\|_4\|\psi_\mathfrak{z}\Pi\Delta u\| \\
&\le C\|u\|_{L^4(D_\mathfrak{z}^*)}\|\nabla(\psi_\mathfrak{z} u)\|^{\frac{1}{2}}\|\psi_\mathfrak{z} u\|_{H^2}^{\frac{1}{2}}\|\psi_\mathfrak{z}\Pi\Delta u\| \\
&\le \mathfrak{a}\left(\|\psi_\mathfrak{z}\Pi\Delta u\|^2 + \|\psi_\mathfrak{z} u\|_{H^2}^2\right) + C_\mathfrak{a}\|u\|_{L^4(D_\mathfrak{z}^*)}^4\|\nabla(\psi_\mathfrak{z} u)\|^2 \\
&\le \mathfrak{a}\left(\|\psi_\mathfrak{z}\Pi\Delta u\|^2 + \|\psi_\mathfrak{z} u\|_{H^2}^2\right) + C_\mathfrak{a}\|u\|_{L^2(D_\mathfrak{z}^*)}^2\|u\|_{H^1(D_\mathfrak{z}^*)}^2\left(\|\psi_\mathfrak{z}\nabla u\|^2 + \|u\|_{L^2(D_\mathfrak{z}^*)}^2\right),
\end{aligned}$$

and

$$\begin{aligned}
|\mathcal{I}_{3,2}| &\le C\|u\|_{L^4(D_\mathfrak{z}^*)}^2\|\psi_\mathfrak{z}\Pi\Delta u\| \le \mathfrak{a}\|\psi_\mathfrak{z}\Pi\Delta u\|^2 + C_\mathfrak{a}\|u\|_{L^4(D_\mathfrak{z}^*)}^4 \\
&\le \mathfrak{a}\|\psi_\mathfrak{z}\Pi\Delta u\|^2 + C_\mathfrak{a}\|u\|_{L^2(D_\mathfrak{z}^*)}^2\|u\|_{H^1(D_\mathfrak{z}^*)}^2.
\end{aligned}$$

The terms $\mathcal{I}_{3,3}, \mathcal{I}_{3,4}$ are bounded by integrating by parts and using (D.39) and the Gagliardo–Nirenberg inequality

$$\|f\|_4 \lesssim \|f\|^{\frac{3}{4}}\|f\|_{H^2}^{\frac{1}{4}},$$

which yields

$$\begin{aligned}
|\mathcal{I}_{3,3}| &= |\langle u\otimes u, \nabla[\Pi, \psi_\mathfrak{z}^2]\Delta u\rangle| \le C\|\theta_{\varepsilon,\mathfrak{z}} u\|_4^2\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2} \\
&\le C\|\theta_{\varepsilon,\mathfrak{z}} u\|^{\frac{3}{2}}\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2}^{\frac{3}{2}} \le \delta\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2}^2 + C_\delta\|\theta_{\varepsilon,\mathfrak{z}} u\|^6,
\end{aligned}$$

and

$$|\mathcal{I}_{3,4}| = \langle u\otimes u, \nabla\Pi(2\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u)\rangle \le C\|\theta_{\varepsilon,\mathfrak{z}} u\|_4^2\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2} \le \delta\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2}^2 + C_\delta\|\theta_{\varepsilon,\mathfrak{z}} u\|^6.$$

Plugging the above estimates into (D.40) gives

$$\begin{aligned}
|\mathcal{I}_3| &\le \mathfrak{a}\left(\|\psi_\mathfrak{z}\Pi\Delta u\|^2 + \|\psi_\mathfrak{z} u\|_{H^2}^2\right) + \delta\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2}^2 \\
&\quad + C_\mathfrak{a}\|u\|_{L^2(D_\mathfrak{z}^*)}^2\|u\|_{H^1(D_\mathfrak{z}^*)}^2\left(1 + \|\psi_\mathfrak{z}\nabla u\|^2 + \|u\|_{L^2(D_\mathfrak{z}^*)}^2\right) + C_\delta\|\theta_{\varepsilon,\mathfrak{z}} u\|^6. \quad \text{(D.41)}
\end{aligned}$$

Next, we estimate

$$\begin{aligned}
\mathcal{I}_4 &:= \langle\Pi(z\cdot\nabla)u, \operatorname{div}(\psi_\mathfrak{z}^2\nabla u)\rangle \\
&= \langle\psi_\mathfrak{z}(z\cdot\nabla)u, \psi_\mathfrak{z}\Pi\Delta u\rangle + \langle(z\cdot\nabla)u, [\Pi, \psi_\mathfrak{z}^2]\Delta u\rangle + \langle(z\cdot\nabla)u, \Pi(2\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u)\rangle \\
&=: \mathcal{I}_{4,1} + \mathcal{I}_{4,2} + \mathcal{I}_{4,3},
\end{aligned}$$

where, by proceeding similarly as in the derivation of (D.41), one has

$$|\mathcal{I}_{4,1}| \le \|z\|_\infty\|\psi_\mathfrak{z}\nabla u\|\|\psi_\mathfrak{z}\Pi\Delta u\| \le \mathfrak{a}\|\psi_\mathfrak{z}\Pi\Delta u\|^2 + C_\mathfrak{a}\|z\|_\infty^2\|\psi_\mathfrak{z}\nabla u\|^2,$$

and

$$\begin{aligned}
|\mathcal{I}_{4,2}| &= \left|\langle u\otimes z, \nabla[\Pi, \psi_\mathfrak{z}^2]\Delta u\rangle\right| \le C\|z\|_\infty\|\theta_{\varepsilon,\mathfrak{z}} u\|\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2} \\
&\le \delta\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2}^2 + C_\delta\|z\|_\infty^2\|\theta_{\varepsilon,\mathfrak{z}} u\|^2,
\end{aligned}$$

and

$$\begin{aligned}
|\mathcal{I}_{4,3}| &= \left|\langle u\otimes z, \nabla\Pi(2\psi_\mathfrak{z}\psi_\mathfrak{z}'\partial_x u)\rangle\right| \le C\|z\|_\infty\|\theta_{\varepsilon,\mathfrak{z}} u\|\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2} \\
&\le \delta\|\theta_{\varepsilon,\mathfrak{z}} u\|_{H^2}^2 + C_\delta\|z\|_\infty^2\|\theta_{\varepsilon,\mathfrak{z}} u\|^2.
\end{aligned}$$

Consequently, it follows

$$|\mathcal{I}_4| \le \delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2 + \mathfrak{a}\|\psi_{\mathfrak{z}}\Pi\Delta u\|^2 + C_\delta\|z\|_\infty^2\|\theta_{\varepsilon,\mathfrak{z}}u\|^2 + C_{\mathfrak{a}}\|z\|_\infty^2\|\psi_{\mathfrak{z}}\nabla u\|^2. \tag{D.42}$$

The term

$$\begin{aligned}\mathcal{I}_5 :&= \langle\Pi(u\cdot\nabla)z, \operatorname{div}(\psi_{\mathfrak{z}}^2\nabla u)\rangle\\ &= \langle\psi_{\mathfrak{z}}(u\cdot\nabla)z, \psi_{\mathfrak{z}}\Pi\Delta u\rangle + \langle(u\cdot\nabla)z, [\Pi,\psi_{\mathfrak{z}}^2]\Delta u\rangle + \langle(u\cdot\nabla)z, \Pi(2\psi_{\mathfrak{z}}\psi_{\mathfrak{z}}'\partial_x u)\rangle\\ &=: \mathcal{I}_{5,1}+\mathcal{I}_{5,2}+\mathcal{I}_{5,3}\end{aligned}$$

is estimated in a similar way, which gives

$$\begin{aligned}|\mathcal{I}_{5,1}| &\le \|\nabla z\|_4\|\psi_{\mathfrak{z}}u\|_4\|\psi_{\mathfrak{z}}\Pi\Delta u\| \le \mathfrak{a}\|\psi_{\mathfrak{z}}\Pi\Delta u\|^2 + C_{\mathfrak{a}}\|\nabla z\|_4^2\|\psi_{\mathfrak{z}}u\|_4^2\\ &\le \mathfrak{a}\|\psi_{\mathfrak{z}}\Pi\Delta u\|^2 + C_{\mathfrak{a}}\|\nabla z\|_4^2\|\psi_{\mathfrak{z}}u\|\left(\|\psi_{\mathfrak{z}}\nabla u\| + \|u\|_{L^2(D_{\mathfrak{z}}^*)}\right),\end{aligned}$$

and

$$\begin{aligned}|\mathcal{I}_{5,2}| &= \left|\langle z\otimes u, \nabla[\Pi,\psi_{\mathfrak{z}}^2]\Delta u\rangle\right| \le C\|z\|_\infty\|\theta_{\varepsilon,\mathfrak{z}}u\|\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}\\ &\le \delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2 + C_\delta\|z\|_\infty^2\|\theta_{\varepsilon,\mathfrak{z}}u\|^2,\end{aligned}$$

and

$$\begin{aligned}|\mathcal{I}_{5,3}| &= \left|\langle z\otimes u, \nabla\Pi(2\psi_{\mathfrak{z}}\psi_{\mathfrak{z}}'\partial_x u)\rangle\right| \le C\|z\|_\infty\|\theta_{\varepsilon,\mathfrak{z}}u\|\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}\\ &\le \delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2 + C_\delta\|z\|_\infty^2\|\theta_{\varepsilon,\mathfrak{z}}u\|^2.\end{aligned}$$

Therefore, one has

$$\begin{aligned}|\mathcal{I}_5| \le \delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2 &+ \mathfrak{a}\|\psi_{\mathfrak{z}}\Pi\Delta u\|^2 + C_\delta\|z\|_\infty^2\|\theta_{\varepsilon,\mathfrak{z}}u\|^2\\ &+ C_{\mathfrak{a}}\|\nabla z\|_4^2\|\psi_{\mathfrak{z}}u\|\left(\|\psi_{\mathfrak{z}}\nabla u\| + \|u\|_{L^2(D_{\mathfrak{z}}^*)}\right).\end{aligned} \tag{D.43}$$

The last term

$$\begin{aligned}\mathcal{I}_6 :&= \langle\Pi(z\cdot\nabla)z, \operatorname{div}(\psi_{\mathfrak{z}}^2\nabla u)\rangle\\ &= \langle\psi_{\mathfrak{z}}(z\cdot\nabla)z, \psi_{\mathfrak{z}}\Pi\Delta u\rangle + \langle(z\cdot\nabla)z, [\Pi,\psi_{\mathfrak{z}}^2]\Delta u\rangle + \langle(z\cdot\nabla)z, \Pi(2\psi_{\mathfrak{z}}\psi_{\mathfrak{z}}'\partial_x u)\rangle\\ &=: \mathcal{I}_{6,1}+\mathcal{I}_{6,2}+\mathcal{I}_{6,3}\end{aligned}$$

is bounded in the same way, which gives

$$|\mathcal{I}_{6,1}| \le \|\psi_{\mathfrak{z}}\nabla z\|\|z\|_\infty\|\psi_{\mathfrak{z}}\Pi\Delta u\| \le \mathfrak{a}\|\psi_{\mathfrak{z}}\Pi\Delta u\|^2 + C_{\mathfrak{a}}\|\psi_{\mathfrak{z}}\nabla z\|^2\|z\|_\infty^2,$$

and

$$|\mathcal{I}_{6,2}| = \left|\langle z\otimes z, \nabla[\Pi,\psi_{\mathfrak{z}}^2]\Delta u\rangle\right| \le C\|\theta_{\varepsilon,\mathfrak{z}}z\|_4^2\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2} \le \delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2 + C_\delta\|\theta_{\varepsilon,\mathfrak{z}}z\|_4^4,$$

and

$$|\mathcal{I}_{6,3}| = \left|\langle z\otimes z, \nabla\Pi(2\psi_{\mathfrak{z}}\psi_{\mathfrak{z}}'\partial_x u)\rangle\right| \le C\|\theta_{\varepsilon,\mathfrak{z}}z\|_4^2\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2} \le \delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2 + C_\delta\|\theta_{\varepsilon,\mathfrak{z}}z\|_4^4.$$

As a result, it follows

$$|\mathcal{I}_6| \le \delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2 + \mathfrak{a}\|\psi_{\mathfrak{z}}\Pi\Delta u\|^2 + C_\delta\|\theta_{\varepsilon,\mathfrak{z}}z\|_4^4 + C_{\mathfrak{a}}\|\psi_{\mathfrak{z}}\nabla z\|^2\|z\|_\infty^2. \tag{D.44}$$

Summing (D.41)–(D.44) and applying (D.34) and (D.35), one gets

$$\begin{aligned}\textstyle\sum_{i=3}^6|\mathcal{I}_i| &\le \mathfrak{a}\left(\|\psi_{\mathfrak{z}}\Pi\Delta u\|^2 + \|\psi_{\mathfrak{z}}u\|_{H^2}^2\right) + \delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2 + C_{\mathfrak{a},\delta}\\ &\quad + C_{\mathfrak{a}}\|u\|_{H^1(D_{\mathfrak{z}}^*)}^2\left(1+\|\psi_{\mathfrak{z}}\nabla u\|^2\right) + C_{\mathfrak{a}}\left(1+\|\psi_{\mathfrak{z}}\nabla u\|^2\right).\end{aligned} \tag{D.45}$$

Plugging (D.38) and (D.45) into (D.36) and choosing $\mathfrak{a}$ sufficiently small imply

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\psi_\mathfrak{z}\nabla u\|^2+\frac{\nu}{4}\left(\|\psi_\mathfrak{z}\Pi\Delta u\|^2+c\|\psi_\mathfrak{z}u\|_{H^2}^2\right)\leq\delta\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2+C_{\nu,\delta}\left(1+\|u\|_{H^1(D_\mathfrak{z}^*)}^2\right)\\+C_\nu\left(1+\|u\|_{H^1(D_\mathfrak{z}^*)}^2\right)\|\psi_\mathfrak{z}\nabla u\|^2,$$

which gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(t\|\psi_\mathfrak{z}\nabla u\|^2\right)+\frac{c\nu t}{4}\|\psi_\mathfrak{z}u\|_{H^2}^2\leq\delta t\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2+C_{\nu,\delta}\left(1+\|u\|_{H^1(D_\mathfrak{z}^*)}^2\right)\\+C_\nu\left(1+\|u\|_{H^1(D_\mathfrak{z}^*)}^2\right)t\|\psi_\mathfrak{z}\nabla u\|^2$$

for any $t\in(0,t_0]$. An application of the Gronwall inequality then yields

$$t\|\psi_\mathfrak{z}\nabla u(t)\|^2+\frac{c\nu}{4}\int_0^t s\|\psi_\mathfrak{z}u\|_{H^2}^2\mathrm{d}s\leq C_1\delta\int_0^t s\|\theta_{\varepsilon,\mathfrak{z}}u\|_{H^2}^2\mathrm{d}s+C_2,\tag{D.46}$$

where C_1 depends only on ν and the bound in (D.35), and C_2 depends additionally on δ. Together with repeating the convolution-type argument used in the derivation of (D.28) and choosing δ sufficiently small, this gives

$$\sup_{s\in(0,t_0]}s\|u(s)\|_{H_b^1}^2+\sup_{\mathfrak{z}\in\mathbb{R}}\int_0^{t_0}s\|u(s)\|_{H^2(D_\mathfrak{z})}^2\mathrm{d}s\leq C.\tag{D.47}$$

Step II. Next, we establish L^∞-bounds. Due to the Gagliardo–Nirenberg inequality

$$\|f\|_\infty\lesssim\|f\|_{L_b^2}^{\frac{1}{5}}\|f\|_{W_b^{3/4,4}}^{\frac{4}{5}},\tag{D.48}$$

it suffices to bound the $W_b^{3/4,4}$-norm. To this end, we apply (C.2) and (C.3) combined with the interpolation inequality

$$\|f\|_{W_b^{3/4,4}}\lesssim\|f\|_{L_b^4}^{\frac{1}{4}}\|f\|_{W_b^{1,4}}^{\frac{3}{4}},\tag{D.49}$$

which yields

$$\|S(t)u\|_{W_b^{3/4,4}}\lesssim_\nu t^{-\frac{3}{8}}\|u\|_{L_b^4},\qquad\|S(t)\Pi\operatorname{div}f\|_{W_b^{3/4,4}}\lesssim_\nu t^{-\frac{7}{8}}\|f\|_{L_b^4}.\tag{D.50}$$

Moreover, by (D.35), (D.47), and the Gagliardo–Nirenberg inequalities

$$\|f\|_{L_b^4}\lesssim\|f\|_{L_b^2}^{\frac{1}{2}}\|f\|_{H_b^1}^{\frac{1}{2}},\qquad\|f\|_{L_b^8}\lesssim\|f\|_{L_b^2}^{\frac{1}{4}}\|f\|_{H_b^1}^{\frac{3}{4}},$$

it follows

$$\|u(t)\|_{L_b^4}\lesssim_\nu t^{-\frac{1}{4}},\qquad\|u(t)\|_{L_b^8}\lesssim_\nu t^{-\frac{3}{8}},\tag{D.51}$$

which, together with (D.34), yields

$$\|(u(t)+z(t))\otimes(u(t)+z(t))\|_{L_b^4}\lesssim_\nu t^{-\frac{3}{4}}.\tag{D.52}$$

Now applying the estimates (D.50)–(D.52) to the Duhamel formula

$$u(t)=S(t/2)u(t/2)-\int_{t/2}^t S(t-s)\Pi\operatorname{div}\big((u(s)+z(s))\otimes(u(s)+z(s))\big)\mathrm{d}s$$

yields

$$\|u(t)\|_{W_b^{3/4,4}}\lesssim t^{-\frac{3}{8}}\|u(t/2)\|_{L_b^4}+t^{-\frac{3}{4}}\int_{t/2}^t(t-s)^{-\frac{7}{8}}\mathrm{d}s\lesssim t^{-\frac{5}{8}}.\tag{D.53}$$

Together with (D.35) and (D.48), this implies

$$\|u(t)\|_\infty \lesssim t^{-\frac{1}{2}}, \qquad \forall t \in (0, t_0].$$

Combining this with (D.34) and (D.35), we obtain (5.13).

Step Ⅲ. It remains to prove (5.15). To this end, we first derive some probability estimates for the Wiener process $W(t) := \int_0^t \eta(s)\mathrm{d}s$. Since $\mathfrak{B}_1 < \infty$, for any $s \le t$, $W(t) - W(s)$ is a Gaussian random variable valued in V. This yields

$$\mathbb{P}\left\{\|W(t) - W(s)\|_V > \rho\right\} \lesssim \exp\left(-\frac{\rho^2}{4\mathfrak{B}_1|t-s|}\right).$$

Applying this estimate to dyadic increments gives

$$\mathbb{P}\left\{[W]_{C^{3/8}([0,1];V)} > \rho\right\} \lesssim \exp\left(-\frac{C\rho^2}{\mathfrak{B}_1}\right). \tag{D.54}$$

We turn to the proof of (5.15). By the stochastic integration by parts formula, see (2.102) in [KS12], one has

$$\mathcal{Z}_t = e^{\nu t L} W(t) - \nu \int_0^t L e^{\nu(t-s)L}(W(t) - W(s))\mathrm{d}s,$$

which, together with the classical Stokes semigroup estimate

$$\|L^\gamma e^{\nu t L}\|_{\mathcal{L}(H)} \le C_\gamma t^{-\gamma},$$

and the embedding $D(L^{3/4}) \hookrightarrow H^{\frac{3}{2}}$, leads to

$$\begin{aligned}
\|\mathcal{Z}_t\|_{H^{\frac{3}{2}}} &\lesssim_\nu \|L^{3/4} e^{\nu t L} W(t)\| + \int_0^t \|L^{7/4} e^{\nu(t-s)L}(W(t) - W(s))\|\mathrm{d}s \\
&\lesssim_\nu t^{-\frac{1}{4}} \|W(t)\|_V + \int_0^t (t-s)^{-\frac{5}{4}} \|W(t) - W(s)\|_V \mathrm{d}s \\
&\lesssim_\nu [W]_{C^{3/8}([0,t];V)} \left(t^{\frac{1}{8}} + \int_0^t (t-s)^{-\frac{7}{8}} \mathrm{d}s\right) \lesssim_\nu t^{\frac{1}{8}} [W]_{C^{3/8}([0,t];V)}.
\end{aligned}$$

Consequently,

$$\sup_{0 \le s \le t} \|\mathcal{Z}_s\|_{H^{\frac{3}{2}}} \lesssim_\nu t^{\frac{1}{8}} [W]_{C^{3/8}([0,t];V)}. \tag{D.55}$$

By rescaling, $[W]_{C^{3/8}([0,t];V)}$ has the same law as $t^{1/8}[W]_{C^{3/8}([0,1];V)}$. Combining this identity with (D.54) and (D.55), one gets (5.15). □

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